Started on	Tuesday, 16 April 2024, 4:12 PM
State	Finished
Completed on	Tuesday, 16 April 2024, 6:56 PM
Time taken	2 hours 44 mins
Grade	**123.00** out of 150.00 (**82**%)

Question 1

Correct

Mark 1.00 out of 1.00

A 34-year-old woman being treated for seizures has irregular menstrual periods, at which time her seizures worsen. It is most likely that she has which of the following?

Select one:

- Catamenial epilepsy
- Interictal anxiety
- Interictal psychosis
- Post-ictal depression
- Rolandic epilepsy

Your answer is correct.

Catamenial (menstrual) epilepsy describes a worsening of seizures in relation to the menstrual cycle and may affect around 40% of women with epilepsy. Catamenial epilepsy has a significant negative impact on quality of life. There is uncertainty regarding which treatment works best and when in the menstrual cycle treatments should be taken.

Ref: Maguire MJ, Nevitt SJ. Treatments for seizures in catamenial (menstrual-related) epilepsy. Cochrane Database of Systematic Reviews. 2019(10).

The correct answer is: Catamenial epilepsy

Question 2

Correct

Mark 1.00 out of 1.00

A 45-year-old patient with bipolar I disorder has recently started taking a new medication. He develops sudden and severe epigastric pain that radiates to his back, as well as vomiting. Which of the following medications is most likely to be responsible?

Select one:

- Sodium valproate
- Lithium
- Lamotrigine
- Carbamazepine
- Topiramate

Your answer is correct.

Pancreatitis presents as sudden and severe epigastric pain that radiates to the back and is often associated with vomiting. Systemic inflammation results in temperature spikes and patients can rapidly lose intravascular volume requiring aggressive fluid resuscitation. Though the most common causes of pancreatitis are gallstones and alcohol, drug-induced pancreatitis has been associated with antipsychotic and mood stabiliser use, in particular sodium valproate.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 702.

The correct answer is: Sodium valproate

Question 3

Correct

Mark 1.00 out of 1.00

A 52-year-old patient on maintenance treatment for bipolar I disorder is noted to have hypercalcemia on routine blood work. Which of the following agents is most likely to be responsible?

Select one:

- Lithium
- Valproate
- Topiramate
- Carbamazepine
- Haloperidol

Your answer is correct.

Barbiturates, carbamazepine, haloperidol, and valproate have all been reported to lower calcium levels, while lithium has been reported to raise them. Lithium increases the risk of hyperparathyroidism, and some recommend that calcium levels should be monitored in patients on long-term treatment. Clinical consequences of chronically increased serum calcium levels include renal stones, osteoporosis, dyspepsia, hypertension, and renal impairment.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 250, 909.

The correct answer is: Lithium

Question 4

Correct

Mark 1.00 out of 1.00

A 58-year-old woman presents with confusion, nystagmus, and unilateral paresis of eye abduction. Though her drug screen is negative, her daughter suspects that the woman is drinking alcohol frequently. Which of the following is the most likely diagnosis?

Select one:

- Wernicke encephalopathy
- Midbrain infarction
- Oculomotor nerve palsy
- Myasthenia gravis
- Multiple sclerosis

Your answer is correct.

Some conditions that affect eye movements also impair cognitive function. Wernicke encephalopathy consists of memory impairment/confusion accompanied by nystagmus and oculomotor or abducens nerve abnormalities. Abducens nerve (sixth cranial nerve) impairment leads to inward deviation (adduction) of the eye.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 28, 31, 360.

The correct answer is: Wernicke encephalopathy

Question 5

Correct

Mark 1.00 out of 1.00

A pregnant woman with bipolar I disorder is scheduled for induction of labour next week. Her lithium dose has been increased in recent weeks, and she has a current level of 0.75 mmol/L. She is not planning to breastfeed. What is the most advisable course of management?

Select one:

- Suspend lithium 24-48 hours before induction and re-check the level 12 hours after the last dose
- Reduce the lithium dose to 400 mg and re-check the level 1 week post-delivery
- Maintain current lithium dosing and re-check the level 4 weeks post-delivery
- Check the lithium level daily until her induction, then discontinue lithium until 4 weeks post-delivery
- As she is not planning to breastfeed, no changes in lithium dosing or further bloodwork are advised

Your answer is correct.

In the third trimester, the use of lithium may be problematic because of changing pharmacokinetics: an increasing dose of lithium is required to maintain the lithium level during pregnancy as total body water increases, but requirements return abruptly to pre-pregnancy levels immediately after delivery. To prevent maternal and neonatal intoxication, lithium should be suspended for 24-48 hours before a planned Caesarean section or induction, and at the onset of labour, and the lithium level should be measured 12 hours after the last dose. If levels are not above the therapeutic range, lithium should be restarted on day 1 postnatal and the level checked again after 1 week.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 690.
McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.
Uguz F, et al. Prophylactic management of women with bipolar disorder during pregnancy and the perinatal period. Journal of clinical psychopharmacology. 2023 Sep 1;43(5):434-52.

The correct answer is: Suspend lithium 24-48 hours before induction and re-check the level 12 hours after the last dose

Question 6

Correct

Mark 1.00 out of 1.00

A 55-year-old man presents with low mood and associated depressive symptoms two weeks after a myocardial infarction. Which of the following would be the best choice for his treatment?

Select one:

- Fluvoxamine
- Sertraline
- Moclobemide
- Amitriptyline
- Citalopram

Your answer is correct.

EXPLANATION: SSRIs may protect against myocardial infarction, and untreated depression worsens prognosis in cardiovascular disease. Post MI, SSRIs and mirtazapine have either a neutral or beneficial effect on mortality. Treatment of depression with SSRIs should not therefore be withheld post-MI. Sertraline is the drug of choice, and it is preferred over TCAs. Fluvoxamine and citalopram/escitalopram should be avoided. If sedation is required, consider the use of mirtazapine or a small dose of trazodone at night.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 386.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 1033.

The correct answer is: Sertraline

Question 7

Incorrect

Mark 0.00 out of 1.00

A man diagnosed with HIV and a history of epilepsy is depressed. Which of the following would be the best choice for treatment of his depression?

Select one:

- Trazodone
- Amitriptyline
- Phenelzine
- Mirtazapine
- Bupropion

Your answer is incorrect.

EXPLANATION: SSRIs are preferrable as first-line agents for the treatment of depression in people living with HIV (PLWH). Escitalopram and citalopram have a low risk of pharmacokinetic interactions. Mirtazapine is also effective, with a relatively low risk of drug interactions, and may be beneficial in coexisting HIV wasting and depression or in reducing methamphetamine use among active users. 'Dual action' antidepressants (duloxetine, venlafaxine) are as effective as SSRIs for depressive symptoms in PLWH. Other agents such as bupropion and trazodone are effective, but their utility is limited by drug interactions and side effects. The side effect burden of TCAs may limit efficacy and compliance; constipation and dry mouth are frequently reported in PLWH on TCAs. MAOIs are not recommended in PLWH. SSRIs, mirtazapine, and duloxetine are all considered low risk and good choices as antidepressants in people with epilepsy. Bupropion and TCAs are considered high risk in epilepsy and should be avoided due to their propensity to be proconvulsive at therapeutic doses.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 778-779, 789.

The correct answer is: Mirtazapine

Question 8

Correct

Mark 1.00 out of 1.00

With respect to prophylaxis in bipolar disorder, the BALANCE study showed which of the following findings?

Select one:

- The combination of lithium and valproate is superior to lithium
- Valproate is superior to lithium
- Lithium is superior to valproate
- The combination of lithium and lamotrigine is superior to lithium
- Lamotrigine is superior to lithium

Your answer is correct.

EXPLANATION: The independent BALANCE study found that valproate as monotherapy was relatively less effective than lithium or the combination of lithium and valproate, casting doubt on valproate's use as a single first-line prophylactic treatment in bipolar I disorder. BALANCE found lithium to be numerically superior to valproate, and the combination of lithium and valproate to be statistically superior to valproate alone. Given this and the fact that valproate is not licenced for prophylaxis, it should be considered a second-line treatment.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 258, 296.

The correct answer is: Lithium is superior to valproate

Question 9

Correct

Mark 1.00 out of 1.00

With respect to first-generation antipsychotics (FGAs) compared to second-generation antipsychotics (SGAs), which of the following was a major finding of the CUtLASS study?

Select one:

FGAs show greater improvement on quality of life

FGAs are superior in efficacy to SGAs

SGAs show greater improvement on quality of life

SGAs are superior in efficacy to FGAs

Efficacy is similar between FGAs and SGAs

Your answer is correct.

EXPLANATION: In CUtLASS (Cost Utility of the Latest Antipsychotic drugs in Schizophrenia Study), a 12-month open-label trial, 277 patients were randomised to receive either an FGA or an SGA. Efficacy was similar between the two groups, with only limited improvement of psychopathology and quality of life. The authors concluded that SGAs do not markedly differ from FGAs regarding compliance, quality of life, or effectiveness.

Ref: Naber D, Lambert M. The CATIE and CUtLASS studies in schizophrenia: results and implications for clinicians. CNS drugs. 2009 Aug;23:649-59.

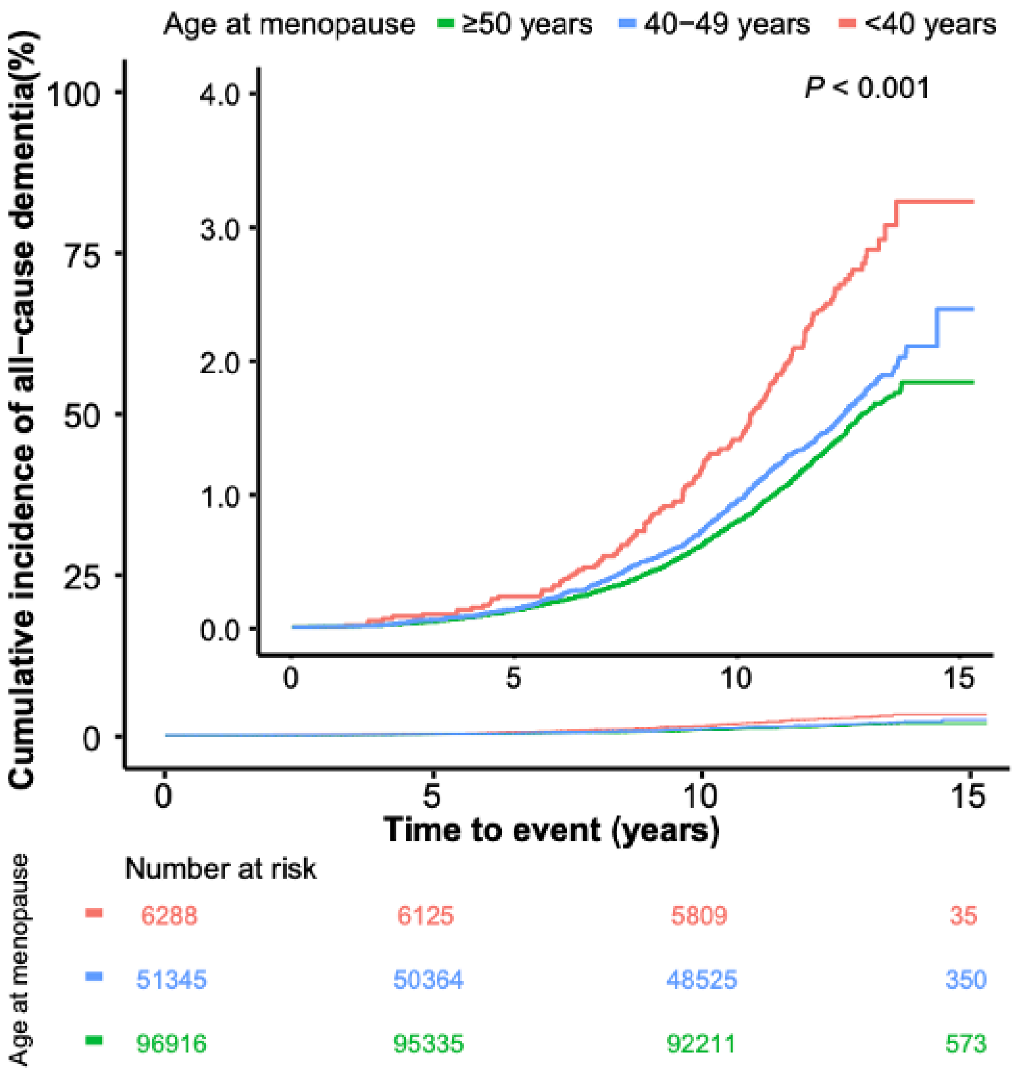
The correct answer is: Efficacy is similar between FGAs and SGAs

Question 10

Correct

Mark 1.00 out of 1.00

In a community-based cohort study, 154,549 postmenopausal women without dementia at enrolment (between 2006 and 2010) from the UK Biobank were followed up until June 2021. Age at menopause (<40, 40-49, and =50 years) was used to predict the primary outcome of all-cause dementia in a time-to-event analysis. The results are shown in the graph below. In total, 2266 incident cases of dementia were observed over a median follow-up period of 12.3 years. After adjusting for confounders, women with earlier menopause showed a higher risk of all-cause dementia compared with those ≥50 years (adjusted-HRs [95% CIs]: 1.21 [1.09-1.34] and 1.71 [1.38-2.11] in the 40-49 years and <40 years groups, respectively; P for trend <0.001). Ref: Liao H, et al. eClinicalMedicine. 2023 Jun 1;60. Which of the following is an accurate interpretation of the reported results?



Select one:

- Age at menopause is more important than one's chronological age for the risk of dementia

Earlier age at menopause is associated with risk of incident dementia

Earlier age at menopause is associated with increased overall mortality rates

Onset of menopause increases the risk of dementia within 5 years

The cumulative risk of dementia plateaus after 12-13 years of observation

Your answer is correct.

From the results presented, we can conclude that earlier age at menopause (premature onset by 40 years) is associated with a higher risk of incident dementia compared to later onset menopause. Nevertheless, the three groups with differing ages of onset of menopause do not differ from each other in the cumulative incident rates over the first 5 years of observation; any deviations appear to occur only with a longer period of observation (around 10 years). We are unable to determine if age at menopause is more important than one's chronological age for the risk of dementia, as no direct comparison has been reported. We cannot infer the differences in mortality rates, as this has not been reported. Finally, the flattening of the survival curves after 13 years is most likely reflective of the lack of further observational data, rather than a plateauing of dementia risk. It is reported in the precis of the study that the median follow-up period was 12.3 years.

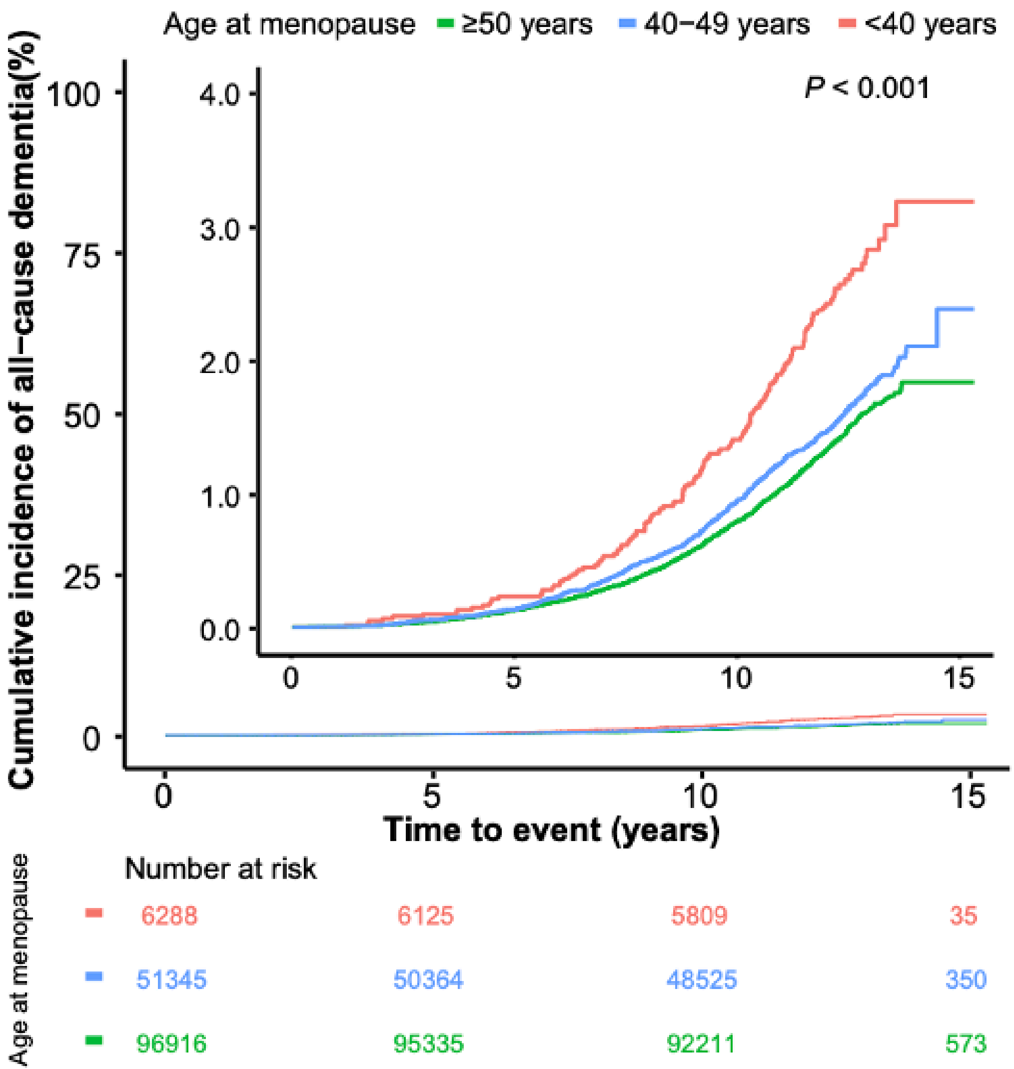
The correct answer is: Earlier age at menopause is associated with risk of incident dementia

Question 11

Correct

Mark 1.00 out of 1.00

In a community-based cohort study, 154,549 postmenopausal women without dementia at enrolment (between 2006 and 2010) from the UK Biobank were followed up until June 2021. Age at menopause (<40, 40-49, and =50 years) was used to predict the primary outcome of all-cause dementia in a time-to-event analysis. The results are shown in the graph below. In total, 2266 incident cases of dementia were observed over a median follow-up period of 12.3 years. After adjusting for confounders, women with earlier menopause showed a higher risk of all-cause dementia compared with those ≥50 years (adjusted-HRs [95% CIs]: 1.21 [1.09-1.34] and 1.71 [1.38-2.11] in the 40-49 years and <40 years groups, respectively; P for trend <0.001). Ref: Liao H, et al. eClinicalMedicine. 2023 Jun 1;60. What type of graph is displayed here?



Select one:

- Growth chart

Kaplan Meier Curve

Regression plot

Scatter-plot

Time trend graph

Your answer is correct.

The survival analysis presented here shows three Kaplan–Meier survival curves, each demonstrating the cumulative incidence over time for all-cause dementia for groups with different age at menopause.

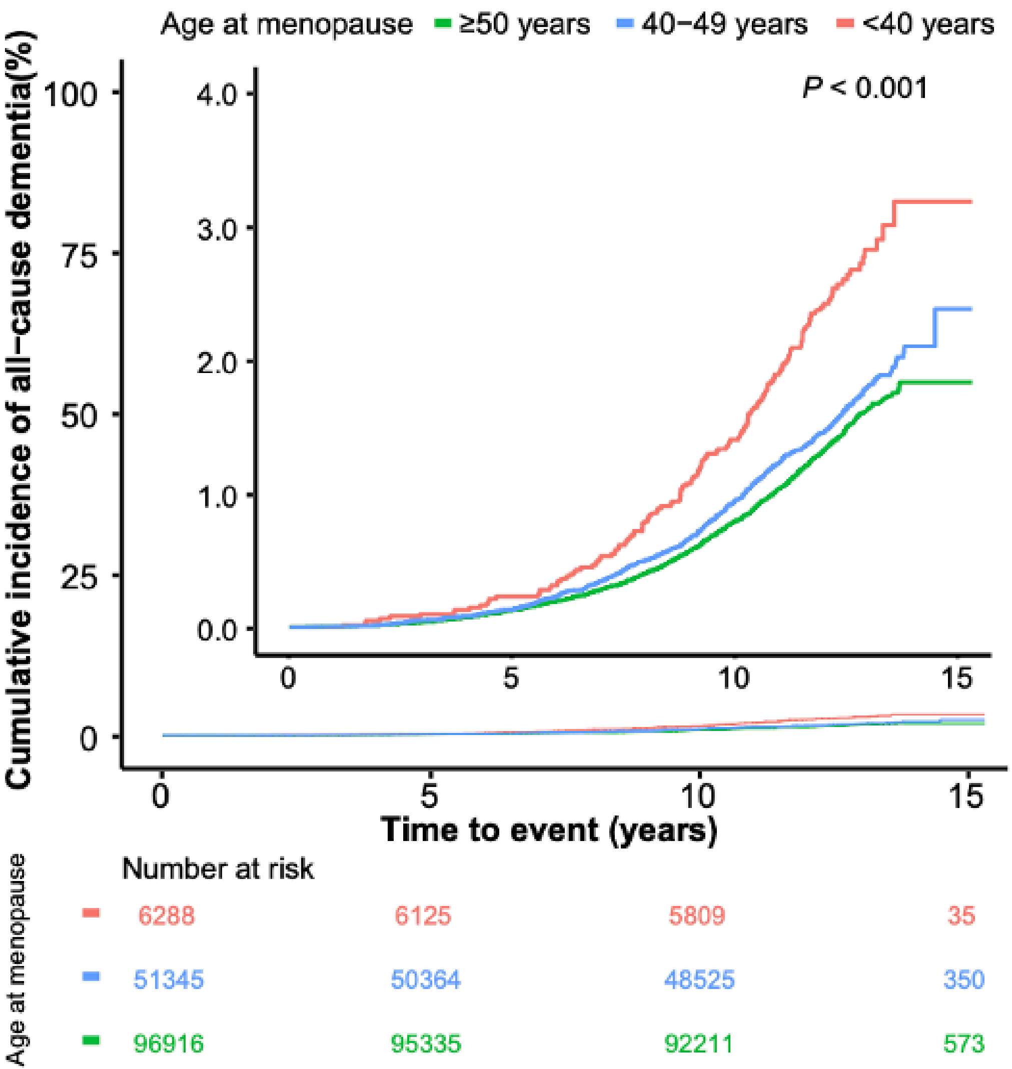
The correct answer is: Kaplan Meier Curve

Question 12

Incorrect

Mark 0.00 out of 1.00

In a community-based cohort study, 154,549 postmenopausal women without dementia at enrolment (between 2006 and 2010) from the UK Biobank were followed up until June 2021. Age at menopause (<40, 40-49, and =50 years) was used to predict the primary outcome of all-cause dementia in a time-to-event analysis. The results are shown in the graph below. In total, 2266 incident cases of dementia were observed over a median follow-up period of 12.3 years. After adjusting for confounders, women with earlier menopause showed a higher risk of all-cause dementia compared with those ≥50 years (adjusted-HRs [95% CIs]: 1.21 [1.09-1.34] and 1.71 [1.38-2.11] in the 40-49 years and <40 years groups, respectively; P for trend <0.001). Ref: Liao H, et al. eClinicalMedicine. 2023 Jun 1;60. Which of the following inferences can be made from the above graph?



Select one:

- 5-10 years after menopause, nearly 25% of all post-menopausal women developed dementia.

- Most women in the cohort had menopause after the age of 49.
- 75-80% of those who had menopause before age 40 developed dementia in the 15 years of observation.
- The highest rate of onset of dementia was in those younger than age 40.
- The lowest rate of onset of dementia was in those younger than age 40.

Your answer is incorrect.

Of the study participants, 96,916 women seems to have experienced menopause =50 years, 51,345 women at 40–49 years, and 6288 women at <40 years. These baseline characteristics are displayed as numerical values in different colours below the graph. It is important to note that the graph itself has 2 different scales; one where y axis is scaled from 0 to 100% of incidence i.e. from no cases of dementia to all of the cohort developing dementia. As the cumulative incidence is low (around 3% even in the highest risk group in red), a rescaled graph with y axis ending at 4% incidence is shown as an insert. It is important to read the correct scale when interpreting the graphs. As per this inserted graph, 5-10 year after menopause, <1% develop dementia; <3% who had menopause before the age of 40 develop dementia over the 15 years of observation. Note that the groups are stratified not according to the age of the individuals per se but the age at which they reached menopause.

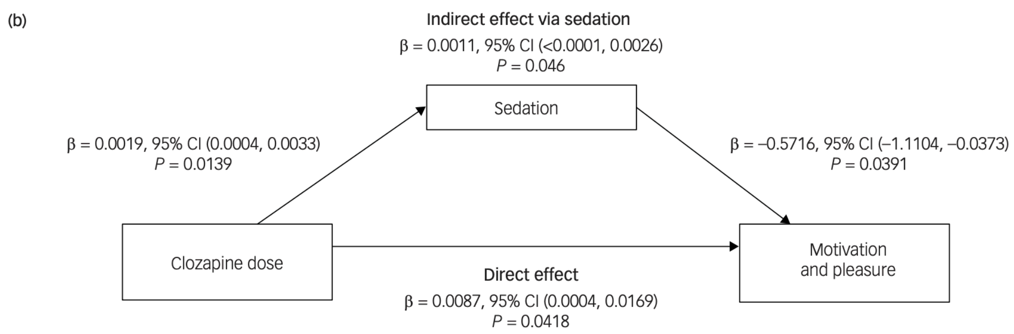
The correct answer is: Most women in the cohort had menopause after the age of 49.

Question **13**

Correct

Mark 1.00 out of 1.00

To understand the longitudinal effects of clozapine on motivation and pleasure (negative symptoms), Wolpe and colleagues studied a cohort of 187 clozapine-treated patients with schizophrenia followed for over 2 years on average. The authors examined the severity of reduced motivation and pleasure (MAP) and the effect of antipsychotic-induced sedation (assessed via the proxy of total daily sleep duration) on MAP. The result of their analysis is summarised in the graph below. They concluded that addressing sedative side-effects of antipsychotics is important to improve clinical outcomes. Which of the following conclusions is appropriate based on the statistical analysis presented here?



Select one:

- Clozapine dose needs to be increased when motivation is high
- Clozapine reduces sedation
- Clozapine directly affects motivation and pleasure
- Being sedated increases motivation and pleasure
- Low motivation causes to sedation

Your answer is correct.

According to the mediation model results, increased sedation levels were linked to reduced MAP levels (Note the negative sign of β= -0.57, P = 0.039)), and clozapine increased sedation levels (β= 0.0019, P = 0.0139), suggesting that clozapine-related sedation worsened MAP within participants (indirect effect). However, after accounting for the clozapine-related sedation (in other words, in the absence of sedation), clozapine improved MAP within participants (direct effect; β= 0.0087, P = 0.042). The results do not support increasing clozapine dose as this can worsen sedation and thus reduce motivation.

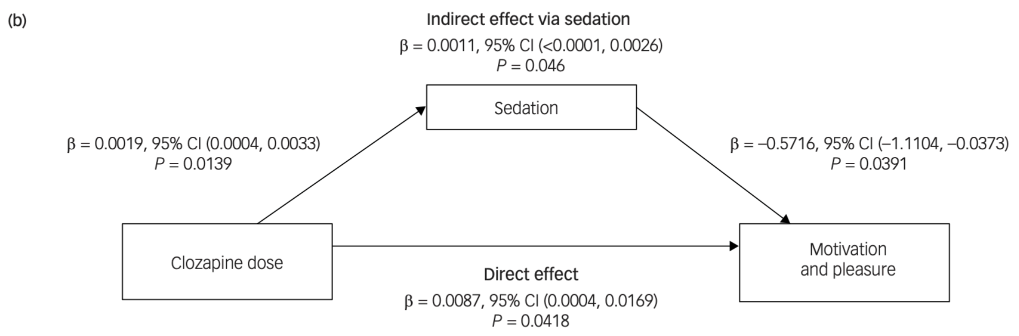
The correct answer is: Clozapine directly affects motivation and pleasure

Question **14**

Correct

Mark 1.00 out of 1.00

To understand the longitudinal effects of clozapine on motivation and pleasure (negative symptoms), Wolpe and colleagues studied a cohort of 187 clozapine-treated patients with schizophrenia followed for over 2 years on average. The authors examined the severity of reduced motivation and pleasure (MAP) and the effect of antipsychotic-induced sedation (assessed via the proxy of total daily sleep duration) on MAP. The result of their analysis is summarised in the graph below. They concluded that addressing sedative side-effects of antipsychotics is important to improve clinical outcomes. Which of the following is the LEAST likely clinical implication of this study?



Select one:

- Regular assessment of sedation in clozapine users is warranted
- Optimisation of clozapine dose may reduce negative symptoms
- Clozapine should not be used in patients with low motivation
- Sedation can have a deleterious effect on motivation
- Sedation is a potentially treatable cause of motivation deficits

Your answer is correct.

There is no observation that is reported here that supports a reduction in the use of clozapine in clinics. In fact, the results support the optimisation of clozapine dose to minimise secondary negative symptoms and highlight the deleterious effect of clozapine- induced sedation on motivation, calling for regular assessment of sedation as a reversible cause of motivation deficits in schizophrenia.

The correct answer is: Clozapine should not be used in patients with low motivation

Question **15**

Correct

Mark 1.00 out of 1.00

A screening instrument has a pre-test probability of 1 in 8 with 80% sensitivity and 80% specificity. Calculate the post-test odds of a positive test result.

Select one:

- 1 in 2
- 1 in 3
- 4 in 7
- 2 in 1
- 7 in 4

Your answer is correct.

EXPLANATION: To calculate the post-test odds, we need to first determine the pre-test odds and then apply the likelihood ratio.

STEP 1: To calculate Pretest odds:

Pre-test Odds = Pre-test Probability / (1 - Pre-test Probability)

Pre-test Odds = (1/8) / (1 - 1/8)

Pre-test Odds = (1/8) / (7/8)

Pre-test Odds = 1/7

STEP 2: To calculate the likelihood ratio (LR+):

LR+ = Sensitivity / (1 - Specificity)

LR+ = 0.80 / (1 - 0.80)

LR+ = 0.80 / 0.20

LR+ = 4

STEP 3: Finally, to calculate the post-test odds:

Post-test Odds = Pre-test Odds x LR+

Post-test Odds = (1/7) x 4

Post-test Odds = 4/7

So, the correct post-test odds are 4 in 7.

Ref: Owen A. Diagnostic Tests. Statistics for Clinicians. 2023 Jun 3:107-20.

The correct answer is: 4 in 7

Question **16**

Correct

Mark 1.00 out of 1.00

A researcher spends one month collecting the published news from all British tabloids since the Covid pandemic started that contained the term "mental health" and analyses the patterns of reporting. What is the appropriate term to describe this type of research?

Select one:

- Textual Scrutiny
- Discourse Analysis
- Lexical Examination
- Content Analysis
- Information Scrutiny

Your answer is correct.

EXPLANATION: Content analysis is a research method in which a researcher systematically collects and examines textual or visual data, such as articles, documents, or media content, to uncover patterns, themes, and trends within the material. In this case, the researcher collected all articles containing the term "mental health" from the media, which is a characteristic of content analysis. Therefore, the correct answer is Content Analysis. The other options are not typically used terms for this specific research method and are less appropriate in describing the process of analyzing media articles for specific content.

Ref: Martin F, Oliver T. A qualitative content analysis of online public mental health resources for COVID-19. Frontiers in Psychiatry. 2022 Feb 24;13:553158.

The correct answer is: Content Analysis

Question 17

Correct

Mark 1.00 out of 1.00

A research team is evaluating the performance of a newly developed diagnostic test for psychotic relapse in established schizophrenia using speech samples. The prevalence of relapses per year)in the observed population is estimated to be 37% of individuals. The researchers recruit a sample of 1,000 individuals, all of whom are at a higher risk of relapse due to recent discontinuation of their treatment. The initial sensitivity of the new speech-based test is found to be 80%. This means that out of those individuals who have relapse, the test correctly identifies 80% as relapsed. However, the specificity of the test is very high, indicating that it accurately identifies individuals without relapse as negative. Using the collected data, the research team wants to assess how an increase in sensitivity of the speech test might impact various statistical measures, including Positive Predictive Value (PPV), Negative Predictive Value (NPV), and statistical power of studies that employ this speech test to select patients that are relapsing. They plan to use a machine learning approach that increases the sensitivity of their test notably, increasing the test's clinical utility and its potential to detect relapse more effectively. When the sensitivity of a test increases

Select one:

- PPV increases, NPV remains relatively stable, and power increases.
- PPV decreases, NPV increases, and power remains relatively stable.
- PPV and NPV both increase, but power decreases.
- PPV, NPV, and power all increase.
- PPV and NPV both decrease, while power increases.

Your answer is correct.

EXPLANATION: When the sensitivity of a test increases, it means that the test is better at correctly identifying true positive cases. As a result, PPV (Positive Predictive Value) tends to increase because the proportion of true positives among positive test results is higher. NPV (Negative Predictive Value) may remain relatively stable because the increase in sensitivity primarily affects true positives and has a lesser impact on true negatives. Power in studies that use the new test increases because the test's ability to detect true positives is enhanced, leading to a higher probability of correctly rejecting the null hypothesis. Other options are less likely to occur when sensitivity increases.

The correct answer is: PPV increases, NPV remains relatively stable, and power increases.

Question **18**

Correct

Mark 1.00 out of 1.00

What minimizes type 2 error?

Select one:

- Decreasing the power
- Reducing variability
- Decreasing sample size
- Reducing the confidence interval levels
- Reducing the P value set as the threshold for significance

Your answer is correct.

Error Type	**Description**	**Symbol**
Type 1 Error *False Positive*	Incorrectly rejecting a null hypothesis when it's true.	α (Alpha)
Type 2 Error *False Negative*	Incorrectly failing to reject a null hypothesis when it's false.	β (Beta)

EXPLANATION: Increasing variability in data increases the spread of observations, making it more difficult to detect differences. Reducing variability will reduce type 2 error (i.e., the rate of failure to find an effect by rejecting the null hypothesis when such an effect truly exists); the other options do not typically minimize Type 2 error, they may indeed inflate it. As sample size increases, type 2 error will decrease.

Increasing the p value (from 0.05 to 0.10, for example) or the widening the confidence interval (from 95% to 90%, correspondingly) will make it easier to find an effect, though at the expense of higher likelihood of committing a Type 1 error, which occurs when you detect a false effect or difference in your sample that does not really exist. The power of a study is a function that decreases directly with type 2 error. Lower the power, higher the type 2 error.

The correct answer is: Reducing variability

Question 19

Correct

Mark 1.00 out of 1.00

In which of the following scenarios is it more appropriate to use the McNemar test instead of the chi-square test?

Select one:

- Investigating the relationship between gender and income in a survey dataset.
- Analysing the frequency of insomnia before and after interpersonal therapy for depression.
- Assessing the independence of two categorical variables in a large sample.
- Evaluating the frequency of voice hearing experience in a sample of children attending a local high school.
- Comparing the response rate for two different medications offered to two different intervention arms in a RCT.

Your answer is correct.

EXPLANATION: The McNemar test is particularly suitable when dealing with paired categorical data, such as in before-after studies in this question. It accounts for the dependency between observations within each pair. None of the other options involve paired data with dependency among observations. When comparing response rates in two different interventional arms, the conventional chi-square test is more appropriate. The conventional chi-square test assumes independence of the cells (frequency counts). Experimental designs such as before-and-after studies, repeated measures of proportions and matched case control studies create dependency among observations either by observing the same categorical outcome more than once in the same patient, or by matching pairs of individuals on a set of variables. The McNemar test (also called the paired or matched chi-square) can be used in these cases for testing the hypotheses in such designs.

Ref: Pembury Smith MQ, Ruxton GD. Effective use of the McNemar test. Behavioral Ecology and Sociobiology. 2020 Nov;74:1-9.

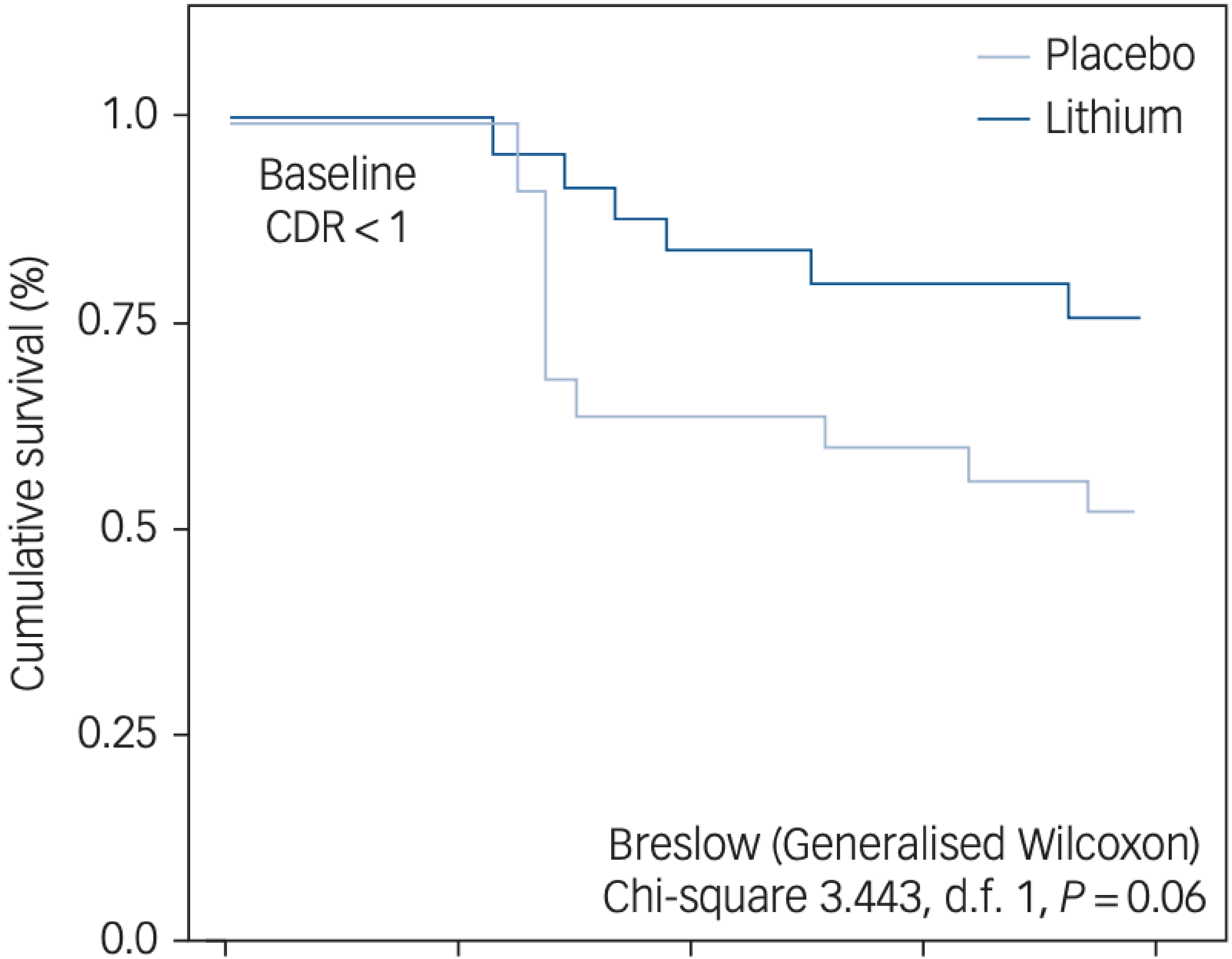
The correct answer is: Analysing the frequency of insomnia before and after interpersonal therapy for depression.

Question 20

Correct

Mark 1.00 out of 1.00

A study aimed at investigating the potential benefits of lithium treatment in individuals with amnestic mild cognitive impairment (MCI) was published recently (Forlenza OV, et al., Clinical and biological effects of long-term lithium treatment in older adults with amnestic mild cognitive impairment: randomised clinical trial. The British Journal of Psychiatry. 2019 Nov;215(5):668-74). The research involved 61 older adults with MCI who received either lithium or a placebo for two years, followed by a 24-month follow-up phase. The results showed that those in the placebo group experienced cognitive and functional decline, while the lithium-treated participants remained stable over the two-year period. Lithium treatment was associated with improved memory and attention test performance after 24 months, as well as increased levels of a biomarker associated with Alzheimer's disease in cerebrospinal fluid after 36 months. These findings suggest that long-term lithium treatment may slow cognitive decline and impact Alzheimer's disease-related biomarkers in individuals with MCI, indicating its potential disease-modifying properties. The following graph was provided by the authors to interpret the study results. What is the rate of conversion to dementia at month 40 for the lithium group?


Placebo
Lithium
1.0
0.75
0.5
0.25
0.0
Cumulative survival (%)
Baseline
CDR < 1
Breslow (Generalised Wilcoxon)
Chi-square 3.443, d.f. 1, P = 0.06

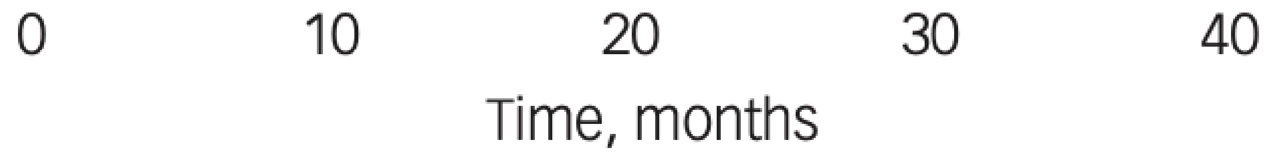


Fig. 2 Kaplan–Meier curves indicating the rate of conversion from mild cognitive impairment (MCI) to dementia, according to exposure or not to lithium treatment. All patients with MCI had baseline scores on the Clinical Dementia Rating Scale (CDR) <1.

Select one:

- 50%
- 25%
- 75%
- Less than 1%
- 80%

Your answer is correct.

EXPLANATION: In this Kaplan Meier curve, the value you read from the y-axis at the chosen time point is the estimated probability of the survival from an event of interest (diagnostic conversion from MCI to dementia) happening up to that time. For example, if you read a value close to 0.75 at 40 months for the Lithium curve, it means there was a 75% estimated cumulative probability of having not experienced dementia yet within the first 40 months. Conversely, there was nearly 25% probability (i.e., 100-75% = 25%) of having converted to dementia during this period.

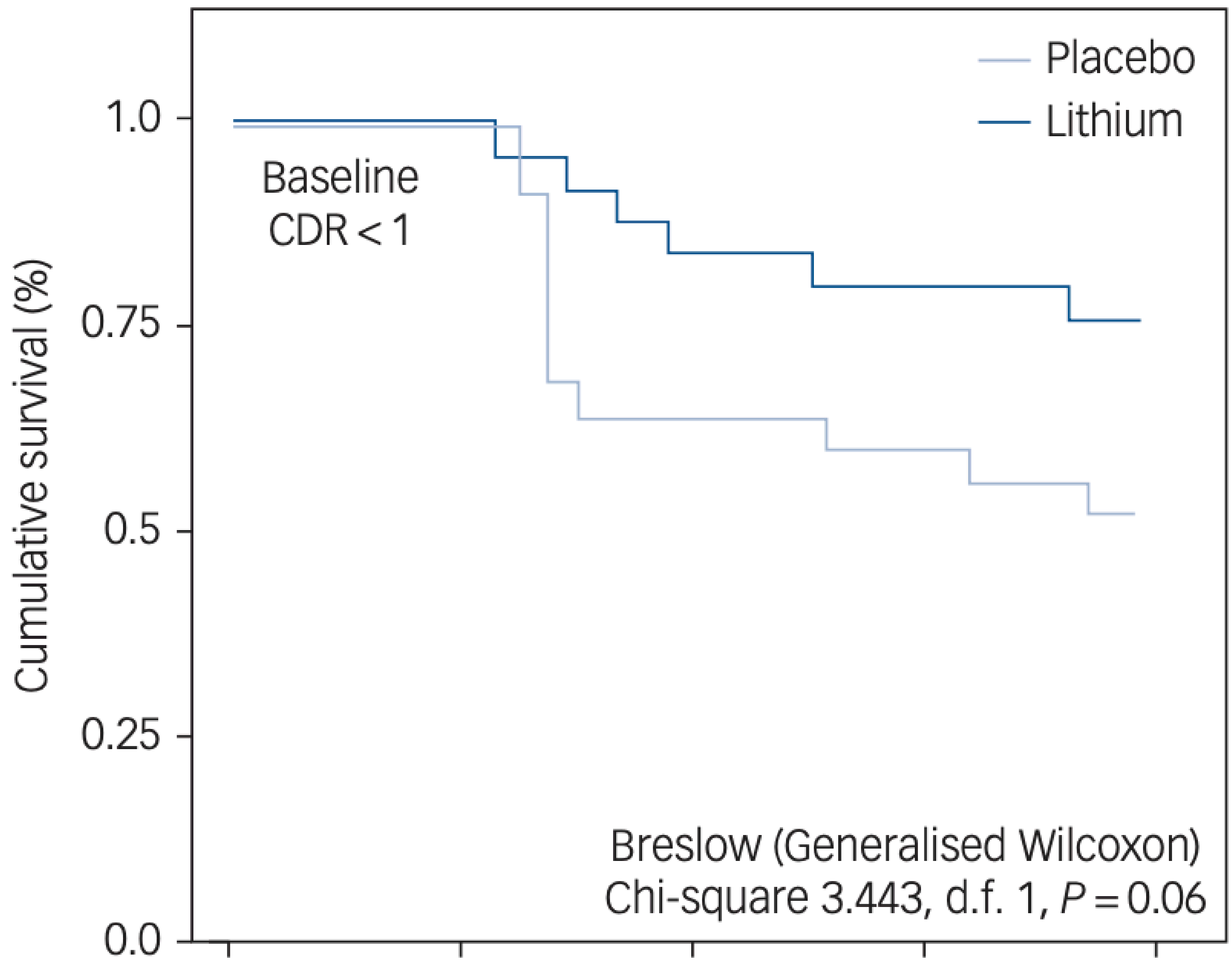
The correct answer is: 25%

Question **21**

Correct

Mark 1.00 out of 1.00

A study aimed at investigating the potential benefits of lithium treatment in individuals with amnestic mild cognitive impairment (MCI) was published recently (Forlenza OV, et al., Clinical and biological effects of long-term lithium treatment in older adults with amnestic mild cognitive impairment: randomised clinical trial. The British Journal of Psychiatry. 2019 Nov;215(5):668-74). The research involved 61 older adults with MCI who received either lithium or a placebo for two years, followed by a 24-month follow-up phase. The results showed that those in the placebo group experienced cognitive and functional decline, while the lithium-treated participants remained stable over the two-year period. Lithium treatment was associated with improved memory and attention test performance after 24 months, as well as increased levels of a biomarker associated with Alzheimer's disease in cerebrospinal fluid after 36 months. These findings suggest that long-term lithium treatment may slow cognitive decline and impact Alzheimer's disease-related biomarkers in individuals with MCI, indicating its potential disease-modifying properties. The following graph was provided by the authors to interpret the study results. From the Kaplan Meier curve given here, what is the absolute benefit increase when using lithium for 3 to 4 years if someone has mild cognitive impairment?


Placebo
Lithium
Baseline
CDR < 1
Cumulative survival (%)
1.0
0.75
0.5
0.25
0.0
Breslow (Generalised Wilcoxon)
Chi-square 3.443, d.f. 1, P = 0.06

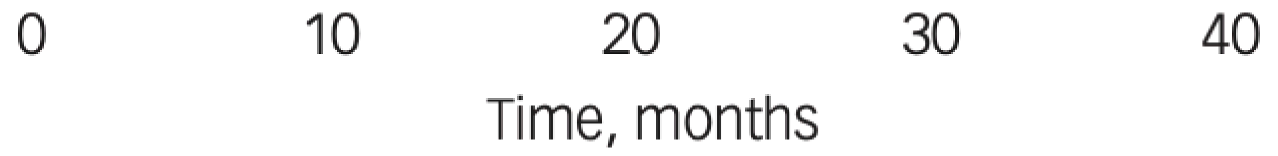


Fig. 2 Kaplan–Meier curves indicating the rate of conversion from mild cognitive impairment (MCI) to dementia, according to exposure or not to lithium treatment. All patients with MCI had baseline scores on the Clinical Dementia Rating Scale (CDR) <1.

Select one:

- 80%
- Less than 1%
- 75%
- 25%
- 50%

Your answer is correct.

EXPLANATION: To calculate the absolute benefit increase, you can subtract the event rate in the placebo group from the event rate in the lithium group. In this case, you are comparing the conversion rates from mild cognitive impairment (MCI) to dementia between the two groups:

Absolute Benefit Increase = Survival Rate in Lithium Group - Survival Rate in Placebo Group

Survival Rate in Lithium Group = 75%

Survival Rate in Placebo Group = 50%

Absolute Benefit Increase = 75% - 50% = 25%

The absolute benefit increase is 25%.

In this context, it means that the lithium treatment group had a 25% lower rate of conversion from MCI to dementia compared to the placebo group. This indicates a potential benefit of lithium in reducing the risk of progression to dementia in individuals with MCI.

The correct answer is: 25%

Question 22

Correct

Mark 1.00 out of 1.00

A study aimed at investigating the potential benefits of lithium treatment in individuals with amnestic mild cognitive impairment (MCI) was published recently (Forlenza OV, et al., Clinical and biological effects of long-term lithium treatment in older adults with amnestic mild cognitive impairment: randomised clinical trial. The British Journal of Psychiatry. 2019 Nov;215(5):668-74). The research involved 61 older adults with MCI who received either lithium or a placebo for two years, followed by a 24-month follow-up phase. The results showed that those in the placebo group experienced cognitive and functional decline, while the lithium-treated participants remained stable over the two-year period. Lithium treatment was associated with improved memory and attention test performance after 24 months, as well as increased levels of a biomarker associated with Alzheimer's disease in cerebrospinal fluid after 36 months. These findings suggest that long-term lithium treatment may slow cognitive decline and impact Alzheimer's disease-related biomarkers in individuals with MCI, indicating its potential disease-modifying properties. The following graph was provided by the authors to interpret the study results. From the Kaplan Meier curve given here, what conclusions can you make?

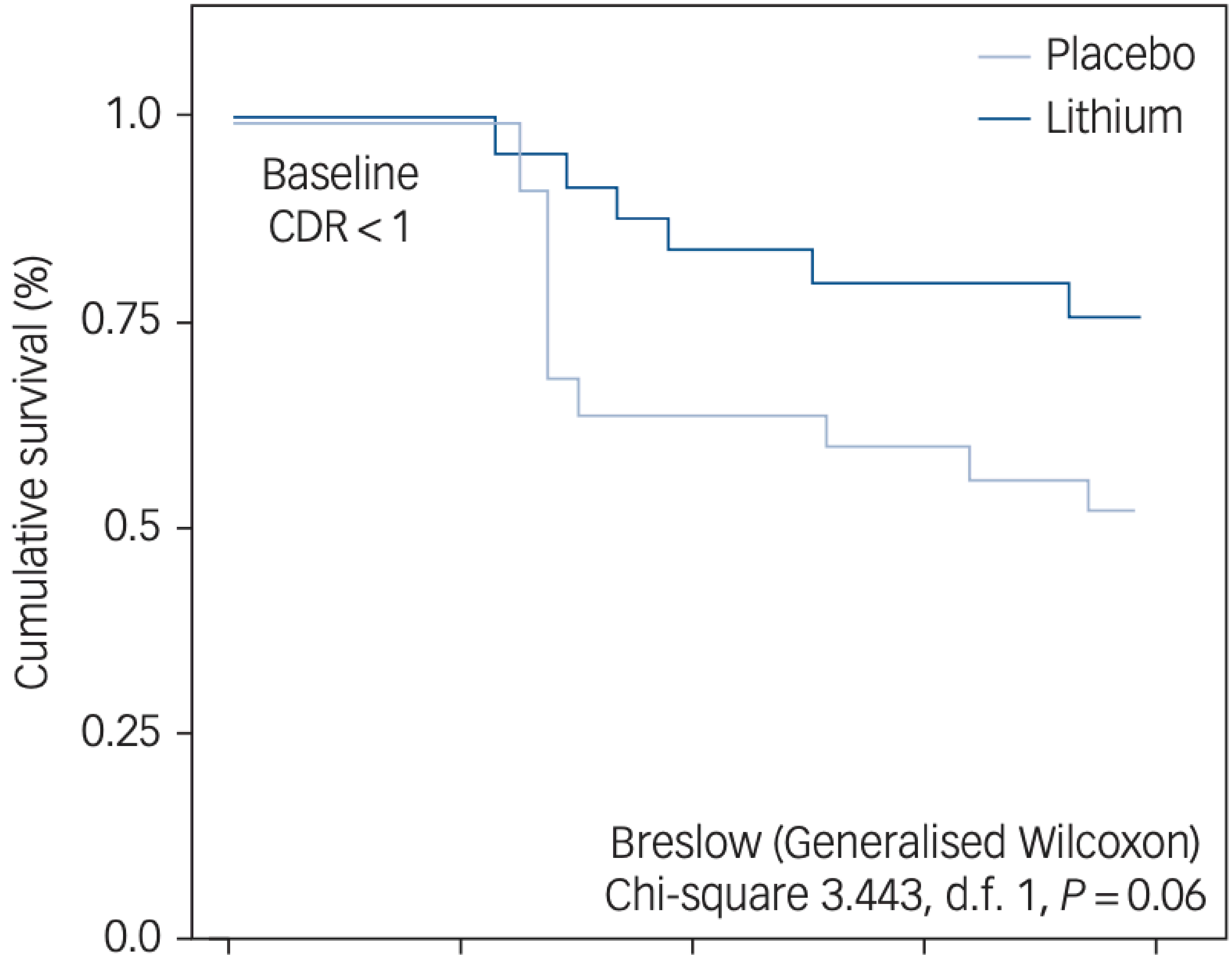

Placebo
Lithium
Baseline
CDR < 1
Cumulative survival (%)
1.0
0.75
0.5
0.25
0.0
Breslow (Generalised Wilcoxon)
Chi-square 3.443, d.f. 1, P = 0.06

Fig. 2 Kaplan–Meier curves indicating the rate of conversion from mild cognitive impairment (MCI) to dementia, according to exposure or not to lithium treatment. All patients with MCI had baseline scores on the Clinical Dementia Rating Scale (CDR) <1.

Select one:

- Lithium has no meaningful clinical effect on reducing conversion to dementia
- The study has sufficient power to demonstrate the effect of lithium in preventing dementia
- Lithium has a statistically significant effect in preventing conversion to dementia
- Lithium significantly worsens the rate of conversion to dementia
- Lithium has a marginal statistical significance in reducing conversion to dementia

Your answer is correct.

EXPLANATION: The p-value of 0.06 suggests that there is some evidence for a difference between the two groups in terms of the conversion rate from MCI to dementia, but this evidence falls just short of the conventional threshold for statistical significance (typically set at 0.05 or lower). This can be termed as a "trend" towards significance, indicating that there may be a meaningful difference between the groups, but the study did not have enough statistical power (sample size) to reach conventional significance. Despite the lack of statistical significance, a difference in survival rates from 50% to 75% between the lithium and placebo groups could have clinical relevance. It may be worthwhile to further investigate this trend in larger studies to determine if lithium has a potential impact on delaying the progression from MCI to dementia.

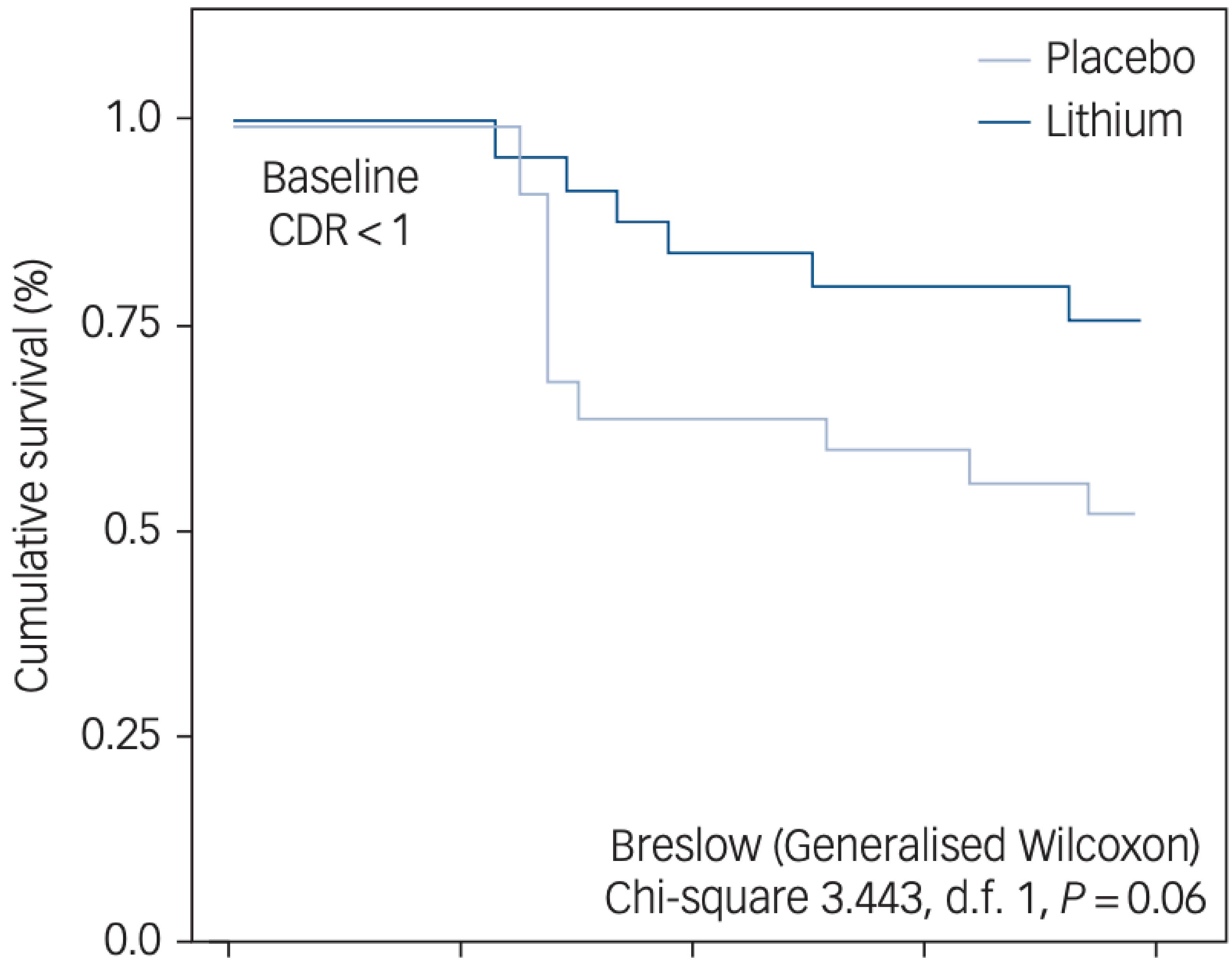
The correct answer is: Lithium has a marginal statistical significance in reducing conversion to dementia

Question 23

Incorrect

Mark 0.00 out of 1.00

A study aimed at investigating the potential benefits of lithium treatment in individuals with amnestic mild cognitive impairment (MCI) was published recently (Forlenza OV, et al., Clinical and biological effects of long-term lithium treatment in older adults with amnestic mild cognitive impairment: randomised clinical trial. The British Journal of Psychiatry. 2019 Nov;215(5):668-74). The research involved 61 older adults with MCI who received either lithium or a placebo for two years, followed by a 24-month follow-up phase. The results showed that those in the placebo group experienced cognitive and functional decline, while the lithium-treated participants remained stable over the two-year period. Lithium treatment was associated with improved memory and attention test performance after 24 months, as well as increased levels of a biomarker associated with Alzheimer's disease in cerebrospinal fluid after 36 months. These findings suggest that long-term lithium treatment may slow cognitive decline and impact Alzheimer's disease-related biomarkers in individuals with MCI, indicating its potential disease-modifying properties. The following graph was provided by the authors to interpret the study results. In the above Kaplan-Meier survival analysis, a chi-square test has been conducted with degree of freedom being 1. What does this degree of freedom represent?


Placebo
Lithium
1.0
0.75
0.5
0.25
0.0
Cumulative survival (%)
Baseline
CDR < 1
Breslow (Generalised Wilcoxon)
Chi-square 3.443, d.f. 1, P = 0.06

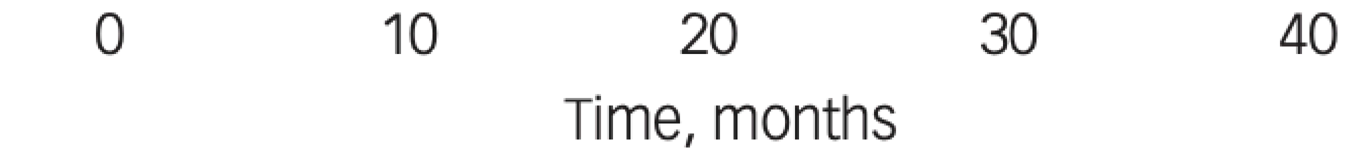


Fig. 2 Kaplan–Meier curves indicating the rate of conversion from mild cognitive impairment (MCI) to dementia, according to exposure or not to lithium treatment. All patients with MCI had baseline scores on the Clinical Dementia Rating Scale (CDR) <1.

Select one:

- The precision of the survival estimates.
- The number of individuals in the study.
- The level of statistical significance.
- The number of time intervals considered in the analysis.
- The number of groups being compared in the analysis.

Your answer is incorrect.

EXPLANATION: In a Kaplan-Meier survival analysis with a chi-square test, the degree of freedom (df) typically corresponds to the number of groups or categories being compared. This indicates how many groups are being assessed for differences in survival or event outcomes over time. Chi-square tests are used to evaluate the independence or association between categorical variables, and the degree of freedom reflects the number of categories minus 1. To be precise, in a chi-square test, we typically set up a contingency table that cross-tabulates the two categorical variables (lithium exposure – yes/no, dementia outcome – yes/no). Each cell in the table represents a combination of categories from the two variables. In this 2x2 table, we have one degree of freedom for making choices within the table. This means that when one cell in the table has an observed value, then the other cell's value is determined based on the total sample size and the value observed for the first cell.

The correct answer is: The number of groups being compared in the analysis.

Question **24**

Correct

Mark 1.00 out of 1.00

Low physical activity and sedentary behaviour significantly impact the mental health and overall health of individuals with schizophrenia. A research study [Pieters LE, et al. Schizophrenia Bulletin. 2021 Jul 1;47(4):906-14] explored the connection between movement disorders (akathisia, dyskinesia, dystonia, and parkinsonism) and physical activity in 216 schizophrenia patients. Adult patients who lived at long-term mental healthcare units for more than 1 year or sheltered housing facilities of a mental health care institution were included, if they had at least 3 days of actigraphy data for 6hrs/day and validated measures of extrapyramidal symptoms and psychopathology at the same time. To investigate the statistical relationship between movement disorders and actigraphy data the authors performed multiple linear regression analysis. Parkinsonism was significantly associated with decreased physical activity ($\beta = -0.21$, $P < .01$) and increased sedentary behaviour ($\beta = 0.26$, $P < .001$). For dystonia, only the relationship with sedentary behaviour was significant ($\beta = 0.15$, $P < .05$). Akathisia was associated with more physical activity ($\beta = 0.14$, $P < .05$) and less sedentary behaviour ($\beta = -0.15$, $P < .05$). For dyskinesia, the relationships were non-significant. In a prediction model, akathisia, dystonia, parkinsonism and age significantly predicted physical activity ($F(5,209) = 16.6$, $P < .001$, R-squared adjusted = 0.27) and sedentary behaviour ($F(4,210) = 13.4$, $P < .001$, R-squared adjusted = 0.19). What is the most likely study design employed here?

Select one:

- Experimental study
- Case-control study
- Qualitative study
- Cross-sectional study
- Cohort study

Your answer is correct.

EXPLANATION: The study described is a cross-sectional study because it examines the relationship between movement disorders (akathisia, dyskinesia, dystonia, and parkinsonism) and physical activity and sedentary behavior at a single point in time. It assesses these variables in 216 patients with schizophrenia and related psychoses using actigraphy, rating scales, and psychopathological ratings. Cross-sectional studies are useful for capturing a snapshot of data at a specific moment and exploring associations between variables at that moment. In this case, it provides some insights into the statistical association between movement disorders and physical activity in the study population but cannot resolve cause-consequence relationships.

The correct answer is: Cross-sectional study

Question 25

Correct

Mark 1.00 out of 1.00

Low physical activity and sedentary behaviour significantly impact the mental health and overall health of individuals with schizophrenia. A research study [Pieters LE, et al. Schizophrenia Bulletin. 2021 Jul 1;47(4):906-14] explored the connection between movement disorders (akathisia, dyskinesia, dystonia, and parkinsonism) and physical activity in 216 schizophrenia patients. Adult patients who lived at long-term mental healthcare units for more than 1 year or sheltered housing facilities of a mental health care institution were included, if they had at least 3 days of actigraphy data for 6hrs/day and validated measures of extrapyramidal symptoms and psychopathology at the same time. To investigate the statistical relationship between movement disorders and actigraphy data the authors performed multiple linear regression analysis. Parkinsonism was significantly associated with decreased physical activity ($\beta = -0.21$, $P < .01$) and increased sedentary behaviour ($\beta = 0.26$, $P < .001$). For dystonia, only the relationship with sedentary behaviour was significant ($\beta = 0.15$, $P < .05$). Akathisia was associated with more physical activity ($\beta = 0.14$, $P < .05$) and less sedentary behaviour ($\beta = -0.15$, $P < .05$). For dyskinesia, the relationships were non-significant. In a prediction model, akathisia, dystonia, parkinsonism and age significantly predicted physical activity ($F(5,209) = 16.6$, $P < .001$, R-squared adjusted = 0.27) and sedentary behaviour ($F(4,210) = 13.4$, $P < .001$, R-squared adjusted = 0.19). In the above question, what does R-squared refer to?

Select one:

- Coefficient of determination
- Coefficient of regression
- Correlation coefficient
- Coefficient of variation
- Spearman's correlation

Your answer is correct.

EXPLANATION: The R-squared (R^2) value, also known as the coefficient of determination, is a statistical measure that provides insights into the goodness of fit of a predictive model. It represents the proportion of the variance in the dependent variable (the outcome being predicted) that is explained by the independent variables (the predictors) in the model. The range of R^2 value is typically from 0 to 1.0. An R^2 of 0 means that none of the variance in the dependent variable is explained by the predictors, indicating a poor fit. An $R^2$of 1.0 means that all of the variance in the dependent variable is explained by the predictors, indicating a perfect fit. A higher R^2 value indicates that a larger proportion of the variance in the dependent variable is explained by the predictors. This suggests that the model provides a good fit to the data and that the predictors are effective in explaining the variation in the outcome. An adjusted R^2 takes into account the number of predictors in the model and adjusts the R^2 value accordingly. This reduces overfitting.

The correct answer is: Coefficient of determination

Question **26**

Correct

Mark 1.00 out of 1.00

Low physical activity and sedentary behaviour significantly impact the mental health and overall health of individuals with schizophrenia. A research study [Pieters LE, et al. Schizophrenia Bulletin. 2021 Jul 1;47(4):906-14] explored the connection between movement disorders (akathisia, dyskinesia, dystonia, and parkinsonism) and physical activity in 216 schizophrenia patients. Adult patients who lived at long-term mental healthcare units for more than 1 year or sheltered housing facilities of a mental health care institution were included, if they had at least 3 days of actigraphy data for 6hrs/day and validated measures of extrapyramidal symptoms and psychopathology at the same time. To investigate the statistical relationship between movement disorders and actigraphy data the authors performed multiple linear regression analysis. Parkinsonism was significantly associated with decreased physical activity ($\beta = -0.21$, $P < .01$) and increased sedentary behaviour ($\beta = 0.26$, $P < .001$). For dystonia, only the relationship with sedentary behaviour was significant ($\beta = 0.15$, $P < .05$). Akathisia was associated with more physical activity ($\beta = 0.14$, $P < .05$) and less sedentary behaviour ($\beta = -0.15$, $P < .05$). For dyskinesia, the relationships were non-significant. In a prediction model, akathisia, dystonia, parkinsonism and age significantly predicted physical activity ($F(5,209) = 16.6$, $P < .001$, R-squared adjusted = 0.27) and sedentary behaviour ($F(4,210) = 13.4$, $P < .001$, R-squared adjusted = 0.19). What can you conclude from this study?

Select one:

[redacted]

Akathisia and parkinsonism have opposing relationship with physical activity

Dystonia was lower in those who are more sedentary

Dyskinesia symptoms strongly influence sedentary behaviour

A predictive model that includes movement disorders and age can predict most of the physical activity and sedentary behavior in patients with schizophrenia

Individuals with higher levels of akathisia may be more sedentary than those without akathisia

Your answer is correct.

EXPLANATION: The reported statistical analysis suggests that specific movement disorders, such as parkinsonism, dystonia, and akathisia, have distinct associations with physical activity and sedentary behavior in individuals with schizophrenia. The relationship between physical activity/sedentary behaviour and the severity of akathisia and parkinsonism are opposite in nature. While a combination of these movement disorders and age has a significant predictive relationship with physical activity/sedentary behavior. In the study, the R^2 values for the predictive models are as follows: For physical activity: R^2Adjusted = 0.27 (explains about 27%, not most of the variance in the observed physical activity). For sedentary behavior: R^2Adjusted = 0.19 (explains about 19%, not most of the observed variance sedentary behaviour).

The correct answer is: Akathisia and parkinsonism have opposing relationship with physical activity

Question **27**

Incorrect

Mark 0.00 out of 1.00

A cross-sectional study examined the relationship between dietary fiber intake and mental health in a large population of adults, using a cross-sectional study among 3,362 Iranian adults working in 50 health centers [Saghafian F, et al. Frontiers in Psychiatry. 2021 Jun 24;12:587468]. After adjustment for potential confounders, participants in the top quartile of total dietary fiber intake had a 33% and 29% lower risk of anxiety and high psychological distress [odds ratio (OR): 0.67; 95% CI: 0.48, 0.95 and OR: 0.71; 95% CI: 0.53, 0.94, respectively] compared to the bottom quartile of intake. The highest total dietary fiber intake was also inversely related to a lower risk of depression in women (OR: 0.63; 95% CI: 0.45, 0.88) but not in men. Among overweight or obese participants, higher intake of dietary fiber was related to a decreased risk of high psychological distress (OR: 0.52; 95% CI: 0.34, 0.79). A high level of dietary fiber intake was related to a lower risk of anxiety in normal-weight individuals (OR: 0.50; 95% CI: 0.31, 0.80). These results highlight the presence of significant inverse associations between total dietary fiber intake with anxiety and high psychological distress. What is the single most important factor that will enable you to establish a causal factor between the dietary fiber and psychiatric outcomes?

Select one:

[redacted]

Replication of this study in a different country and healthcare setting

Using a different statistical analysis method that does not rely on proportions

Implementing a randomised controlled trial

A linear relationship between fiber dose and symptom severity

An inverse association that is not confounded by gender or obesity

Your answer is incorrect.

EXPLANATION: The single most important factor for establishing a causal relationship between dietary fiber intake and psychiatric outcomes, as observed in the cross-sectional study, is implementing a randomized controlled trial (RCT). While observational studies, like the one presented here, can provide valuable insights and generate hypotheses regarding associations, RCTs are the gold standard for establishing causality in the relationship between a cause (dietary fiber intake) and outcomes (psychiatric symptoms in this case). Only an RCT can rigorously manipulate the independent variable (dietary fiber intake), control for potential confounding variables (weight, gender etc.) and establish causality. Replicating the study in different settings, using alternative statistical methods, or emphasizing specific associations may improve understanding but would not provide the level of evidence for causation that an RCT can offer.

The correct answer is: Implementing a randomised controlled trial

Question 28

Correct

Mark 1.00 out of 1.00

Taipale et al. [Lancet Psychiatry 2021; 8: 883–91] examined Finland's health databases to investigate the association between prolactin-increasing antipsychotic use in women with schizophrenia and breast cancer risk, while considering established risk factors. Using a case-control design within a patient cohort, the study included women diagnosed with schizophrenia or schizoaffective disorder between 1972 and 2014. "Cases" were those diagnosed with breast cancer from 2000 to 2017, matched with "controls" without breast cancer in this patient cohort. Both groups had a history of inpatient mental health care, and matching considered age and time since schizophrenia diagnosis. 1,069 women were identified as breast cancer cases and matched with 5,339 controls. From the given table, which of the following interpretation is most accurate?

	Control N=5339	Case N=1069	Unadjusted OR (95% CI)	Adjusted OR (95% CI)*
Duration of any antipsychotic use				
<1 year	917 (17·2%)	131 (12·3%)	..	..
1–4 years	646 (12·1%)	108 (10·1%)	1·16 (0·85–1·58)	1·18 (0·86–1·62)
≥5 years	3776 (70·7)	830 (77·6%)	1·75 (1·39–2·19)	1·74 (1·38–2·21)
Duration of prolactin-increasing antipsychotic use†				
<1 year	1134 (21·2%)	179 (16·7%)	..	..
1–4 years	772 (14·5%)	127 (11·9%)	1·03 (0·79–1·35)	1·04 (0·79–1·36)
≥5 years	3433 (64·3%)	763 (71·4%)	1·53 (1·26–1·86)	1·56 (1·27–1·92)
Duration of prolactin-sparing antipsychotic use‡				
<1 year	4521 (84·7%)	907 (84·8%)	..	..
1–4 years	382 (7·2%)	73 (6·8%)	0·95 (0·73–1·24)	0·95 (0·73–1·25)
≥5 years	436 (8·2%)	89 (8·3%)	1·02 (0·79–1·31)	1·19 (0·90–1·58)

Data are n (%) unless otherwise stated. *Adjusted for: substance misuse, previous suicide attempt, cardiovascular disease, asthma or chronic obstructive pulmonary disease, diabetes; number of children; use of opioids, paracetamol, non-steroidal anti-inflammatory drugs, digoxin, spironolactone, statins, loop diuretics, beta blockers, calcium channel blockers, angiotensin system drugs, anti-parkinson drugs, tricyclic antidepressants, selective serotonin reuptake inhibitors, verapamil; and duration of systemic hormone-replacement therapy use. Prolactin-increasing and prolactin-

inhibitors, verapamil, and duration of systemic hormone replacement therapy use. Prolactin-increasing and prolactin-sparing antipsychotics use analyses additionally adjusted for each other. †When excluding time with concomitant use of aripiprazole, additionally adjusted for aripiprazole use time (with and without concomitant prolactin-increasing antipsychotic use). ‡Without concomitant prolactin-increasing antipsychotic use.

***Table* 2: Association between duration of exposure to any antipsychotics, prolactin-increasing antipsychotics, and prolactin-sparing antipsychotics and risk of breast cancer with 1-year lag window for exposure**

Select one:

- Prolactin-increasing antipsychotic use for 5 years or more significantly increases the risk of breast cancer.
- Prolactin-increasing antipsychotic use for 1–4 years significantly reduces the risk of breast cancer.
- Prolactin-sparing antipsychotic use for 1–4 years significantly decreases the risk of breast cancer.
- Prolactin-increasing antipsychotic use for less than 1 year significantly increases the risk of breast cancer.
- Prolactin-sparing antipsychotic use for 5 years or more significantly decreases the risk of breast cancer.

Your answer is correct.

EXPLANATION: According to the table, the adjusted odds ratio (OR) for breast cancer risk is significantly increased (adjusted OR 1.56) for women who used prolactin-increasing antipsychotics for 5 years or more compared to those with shorter durations of use. This suggests that prolonged use of prolactin-increasing antipsychotics is associated with a higher risk of breast cancer.

The correct answer is: Prolactin-increasing antipsychotic use for 5 years or more significantly increases the risk of breast cancer.

Question 29

Correct

Mark 1.00 out of 1.00

Taipale et al. [Lancet Psychiatry 2021; 8: 883–91] examined Finland's health databases to investigate the association between prolactin-increasing antipsychotic use in women with schizophrenia and breast cancer risk, while considering established risk factors. Using a case-control design within a patient cohort, the study included women diagnosed with schizophrenia or schizoaffective disorder between 1972 and 2014. "Cases" were those diagnosed with breast cancer from 2000 to 2017, matched with "controls" without breast cancer in this patient cohort. Both groups had a history of inpatient mental health care, and matching considered age and time since schizophrenia diagnosis. 1,069 women were identified as breast cancer cases and matched with 5,339 controls. In the context of the information presented here, which of the following description captures the 5,339 individuals defined as "controls"?

	Control N=5339	Case N=1069	Unadjusted OR (95% CI)	Adjusted OR (95% CI)*
Duration of any antipsychotic use				
<1 year	917 (17·2%)	131 (12·3%)	..	..
1–4 years	646 (12·1%)	108 (10·1%)	1·16 (0·85–1·58)	1·18 (0·86–1·62)
≥5 years	3776 (70·7)	830 (77·6%)	1·75 (1·39–2·19)	1·74 (1·38–2·21)
Duration of prolactin-increasing antipsychotic use†				
<1 year	1134 (21·2%)	179 (16·7%)	..	..
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Duration of prolactin-sparing antipsychotic use‡				
<1 year	4521 (84·7%)	907 (84·8%)	..	..
1–4 years	382 (7·2%)	73 (6·8%)	0·95 (0·73–1·24)	0·95 (0·73–1·25)
≥5 years	436 (8·2%)	89 (8·3%)	1·02 (0·79–1·31)	1·19 (0·90–1·58)

Data are n (%) unless otherwise stated. *Adjusted for: substance misuse, previous suicide attempt, cardiovascular disease, asthma or chronic obstructive pulmonary disease, diabetes; number of children; use of opioids, paracetamol, non-steroidal anti-inflammatory drugs, digoxin, spironolactone, statins, loop diuretics, beta blockers, calcium channel blockers, angiotensin system drugs, anti-parkinson drugs, tricyclic antidepressants, selective serotonin reuptake inhibitors, verapamil; and duration of systemic hormone-replacement therapy use. Prolactin-increasing and prolactin-

inhibitors, verapamil, and duration of systemic hormone replacement therapy use. Prolactin-increasing and prolactin-sparing antipsychotics use analyses additionally adjusted for each other. †When excluding time with concomitant use of aripiprazole, additionally adjusted for aripiprazole use time (with and without concomitant prolactin-increasing antipsychotic use). ‡Without concomitant prolactin-increasing antipsychotic use.

Table 2: Association between duration of exposure to any antipsychotics, prolactin-increasing antipsychotics, and prolactin-sparing antipsychotics and risk of breast cancer with 1-year lag window for exposure

Select one:

Individuals diagnosed with schizophrenia but without breast cancer

Individuals aged 16 and above who did not use antipsychotic medications

Individuals without any mental health diagnoses

Individuals without a history of hospital care for mental health conditions

Individuals with a history of antipsychotic use for conditions other than schizophrenia

Your answer is correct.

EXPLANATION: In the table, "controls" are defined as women aged 16 and above who have been diagnosed with schizophrenia or schizoaffective disorder (patient cohort) but do not have a breast cancer diagnosis during the specified period. This group serves as the comparison or reference group for assessing the breast cancer risk associated with antipsychotic use.

The correct answer is: Individuals diagnosed with schizophrenia but without breast cancer

Question 30

Correct

Mark 1.00 out of 1.00

Taipale et al. [Lancet Psychiatry 2021; 8: 883–91] examined Finland's health databases to investigate the association between prolactin-increasing antipsychotic use in women with schizophrenia and breast cancer risk, while considering established risk factors. Using a case-control design within a patient cohort, the study included women diagnosed with schizophrenia or schizoaffective disorder between 1972 and 2014. "Cases" were those diagnosed with breast cancer from 2000 to 2017, matched with "controls" without breast cancer in this patient cohort. Both groups had a history of inpatient mental health care, and matching considered age and time since schizophrenia diagnosis. 1,069 women were identified as breast cancer cases and matched with 5,339 controls. In the context of the information presented in the table, what does the adjusted ORs and unadjusted ORs suggest?

	Control N=5339	Case N=1069	Unadjusted OR (95% CI)	Adjusted OR (95% CI)*
Duration of any antipsychotic use				
<1 year	917 (17·2%)	131 (12·3%)	··	··
1–4 years	646 (12·1%)	108 (10·1%)	1·16 (0·85–1·58)	1·18 (0·86–1·62)
≥5 years	3776 (70·7)	830 (77·6%)	1·75 (1·39–2·19)	1·74 (1·38–2·21)
Duration of prolactin-increasing antipsychotic use†				
<1 year	1134 (21·2%)	179 (16·7%)	··	··
1–4 years	772 (14·5%)	127 (11·9%)	1·03 (0·79–1·35)	1·04 (0·79–1·36)
≥5 years	3433 (64·3%)	763 (71·4%)	1·53 (1·26–1·86)	1·56 (1·27–1·92)
Duration of prolactin-sparing antipsychotic use‡				
<1 year	4521 (84·7%)	907 (84·8%)	··	··
1–4 years	382 (7·2%)	73 (6·8%)	0·95 (0·73–1·24)	0·95 (0·73–1·25)
≥5 years	436 (8·2%)	89 (8·3%)	1·02 (0·79–1·31)	1·19 (0·90–1·58)

Data are n (%) unless otherwise stated. *Adjusted for: substance misuse, previous suicide attempt, cardiovascular disease, asthma or chronic obstructive pulmonary disease, diabetes; number of children; use of opioids, paracetamol, non-steroidal anti-inflammatory drugs, digoxin, spironolactone, statins, loop diuretics, beta blockers, calcium channel blockers, angiotensin system drugs, anti-parkinson drugs, tricyclic antidepressants, selective serotonin reuptake inhibitors, verapamil; and duration of systemic hormone-replacement therapy use. Prolactin-increasing and prolactin-

inhibitors, verapamil, and duration of systemic hormone replacement therapy use. Prolactin-increasing and prolactin-sparing antipsychotics use analyses additionally adjusted for each other. †When excluding time with concomitant use of aripiprazole, additionally adjusted for aripiprazole use time (with and without concomitant prolactin-increasing antipsychotic use). ‡Without concomitant prolactin-increasing antipsychotic use.

***Table 2:* Association between duration of exposure to any antipsychotics, prolactin-increasing antipsychotics, and prolactin-sparing antipsychotics and risk of breast cancer with 1-year lag window for exposure**

Select one:

[redacted]

The study included a very small sample size, leading to negligible adjustments.

The study did not consider any potential confounding factors in its analysis.

Adjusting for various factors had a minimal impact on the association between antipsychotic use and breast cancer risk.

Adjusting for factors unrelated to the study's objectives led to inconclusive results.

Unadjusted ORs are more reliable indicators of the true association than adjusted ORs.

Your answer is correct.

EXPLANATION: The observation that adjusted ORs are not notably different from unadjusted ORs suggests that after considering various potential confounding factors, the association between antipsychotic use and breast cancer risk remained relatively stable. This indicates that adjusting for these factors had a minimal impact on the observed association, implying a robust relationship between antipsychotic use and breast cancer risk in the study population.

The correct answer is: Adjusting for various factors had a minimal impact on the association between antipsychotic use and breast cancer risk.

Question 31

Correct

Mark 1.00 out of 1.00

Taipale et al. [Lancet Psychiatry 2021; 8: 883–91] examined Finland's health databases to investigate the association between prolactin-increasing antipsychotic use in women with schizophrenia and breast cancer risk, while considering established risk factors. Using a case-control design within a patient cohort, the study included women diagnosed with schizophrenia or schizoaffective disorder between 1972 and 2014. "Cases" were those diagnosed with breast cancer from 2000 to 2017, matched with "controls" without breast cancer in this patient cohort. Both groups had a history of inpatient mental health care, and matching considered age and time since schizophrenia diagnosis. 1,069 women were identified as breast cancer cases and matched with 5,339 controls. In the provided table, why are odds ratios not presented for the <1 year duration of antipsychotic use category?

	Control N=5339	Case N=1069	Unadjusted OR (95% CI)	Adjusted OR (95% CI)*
Duration of any antipsychotic use				
<1 year	917 (17·2%)	131 (12·3%)	..	..
1–4 years	646 (12·1%)	108 (10·1%)	1·16 (0·85–1·58)	1·18 (0·86–1·62)
≥5 years	3776 (70·7)	830 (77·6%)	1·75 (1·39–2·19)	1·74 (1·38–2·21)
Duration of prolactin-increasing antipsychotic use†				
<1 year	1134 (21·2%)	179 (16·7%)	..	..
1–4 years	772 (14·5%)	127 (11·9%)	1·03 (0·79–1·35)	1·04 (0·79–1·36)
≥5 years	3433 (64·3%)	763 (71·4%)	1·53 (1·26–1·86)	1·56 (1·27–1·92)
Duration of prolactin-sparing antipsychotic use‡				
<1 year	4521 (84·7%)	907 (84·8%)	..	..
1–4 years	382 (7·2%)	73 (6·8%)	0·95 (0·73–1·24)	0·95 (0·73–1·25)
≥5 years	436 (8·2%)	89 (8·3%)	1·02 (0·79–1·31)	1·19 (0·90–1·58)

Data are n (%) unless otherwise stated. *Adjusted for: substance misuse, previous suicide attempt, cardiovascular disease, asthma or chronic obstructive pulmonary disease, diabetes; number of children; use of opioids, paracetamol, non-steroidal anti-inflammatory drugs, digoxin, spironolactone, statins, loop diuretics, beta blockers, calcium channel blockers, angiotensin system drugs, anti-parkinson drugs, tricyclic antidepressants, selective serotonin reuptake inhibitors, verapamil; and duration of systemic hormone-replacement therapy use. Prolactin-increasing and prolactin-

inhibitors, verapamil, and duration of systemic hormone replacement therapy use. Prolactin-increasing and prolactin-sparing antipsychotics use analyses additionally adjusted for each other. †When excluding time with concomitant use of aripiprazole, additionally adjusted for aripiprazole use time (with and without concomitant prolactin-increasing antipsychotic use). ‡Without concomitant prolactin-increasing antipsychotic use.

***Table 2:* Association between duration of exposure to any antipsychotics, prolactin-increasing antipsychotics, and prolactin-sparing antipsychotics and risk of breast cancer with 1-year lag window for exposure**

Select one:

- There were too many cases of breast cancer within the <1 year exposure group.
- The odds ratios for <1 year use were not statistically significant.
- Individuals with <1 year exposure are used as the reference category to calculate odds ratios.
- The study did not consider individuals with <1 year of antipsychotic use.
- There were too few individuals in the <1 year exposure group to calculate meaningful odds ratios.

Your answer is correct.

The authors have used <1 year exposure to antipsychotics (including likely non-users of antipsychotics) as the reference category for the exposure. This is perhaps because shorter exposures, just like non-exposure, are unlikely to contribute to the risk of breast cancer. Note that there were sufficient cases of breast cancer within the <1 year exposure group as well.

The correct answer is: Individuals with <1 year exposure are used as the reference category to calculate odds ratios.

Question 32

Correct

Mark 1.00 out of 1.00

A research study [Coupland C, et al. Antidepressant use and risk of cardiovascular outcomes in people aged 20 to 64: cohort study using primary care database. BMJ. 2016 Mar 22;352] based on the QResearch primary care database used by UK general practices aimed to investigate associations between different antidepressant treatments and the incidence of three cardiovascular outcomes (myocardial infarction, stroke or transient ischaemic attack, and arrhythmia) in individuals diagnosed with depression. The study included 238,963 patients aged 20 to 64 years with an initial diagnosis of depression between January 1, 2000, and July 31, 2011. The research assessed various factors, including antidepressant class, dose, duration of use, and individual antidepressant drugs. Cox proportional hazards models were used to estimate hazard ratios while adjusting for potential confounding variables. The results indicated that there were no significant associations between antidepressant class and myocardial infarction over five years of follow-up. However, during the first year of follow-up, patients treated with selective serotonin reuptake inhibitors had a significantly reduced risk of myocardial infarction (adjusted hazard ratio 0.58, 95% confidence interval 0.42 to 0.79) compared with no use of antidepressants, particularly fluoxetine (0.44, 0.27 to 0.72). Lofepramine, on the other hand, was associated with an increased risk (3.07, 1.50 to 6.26). No significant associations were found between antidepressant class or individual drugs and the risk of stroke or transient ischaemic attack. Antidepressant class was not significantly associated with arrhythmia, although there was an increased risk during the first 28 days of treatment with tricyclic and related antidepressants (adjusted hazard ratio 1.99, 1.27 to 3.13). Fluoxetine was associated with a reduced risk of arrhythmia over five years (0.74, 0.59 to 0.92), while citalopram was not significantly associated with arrhythmia risk, even at high doses (1.11, 0.72 to 1.71 for doses =40 mg/day). What type of study design was employed in this research?

Select one:

- Longitudinal study
- Case-control study
- Randomized controlled trial
- Cohort study
- Cross-sectional study

Your answer is correct.

EXPLANATION: This study utilised a register-based cohort study design, which involves following a group of individuals over time to assess the association between exposure (antidepressants) and outcomes (cardiac events).

The correct answer is: Cohort study

Question 33

Correct

Mark 1.00 out of 1.00

A research study [Coupland C, et al. Antidepressant use and risk of cardiovascular outcomes in people aged 20 to 64: cohort study using primary care database. BMJ. 2016 Mar 22;352] based on the QResearch primary care database used by UK general practices aimed to investigate associations between different antidepressant treatments and the incidence of three cardiovascular outcomes (myocardial infarction, stroke or transient ischaemic attack, and arrhythmia) in individuals diagnosed with depression. The study included 238,963 patients aged 20 to 64 years with an initial diagnosis of depression between January 1, 2000, and July 31, 2011. The research assessed various factors, including antidepressant class, dose, duration of use, and individual antidepressant drugs. Cox proportional hazards models were used to estimate hazard ratios while adjusting for potential confounding variables. The results indicated that there were no significant associations between antidepressant class and myocardial infarction over five years of follow-up. However, during the first year of follow-up, patients treated with selective serotonin reuptake inhibitors had a significantly reduced risk of myocardial infarction (adjusted hazard ratio 0.58, 95% confidence interval 0.42 to 0.79) compared with no use of antidepressants, particularly fluoxetine (0.44, 0.27 to 0.72). Lofepramine, on the other hand, was associated with an increased risk (3.07, 1.50 to 6.26). No significant associations were found between antidepressant class or individual drugs and the risk of stroke or transient ischaemic attack. Antidepressant class was not significantly associated with arrhythmia, although there was an increased risk during the first 28 days of treatment with tricyclic and related antidepressants (adjusted hazard ratio 1.99, 1.27 to 3.13). Fluoxetine was associated with a reduced risk of arrhythmia over five years (0.74, 0.59 to 0.92), while citalopram was not significantly associated with arrhythmia risk, even at high doses (1.11, 0.72 to 1.71 for doses =40 mg/day). One of your patients who come across this study asks you if this means that fluoxetine does not cause adverse cardiovascular events in those with depression. What issue pertaining to the study design in this research limits your ability to endorse her statement?

Select one:

- Difficulty in recruiting participants
- Limited generalizability
- Inability to establish causation
- Lack of statistical power
- High risk of selection bias

Your answer is correct.

EXPLANATION: Cohort studies can identify associations, and temporal relationships but cannot establish causation definitively, as they do not involve experimental manipulation. Randomised design where treatment is manipulated independent of other confounds will be better suited for causal inferences for the question posed here. While statistical power can be a limitation in some studies, it is not the primary issue in this case. While a formal power calculation is not available in the precis, the study's significant findings (e.g. pertaining to 1st year, and individual antidepressants) imply that they had sufficient power and sample size (n= 238,963 to answer these questions. This study did not directly recruit any participant, but used data from the QResearch primary care database, which included a large number of participants, so difficulty in recruiting participants is not a relevant issue. Selection bias can affect the validity of study results, but the inclusion of a representative sample of primary care service users improves generalisability and mitigates this concern.

The correct answer is: Inability to establish causation

Question **34**

Correct

Mark 1.00 out of 1.00

A research study [Coupland C, et al. Antidepressant use and risk of cardiovascular outcomes in people aged 20 to 64: cohort study using primary care database. BMJ. 2016 Mar 22;352] based on the QResearch primary care database used by UK general practices aimed to investigate associations between different antidepressant treatments and the incidence of three cardiovascular outcomes (myocardial infarction, stroke or transient ischaemic attack, and arrhythmia) in individuals diagnosed with depression. The study included 238,963 patients aged 20 to 64 years with an initial diagnosis of depression between January 1, 2000, and July 31, 2011. The research assessed various factors, including antidepressant class, dose, duration of use, and individual antidepressant drugs. Cox proportional hazards models were used to estimate hazard ratios while adjusting for potential confounding variables. The results indicated that there were no significant associations between antidepressant class and myocardial infarction over five years of follow-up. However, during the first year of follow-up, patients treated with selective serotonin reuptake inhibitors had a significantly reduced risk of myocardial infarction (adjusted hazard ratio 0.58, 95% confidence interval 0.42 to 0.79) compared with no use of antidepressants, particularly fluoxetine (0.44, 0.27 to 0.72). Lofepramine, on the other hand, was associated with an increased risk (3.07, 1.50 to 6.26). Fluoxetine was associated with a reduced risk of arrhythmia over five years (0.74, 0.59 to 0.92), while citalopram was not significantly associated with arrhythmia risk, even at high doses (1.11, 0.72 to 1.71 for doses =40 mg/day). What can be concluded regarding the risk of arrhythmia associated with citalopram use at high doses?

Select one:

- Increased risk
- Higher doses are safer than lower doses
- No conclusion can be drawn
- No significant change in risk
- Reduced risk

Your answer is correct.

EXPLANATION: The study did not find a significant association between citalopram use, even at high doses, and the risk of arrhythmia. This does not mean higher doses are safer, as there is no comparison between high and low-dose use that is presented here.

The correct answer is: No significant change in risk

Question 35

Incorrect

Mark 0.00 out of 1.00

A research study [Coupland C, et al. Antidepressant use and risk of cardiovascular outcomes in people aged 20 to 64: cohort study using primary care database. BMJ. 2016 Mar 22;352] based on the QResearch primary care database used by UK general practices aimed to investigate associations between different antidepressant treatments and the incidence of three cardiovascular outcomes (myocardial infarction, stroke or transient ischaemic attack, and arrhythmia) in individuals diagnosed with depression. The study included 238,963 patients aged 20 to 64 years with an initial diagnosis of depression between January 1, 2000, and July 31, 2011. The research assessed various factors, including antidepressant class, dose, duration of use, and individual antidepressant drugs. Cox proportional hazards models were used to estimate hazard ratios while adjusting for potential confounding variables. The results indicated that there were no significant associations between antidepressant class and myocardial infarction over five years of follow-up. However, during the first year of follow-up, patients treated with selective serotonin reuptake inhibitors had a significantly reduced risk of myocardial infarction (adjusted hazard ratio 0.58, 95% confidence interval 0.42 to 0.79) compared with no use of antidepressants, particularly fluoxetine (0.44, 0.27 to 0.72). Lofepramine, on the other hand, was associated with an increased risk (3.07, 1.50 to 6.26). No significant associations were found between antidepressant class or individual drugs and the risk of stroke or transient ischaemic attack. Antidepressant class was not significantly associated with arrhythmia, although there was an increased risk during the first 28 days of treatment with tricyclic and related antidepressants (adjusted hazard ratio 1.99, 1.27 to 3.13). Fluoxetine was associated with a reduced risk of arrhythmia over five years (0.74, 0.59 to 0.92), while citalopram was not significantly associated with arrhythmia risk, even at high doses (1.11, 0.72 to 1.71 for doses =40 mg/day). Which of the following conclusions is valid?

Select one:

Fluoxetine is best prescribed for only one year to reduce the risk of myocardial infarctions

Lofepramine increases the risk of stroke in the first year of its use

Antidepressants do not increase the risk of stroke in patients with depression

Antidepressants do not increase the risk of arrhythmia in patients with depression

When using tricyclics, combining them with fluoxetine reduces the risk of arrythmia

Your answer is incorrect.

EXPLANATION: From the results presented, it is clear that there was no significant association between antidepressants (as a class) or individual drugs and the risk of stroke or transient ischemic attack. Additionally, citalopram was not significantly associated with an increased risk of arrhythmia. There was some indication of a reduced risk of myocardial infarction with selective serotonin reuptake inhibitors in year 1, particularly fluoxetine, and an increased risk with lofepramine.

The correct answer is: Antidepressants do not increase the risk of stroke in patients with depression

Question 36

Correct

Mark 1.00 out of 1.00

What statistical test is most appropriate for analyzing the relationship between PANSS item P2 scores (conceptual disorganisation) and Thought and Language Index scale scores (positive and negative formal thought disorder) in a clinical research study?

Select one:

- Pearson's correlation coefficient
- Student's t-test
- Spearman's rank correlation coefficient
- Chi-squared test for independence
- Analysis of Variance (ANOVA)

Your answer is correct.

EXPLANATION: In this scenario, you are dealing with ordinal data (PANSS item P2 scores, varies on a Likert scale of 1 to 7). To assess the relationship between these two sets of data where both variables are likely to be not continuous (at least one being ordinal), the most appropriate statistical test is Spearman's rank correlation coefficient. Spearman's rank correlation is a non-parametric test used to assess the strength and direction of the monotonic relationship between ordinal or non-normally distributed continuous variables. Chi-squared test for independence and ANOVA are more appropriate for categorical or continuous data, respectively, and not ordinal data. Student's t-test is typically used for comparing means of two continuous variables, which is not applicable to your ordinal data. Pearson's correlation coefficient assumes a linear relationship between two continuous variables and is not well-suited for ordinal data.

The correct answer is: Spearman's rank correlation coefficient

Question **37**

Correct

Mark 1.00 out of 1.00

You are conducting a study to understand the relationship between cortisol levels in saliva and melatonin levels in a help-seeking adolescent population. You recruit a group of participants and collect morning salivary levels for both hormones. What statistical test is most appropriate for analyzing the relationship between these two measures?

Select one:

- Spearman's rank correlation coefficient
- Student's t-test
- Pearson's correlation coefficient
- Analysis of Variance (ANOVA)
- Chi-squared test for independence

Your answer is correct.

EXPLANATION: Both cortisol levels in saliva and melatonin levels are continuous variables. Continuous variables are measured on a scale and can take on any numerical value within a range. In this case, the hormone levels are measured quantitatively, making them continuous. As you are dealing with two continuous variables and want to assess the strength and direction of their linear relationship, Pearson's correlation coefficient is the most appropriate statistical test for this study. Other tests, such as chi-squared tests or ANOVA, are designed for different types of data or research questions and would not be suitable for analyzing the relationship between continuous hormone levels.

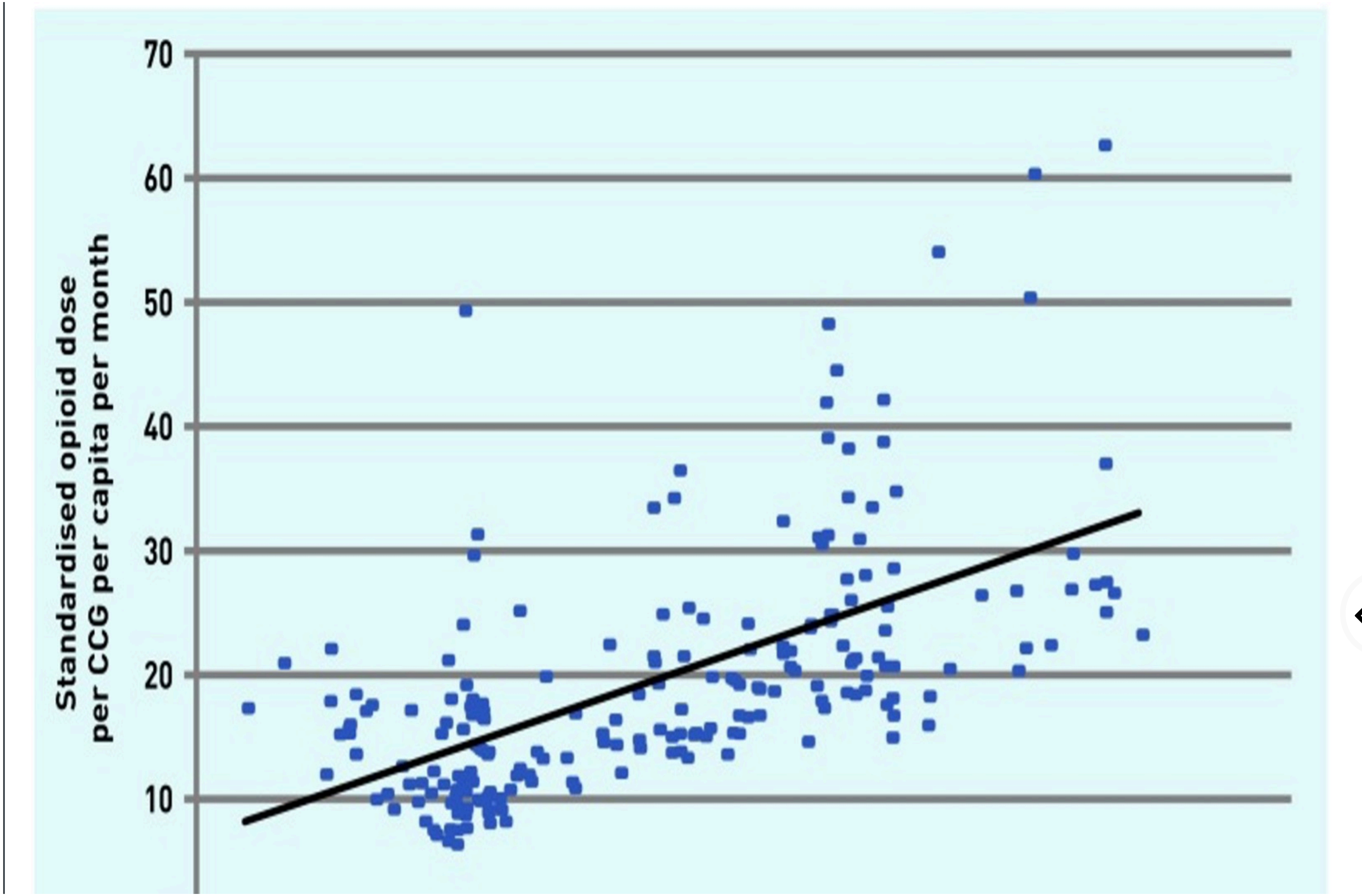
The correct answer is: Pearson's correlation coefficient

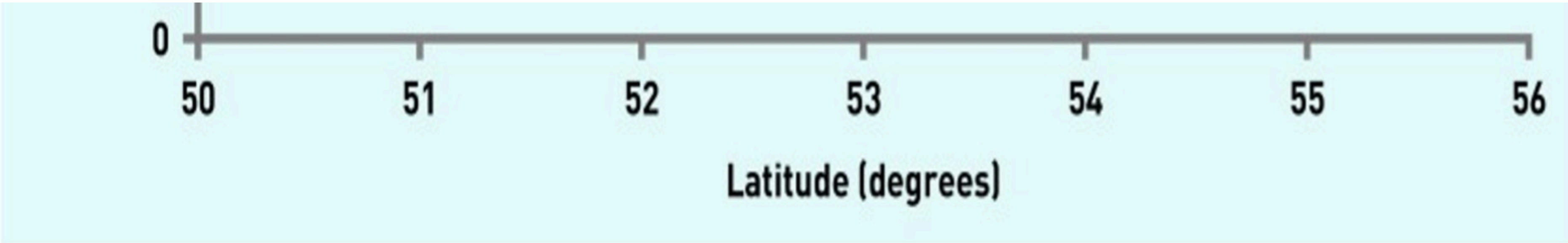
Question 38

Correct

Mark 1.00 out of 1.00

A retrospective observational study [Mordecai et al., British Journal of General Practice. 2018 Mar 1;68(668):e225-33] conducted in England aimed to evaluate the quantity and types of opioids prescribed in primary care and to analyse regional variations in prescribing patterns. Analysing government data from August 2010 to February 2014, the study revealed an increase in total opioid prescriptions, standardized for geographical comparisons, during this period. The relationship between the amount of opioid prescribed and the latitude of location of clinical commissioning groups (CCGs) is shown in the attached figure. Despite concerns about the efficacy and potential harm associated with long-term opioid use for non-cancer pain, the study highlights a rising trend in opioid prescribing, suggesting a need for better patient safety measures and epidemiological data through a national registry for patients with high opioid use. What kind of graph is this?


Standardised opioid dose per CCG per capita per month
70
60
50
40
30
20
10



Select one:

Bubble chart

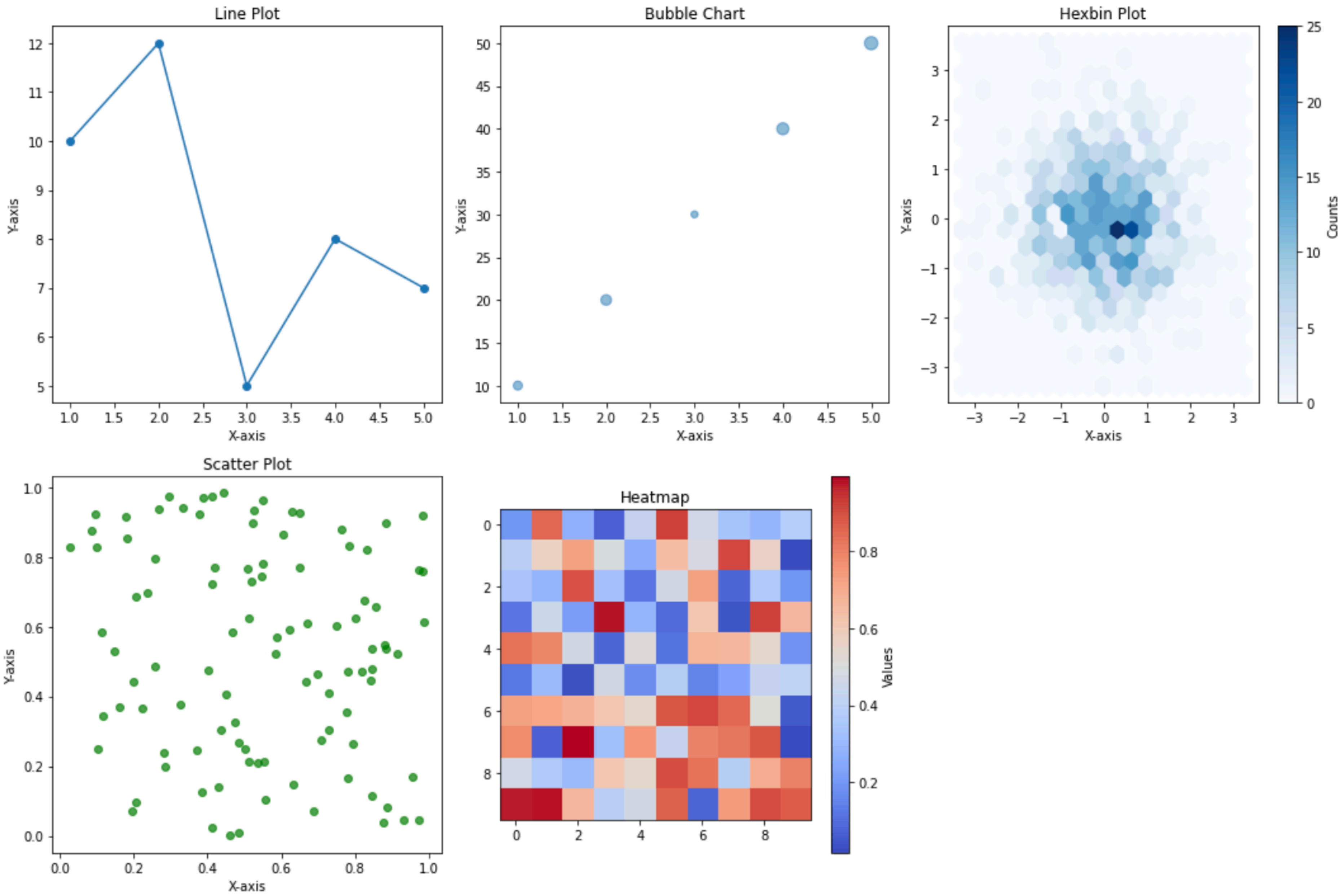
Scatter plot

Line chart

Hexbin plot

Heatmap

Your answer is correct.


Line Plot
Y-axis
X-axis
Bubble Chart
Y-axis
X-axis
Hexbin Plot
Y-axis
X-axis
Counts
Scatter Plot
Y-axis
X-axis
Heatmap
Values

EXPLANATION: A bubble chart is similar to a scatterplot, but it adds a third dimension by varying the size of data points (bubbles) to represent a third variable.

Heatmaps are grids where each cell's color intensity represents a value, which can be the concentration of data points (as shown in the sample figure above).

A hexbin plot is a density plot where data points of a scattergram are aggregated into hexagonal bins, showing the density of points in different areas of the plot.

Line charts are primarily used to show trends over time, but they can also resemble scatterplots when data points are not connected by lines.

Note that in a scatter plot the trend line shown is generally a regression line, determined by the best fit of the observed data points. This is a straight line that best fits the data points in the scatterplot. It's generally constructed using a linear regression analysis, which minimizes the sum of the squared differences between the observed data points and the predicted values on the line.

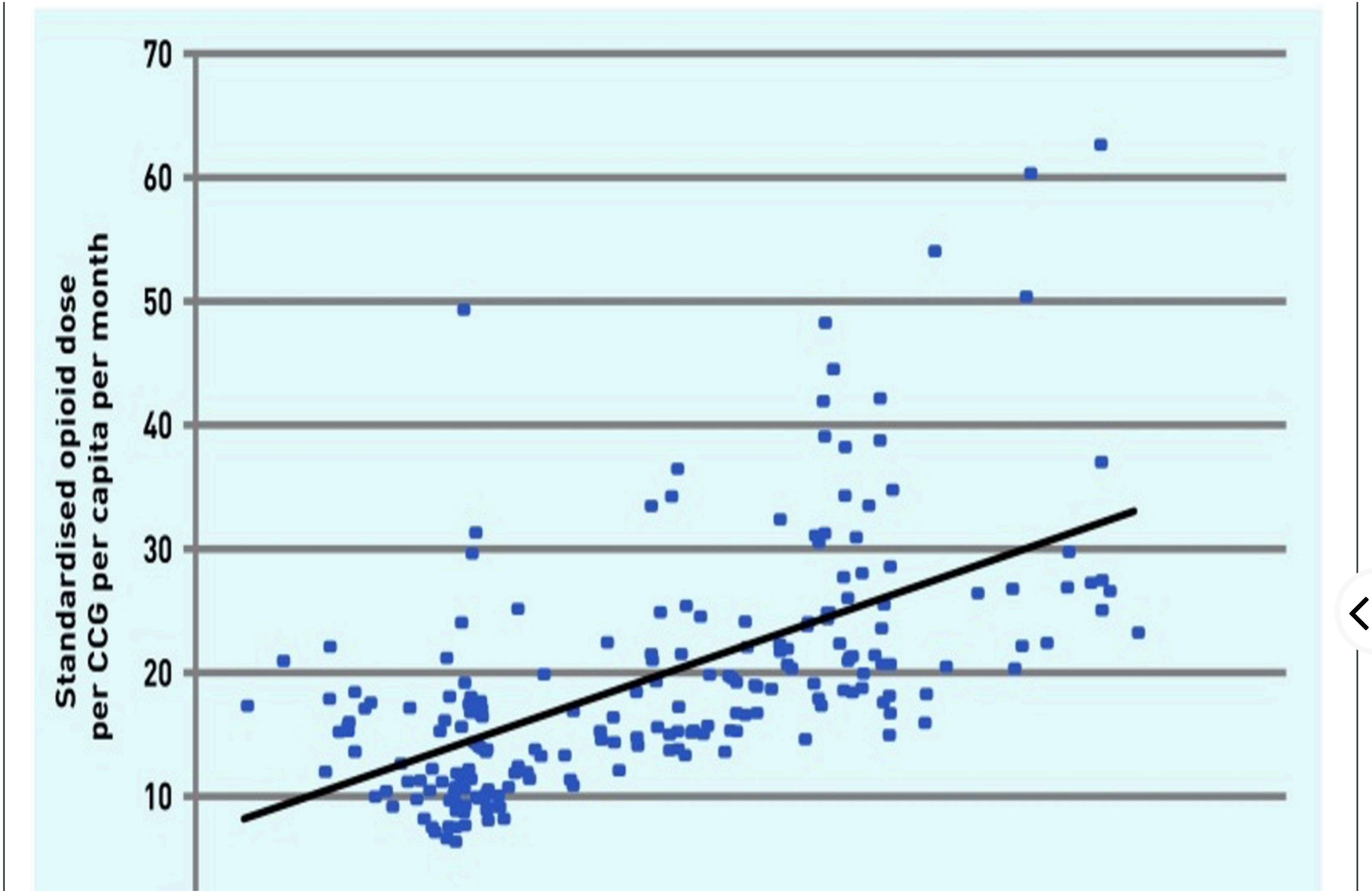
The correct answer is: Scatter plot

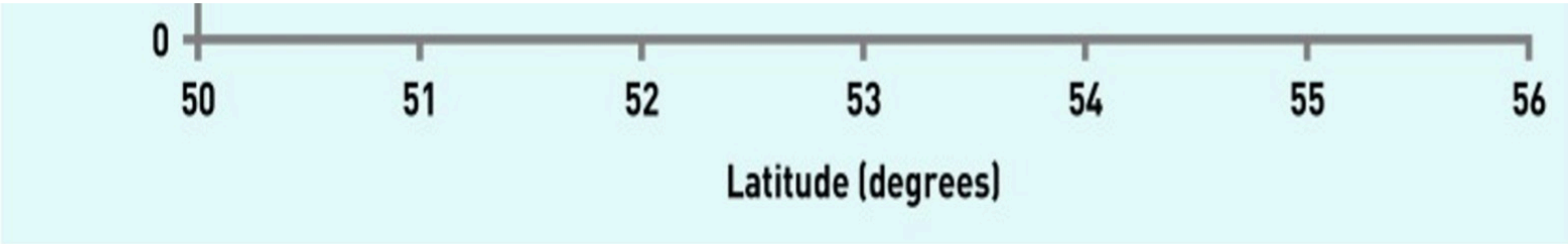
Question 39

Incorrect

Mark 0.00 out of 1.00

A retrospective observational study [Mordecai et al., British Journal of General Practice. 2018 Mar 1;68(668):e225-33] conducted in England aimed to evaluate the quantity and types of opioids prescribed in primary care and to analyse regional variations in prescribing patterns. Analysing government data from August 2010 to February 2014, the study revealed an increase in total opioid prescriptions, standardized for geographical comparisons, during this period. The relationship between the amount of opioid prescribed and the latitude of location of clinical commissioning groups (CCGs) is shown in the attached figure. Despite concerns about the efficacy and potential harm associated with long-term opioid use for non-cancer pain, the study highlights a rising trend in opioid prescribing, suggesting a need for better patient safety measures and epidemiological data through a national registry for patients with high opioid use. What can be inferred from the attached plot?


Standardised opioid dose per CCG per capita per month
70
60
50
40
30
20
10



Select one:

[redacted]

Immigration to northern regions is associated with higher use of opioids.

Stronger opioids are used in the northern than in the southern primary care units

There is no difference in the distribution of opioid prescription across the country.

There is a higher prescription of opioids in the south of the country.

Primary care units in northern regions are prescribing more opioids.

Your answer is incorrect.

EXPLANATION: The attached figure suggests a substantial linear correlation between the amount of opioid prescribed and higher latitude of CCGs. As higher latitudes refer to northerly locations of the prescribing units (CCGs), this suggests a much higher use of opioid prescriptions in the north of the country.

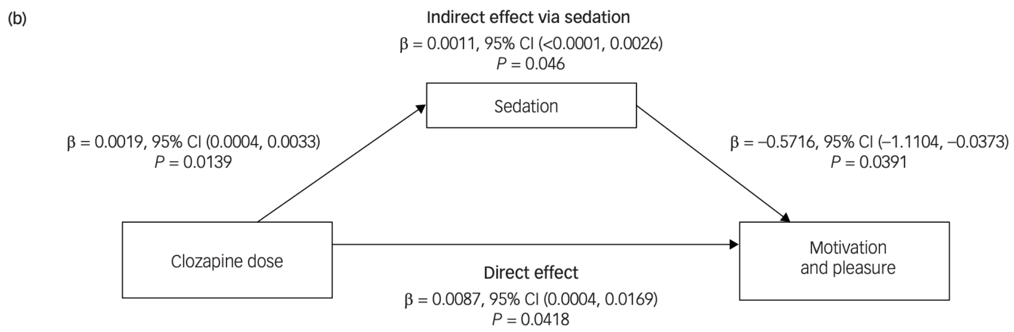
The correct answer is: Primary care units in northern regions are prescribing more opioids.

Question **40**

Correct

Mark 1.00 out of 1.00

To understand the longitudinal effects of clozapine on motivation and pleasure (negative symptoms), Wolpe and colleagues studied a cohort of 187 clozapine-treated patients with schizophrenia followed for over 2 years on average. The authors examined the severity of reduced motivation and pleasure (MAP) and the effect of antipsychotic-induced sedation (assessed via the proxy of total daily sleep duration) on MAP. The result of their analysis is summarised in the graph below. They concluded that addressing sedative side-effects of antipsychotics is important to improve clinical outcomes. Which of the following statistical analysis is most likely conducted in this study?



Select one:

- Triangulation analysis
- Reflexive analysis
- Survival analysis
- Mediation analysis
- Meta-analysis

Your answer is correct.

To explore the specific effect of an independent variable (clozapine) on the outcome (reduced motivation and pleasure) we can model the effect of clozapine to be direct (i.e., in the absence of mediating effects of sedation) or indirect (i.e., mediated by clozapine- indiced sedation). Such ana analysis is called mediation analysis, which is one form of Structural Equation Modelling.

Ref: Rijnhart JJ, et al. Mediation analysis methods used in observational research: a scoping review and recommendations. BMC medical research methodology. 2021 Dec;21(1):1-7.

The correct answer is: Mediation analysis

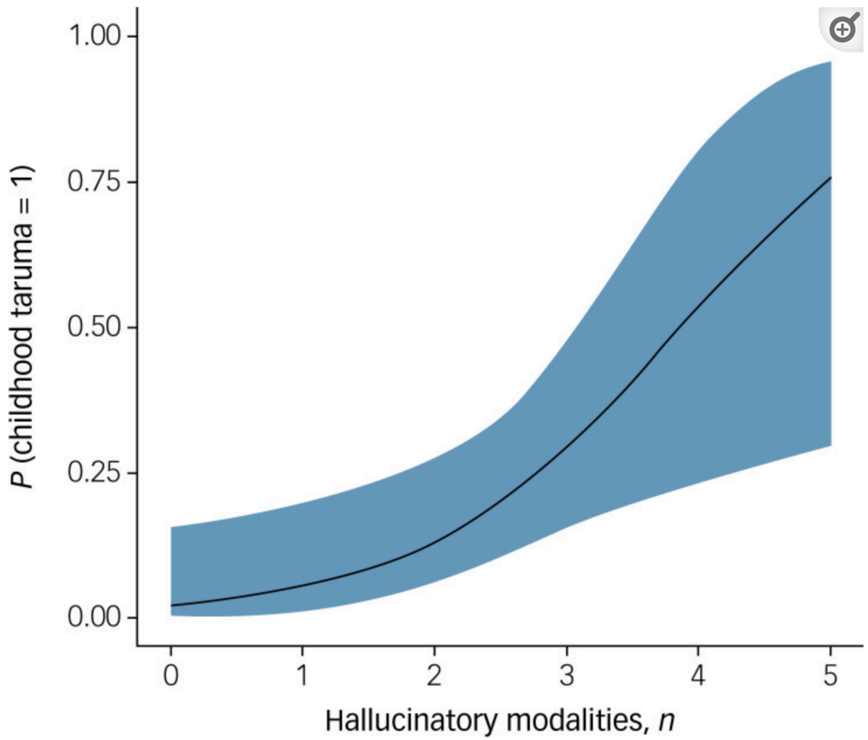
Question 41

Incorrect

Mark 0.00 out of 1.00

To understand the relationship between childhood traumatic experiences and the sensory complexity of hallucinations, a cohort of 75 children and adolescents seeking help for hallucinations were studied (Ref: Medjkane F, et al. The British Journal of Psychiatry. 2020 Mar;216(3):156-8). To determine the extent by which the various sensory modalities could predict the probability of a previous early trauma, the authors employed a hierarchical binomial logistic regression with history of childhood trauma as the binary dependent variable with control variables (i.e., age, educational level and number of ICD-10 diagnoses, the presence of suicidal ideation), and the number of hallucinatory modalities as predictors. The number of hallucinatory modalities appeared to be the only significant predictor of childhood trauma (estimate (i.e. the log odds of 'history of childhood trauma' = 1 v. 0) 0.81, s.e. = 0.34, z = 2.39, P = 0.017, OR = 2.24 (95% CI 1.16–4.33). The predictor that emerged as statistically significant variable is most likely a/an

Fig. 1



Estimated marginal means with 95% confidence interval showing the positive relationship between the number of sensory modalities in hallucinatory experiences and the probability of a childhood trauma (OR = 2.24 (95% CI 1.16–4.33), $P = 0.017$, $n = 75$).

Select one:

- Ordinal variable
- Continuous variable
- Ratio variable
- Categorical variable
- Probability measure

Your answer is incorrect.

In the current study, the dependent variable (childhood trauma) is a categorical one (yes/no type of binary variable) while the independent variable (predictor) is an ordinal variable (no. of modalities in an increasing order of 1 to 5).

The correct answer is: Categorical variable

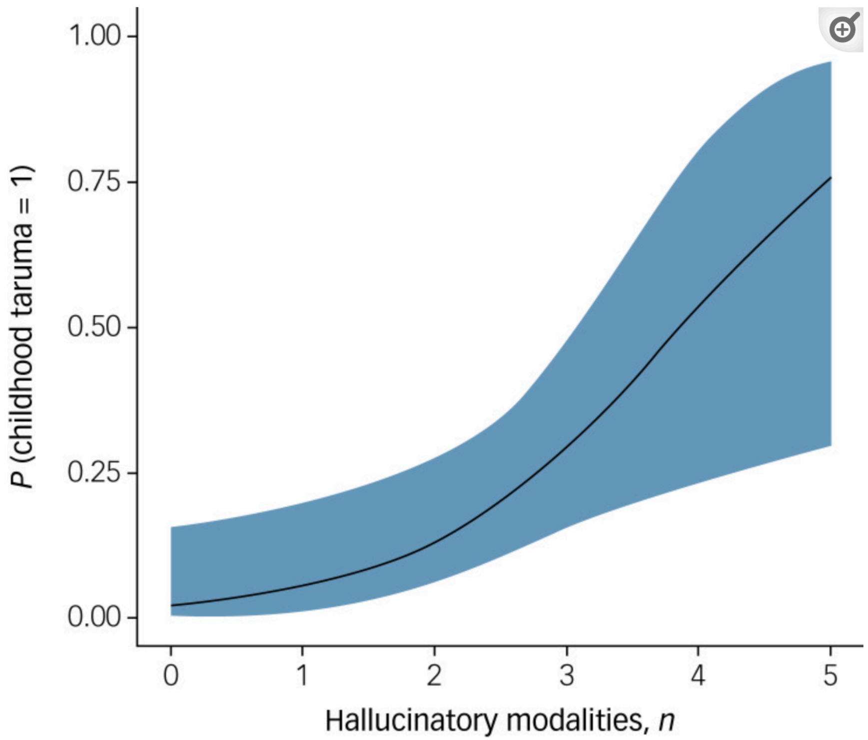
Question 42

Correct

Mark 1.00 out of 1.00

To understand the relationship between childhood traumatic experiences and the sensory complexity of hallucinations, a cohort of 75 children and adolescents seeking help for hallucinations were studied (Ref: Medjkane F, et al. The British Journal of Psychiatry. 2020 Mar;216(3):156-8). To determine the extent by which the various sensory modalities could predict the probability of a previous early trauma, the authors employed a hierarchical binomial logistic regression with history of childhood trauma as the binary dependent variable with control variables (i.e., age, educational level and number of ICD-10 diagnoses, the presence of suicidal ideation), and the number of hallucinatory modalities as predictors. The number of hallucinatory modalities appeared to be the only significant predictor of childhood trauma (estimate (i.e. the log odds of 'history of childhood trauma' = 1 v. 0) 0.81, s.e. = 0.34, z = 2.39, P = 0.017, OR = 2.24 (95% CI 1.16–4.33). In the attached graph, the shaded area represents:

Fig. 1



Estimated marginal means with 95% confidence interval showing the positive relationship between the number of sensory modalities in hallucinatory experiences and the probability of a childhood trauma (OR = 2.24 (95% CI 1.16–4.33), $P = 0.017$, $n = 75$).

Select one:

Confidence interval around the probability of multimodal hallucinations

The number of patients with a given number of hallucinatory modalities

Number of different trauma events for each given number of hallucinatory modalities

Confidence interval around the estimated probability of trauma

Confidence interval around the odds ratio

Your answer is correct.

In a logistic regression plot, the main curve is the best fitting line through the points of estimated probability for each value of the predictor. The shaded area around the curve is the confidence interval (generally with 95% bounds) of the estimated probability of the outcome based on the values of the independent predictor

The correct answer is: Confidence interval around the estimated probability of trauma

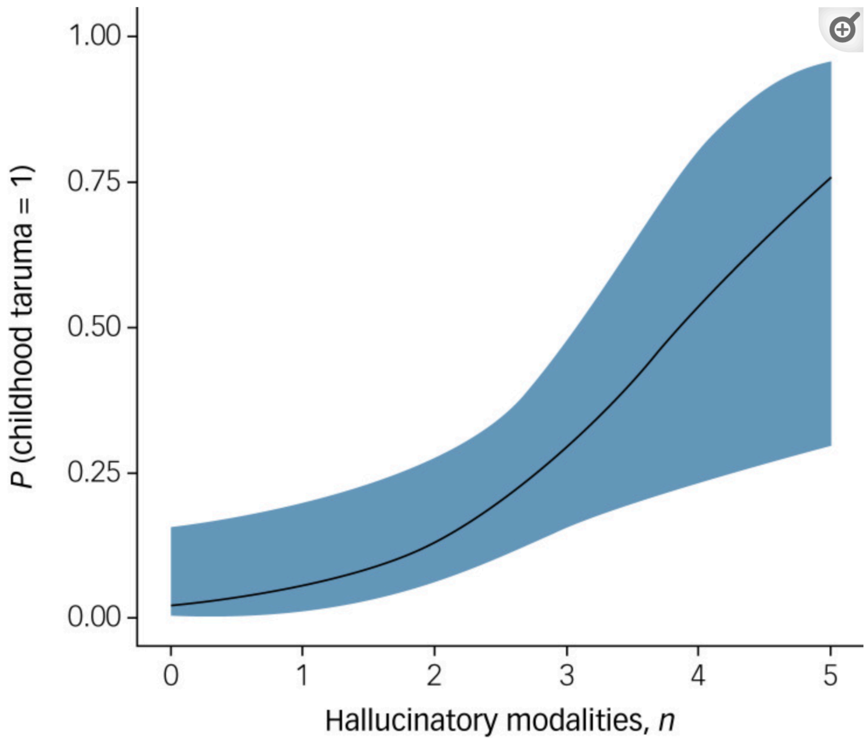
Question 43

Correct

Mark 1.00 out of 1.00

To understand the relationship between childhood traumatic experiences and the sensory complexity of hallucinations, a cohort of 75 children and adolescents seeking help for hallucinations were studied (Ref: Medjkane F, et al. The British Journal of Psychiatry. 2020 Mar;216(3):156-8). To determine the extent by which the various sensory modalities could predict the probability of a previous early trauma, the authors employed a hierarchical binomial logistic regression with history of childhood trauma as the binary dependent variable with control variables (i.e., age, educational level and number of ICD-10 diagnoses, the presence of suicidal ideation), and the number of hallucinatory modalities as predictors. The number of hallucinatory modalities appeared to be the only significant predictor of childhood trauma (estimate (i.e. the log odds of 'history of childhood trauma' = 1 v. 0) 0.81, s.e. = 0.34, z = 2.39, P = 0.017, OR = 2.24 (95% CI 1.16–4.33). The attached graph is most likely a

Fig. 1



Estimated marginal means with 95% confidence interval showing the positive relationship between the number of sensory modalities in hallucinatory experiences and the probability of a childhood trauma (OR = 2.24 (95% CI 1.16–4.33), $P = 0.017$, $n = 75$).

Select one:

- Growth curve
- Correlation plot
- Violin plot
- Logistic Regression plot
- Linear Regression plot

Your answer is correct.

When relating two continuous variables, a scatter plot (correlation plot) is often used. When predicting one variable (dependent) from another (independent) variable, a regression plot is used. In the current study, the dependent is a categorical one (yes/no type of binary variable) while the independent variables is an ordinal variable. In this case, a logistic regression is suitable. The resulting graph fits a regression line such that the estimated probability of the dependent variable (y=1) is shown against an 'increasing dose' of the independent predictor variable (values of x).

The correct answer is: Logistic Regression plot

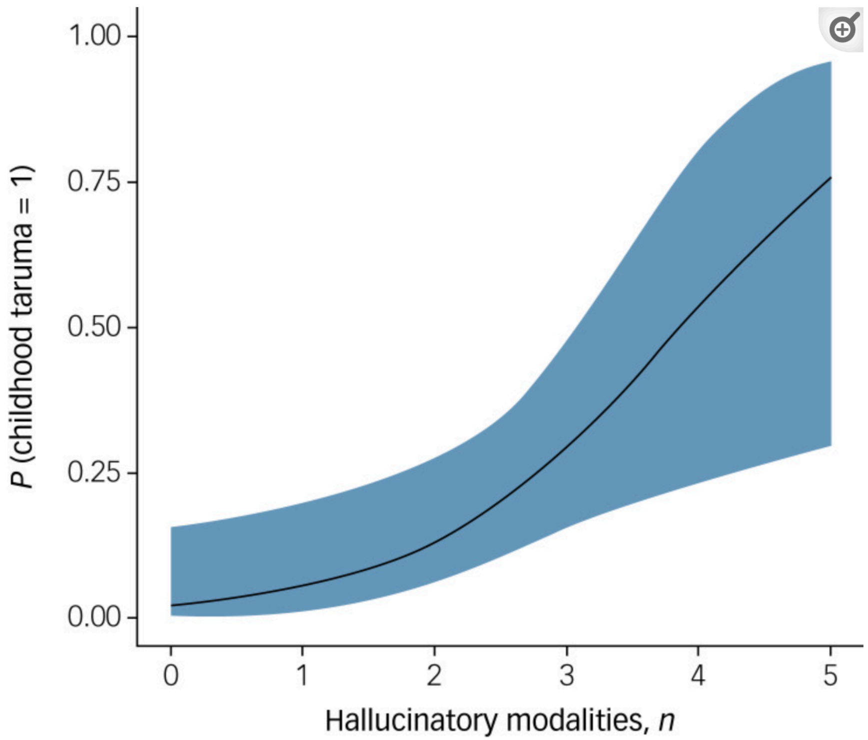
Question 44

Correct

Mark 1.00 out of 1.00

To understand the relationship between childhood traumatic experiences and the sensory complexity of hallucinations, a cohort of 75 children and adolescents seeking help for hallucinations were studied (Ref: Medjkane F, et al. The British Journal of Psychiatry. 2020 Mar;216(3):156-8). To determine the extent by which the various sensory modalities could predict the probability of a previous early trauma, the authors employed a hierarchical binomial logistic regression with history of childhood trauma as the binary dependent variable with control variables (i.e., age, educational level and number of ICD-10 diagnoses, the presence of suicidal ideation), and the number of hallucinatory modalities as predictors. The number of hallucinatory modalities appeared to be the only significant predictor of childhood trauma (estimate (i.e. the log odds of 'history of childhood trauma' = 1 v. 0) 0.81, s.e. = 0.34, z = 2.39, P = 0.017, OR = 2.24 (95% CI 1.16–4.33). In the attached figure, the term 'p' most likely refers to:

Fig. 1



Estimated marginal means with 95% confidence interval showing the positive relationship between the number of sensory modalities in hallucinatory experiences and the probability of a childhood trauma (OR = 2.24 (95% CI 1.16–4.33), $P = 0.017$, $n = 75$).

Select one:

The probability of not having childhood trauma

Significance of the statistical test

The probability of having multimodal hallucinations

The probability of the dependent variable

The probability of suicidal ideation if childhood trauma is present

Your answer is correct.

The y axis of this graph refers to the probability of having a history of childhood trauma (dependent variable). If the value is 1, it means the probability of having had an exposure to trauma is 100%. The x axis of this graph refers to the number of modalities in which hallucinations occur in the patients who were recruited for this research study. When 5 different sensory modalities are affected, the probability of having had childhood trauma is likely to be quite high.

The correct answer is: The probability of the dependent variable

Question **45**

Correct

Mark 1.00 out of 1.00

When conducting qualitative research, a researcher makes active attempts to withhold her biases and individual ideas regarding the phenomena of interest. This is known as

Select one:

- Reflexivity
- Conservatism
- Liberalisation
- Bracketing
- Exclusivity

Your answer is correct.

The term “bracketing”, first used by Husserl, is employed in qualitative research to mean the suspension of judgment and the setting aside of the researcher’s own bias, assumptions and prior notions, with regard to the object of their study.

Ref: Squires V. Reflexive bracketing. In Varieties of qualitative research methods: selected contextual perspectives 2023 Jan 2 (pp. 425-430). Cham: Springer International Publishing.

The correct answer is: Bracketing

Question **46**

Correct

Mark 1.00 out of 1.00

The Declaration of Helsinki is a set of ethical guidelines relating to

Select one:

- Rights of children and adolescents
- Confidentiality of patient information
- Research on human subjects
- Determination of capacity
- Third-party consent

Your answer is correct.

The Declaration of Helsinki was adopted by the World Medical Association in 1964 at its 18th General Assembly. It is a set of ethical guidelines relating to research on human subjects and is widely regarded as authoritative. It is a living instrument and is reviewed and revised regularly. It both reflects and shapes the ethos of international research ethics.

Ref: Malik AY, Foster C. The revised Declaration of Helsinki: cosmetic or real change?. Journal of the Royal Society of Medicine. 2016 May;109(5):184-9.

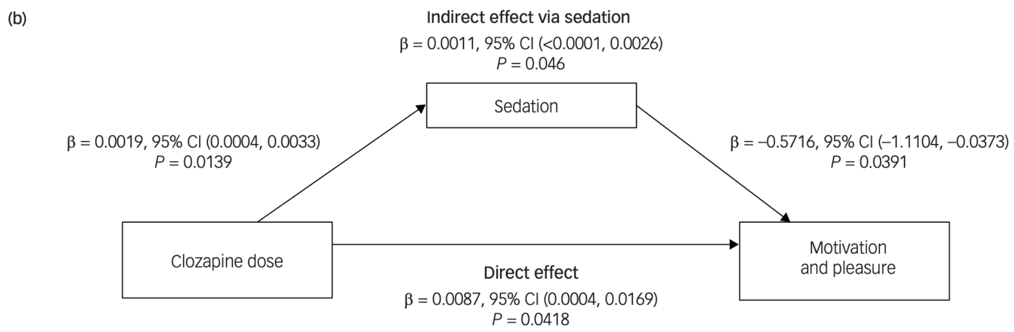
The correct answer is: Research on human subjects

Question **47**

Correct

Mark 1.00 out of 1.00

To understand the longitudinal effects of clozapine on motivation and pleasure (negative symptoms), Wolpe and colleagues studied a cohort of 187 clozapine-treated patients with schizophrenia followed for over 2 years on average. The authors examined the severity of reduced motivation and pleasure (MAP) and the effect of antipsychotic-induced sedation (assessed via the proxy of total daily sleep duration) on MAP. The result of their analysis is summarised in the graph below. They concluded that addressing the sedative side effects of antipsychotics is important to improve clinical outcomes. Which of the following inferences is accurate based on the reported results?



Select one:

- Clozapine mediates the effect of sedation on motivation
- After accounting for sedation, higher clozapine dose reduces motivation
- After accounting for sedation, higher clozapine dose improves motivation
- In the absence of sedation, clozapine has no effect on motivation
- The effect of clozapine on sedation is insignificant

Your answer is correct.

Sedation, not clozapine, is the mediator in this model. According to the mediation model results, increased sedation levels were linked to reduced MAP levels (Note the negative sign of $\beta = -0.57$, $P = 0.039$)), and clozapine increased sedation levels ($\beta = 0.0019$, $P = 0.0139$), suggesting that clozapine-related sedation worsened MAP within participants (Indirect effect). However, after accounting for the clozapine-related sedation (in other words, in the absence of sedation), clozapine improved MAP within participants (Direct effect; $\beta = 0.0087$, $P = 0.042$).

The correct answer is: After accounting for sedation, higher clozapine dose improves motivation

Question **48**

Correct

Mark 1.00 out of 1.00

In ICD-11, which of the following are trait domain qualifiers that may be applied to personality disorders or personality difficulty?

Select one:

- Positive affectivity, Anankastia, and Detachment
- Dissociality, Disinhibition, and Detachment
- Antisociality, Borderline pattern, and Disinhibition
- Anankastia, Negative affectivity, and Asociality
- Negative affectivity, Attachment, and Borderline pattern

Your answer is correct.

In ICD-11, trait domain qualifiers may be applied to personality disorders or personality difficulty to describe the characteristics of the individual's personality that are most prominent and that contribute to personality disturbance. ICD-11 trait domain specifiers include: Negative affectivity, Detachment, Dissociality, Disinhibition, Anankastia, and Borderline pattern.

Ref: Bach B, et al. The ICD-11 classification of personality disorders: a European perspective on challenges and opportunities. Borderline Personality Disorder and Emotion Dysregulation. 2022 Dec;9(1):1-1.

The correct answer is: Dissociality, Disinhibition, and Detachment

Question 49

Correct

Mark 1.00 out of 1.00

In ICD-11, which of the following differentiates anorexia nervosa from bulimia nervosa?

Select one:

- Binge eating
- Very low body weight
- Purging
- Suicidality
- Excessive exercise

Your answer is correct.

Individuals with anorexia nervosa may engage in binge eating and purging but can be distinguished from individuals with bulimia nervosa by their very low body weight. If binge eating and purging are associated with very low body weight and all the other diagnostic requirements are met, a diagnosis of anorexia nervosa, binge-purge pattern, rather than bulimia nervosa should be assigned.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Very low body weight

Question **50**

Incorrect

Mark 0.00 out of 1.00

In the UK, what is the most common type of mental disorder in patients on mental health rehabilitation units?

Select one:

- Intellectual disability
- OCD
- Psychotic disorder
- Anxiety disorder
- Mood disorder

Your answer is incorrect.

The most common disorders in patients on mental health rehabilitation disorders are psychotic disorders, including schizophrenia and schizoaffective disorder. Bipolar disorder is also relatively common.

Ref: https://www.rcpsych.ac.uk/mental-health/treatments-and-wellbeing/mental-health-rehabilitation-services

The correct answer is: Psychotic disorder

Question **51**

Correct

Mark 1.00 out of 1.00

The male to female ratio of the prevalence of obsessive-compulsive disorder (OCD) in adults is approximately

Select one:

- 1 to 2
- 2 to 1
- 1 to 1.5
- 3 to 1
- 1 to 3

Your answer is correct.

The mean age of onset of OCD is 20 years, with 70% onset before age 25 years, and 15% onset after age 35 years. The male to female ratio is approximately 1 to 1.5 in adults, while paediatric OCD appears to be slightly more common in males. As adult males are more treatment seeking, in clinical samples the sex ratio often appears to be 1:1.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 385.
Fawcett EJ, Power H, Fawcett JM. Women are at greater risk of OCD than men: a meta-analytic review of OCD prevalence worldwide. The Journal of clinical psychiatry. 2020 Jun 23;81(4):13075.

The correct answer is: 1 to 1.5

Question **52**

Correct

Mark 1.00 out of 1.00

What is the most common mental disorder in stalking offenders?

Select one:

Anxiety disorder

Psychotic disorder

Mood disorder

Personality disorder

Substance use disorder

Your answer is correct.

Stalking is a behaviour, not a mental disorder, but research suggests that up to 50% of offenders experience some sort of mental disorder. The most common is personality disorder. Other common diagnoses include a psychotic disorder, mood or anxiety disorder, and substance use disorder.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019. p. 742. Nijdam-Jones A et al. Psychopathology of stalking offenders: Examining the clinical, demographic, and stalking characteristics of a community-based sample. Criminal Justice and Behavior. 2018 May;45(5):712-31.

The correct answer is: Personality disorder

Question **53**

Incorrect

Mark 0.00 out of 1.00

Which of the following behaviours is most unlikely to be involved in factitious disorder?

Select one:

- Ingesting a substance to produce illness
- Repeatedly simulating seizures
- Manipulating laboratory tests
- Falsifying medical records
- Reporting vague abdominal discomfort

Your answer is incorrect.

Examples of behaviours involved in factitious disorder include falsely reporting or simulating episodes of neurological or mental symptoms (e.g., seizures, hearing voices); manipulating laboratory tests to falsely indicate an abnormality; falsifying past or current medical records to indicate an illness; ingesting a substance to produce an abnormal laboratory result or illness; and physically injuring or intentionally inducing illness in oneself (e.g., intentional exposure to infectious or toxic agents).

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Reporting vague abdominal discomfort

Question **54**

Incorrect

Mark 0.00 out of 1.00

What is the likelihood of an individual completing suicide within two years following an initial act of deliberate self-harm?

Select one:

- 10%
- 5%
- 3%
- 1%
- 15%

Your answer is incorrect.

In this population of patients, an estimated 1.49% will die by completed suicide in the 24 months after the initial act, with the risk highest in the weeks following the original act, in males and in older cohorts. This represents a mortality by suicide 50-100 times that of the general population.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 848.
Liu BP, Lunde KB, Jia CX, Qin P. The short-term rate of non-fatal and fatal repetition of deliberate self-harm: A systematic review and meta-analysis of longitudinal studies. Journal of affective disorders. 2020 Aug 1;273:597-603.

The correct answer is: 1%

Question **55**

Correct

Mark 1.00 out of 1.00

To assess if schizophrenia, mood disorder, or anxiety disorder are associated with mortality in patients with COVID-19, a retrospective cohort study of 7348 consecutive adult patients observed for 45 days following COVID-19 infection was conducted between March 3 and May 31, 2020, in New York (Ref: Nemani K, et al. JAMA psychiatry. 2021 Apr 1;78(4):380-6). The attached table reports on the outcome of mortality, defined as death or discharge to hospice within 45 days following a positive severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) test result. What can be inferred from the table?

Odds Ratios and Rates of 45-Day Case Fatality by Lifetime Psychiatric Diagnosis

SARS-CoV-2– Positive	Mortality or hospice, No. (%)	OR (95% CI)		
		Unadjusted	Demographically adjusted[a]	Fully adjusted[b]
All patients (n = 7348)	864 (11.8)	NA	NA	NA
Schizophrenia spectrum (n = 75)	20 (26.7)	2.93 (1.75, 4.92)	2.87 (1.62-5.08)	2.67 (1.48-4.80)
Mood disorders (n = 564)	104 (18.4)	1.82 (1.45, 2.29)	1.25 (0.98-1.61)	1.14 (0.87-1.49)
Anxiety disorders (n = 360)	39 (10.8)	0.98 (0.70, 1.38)	0.97 (0.67-1.41)	0.96 (0.65-1.40)
Reference[c] (n = 6349)	701 (11.0)	1 [Reference]	1 [Reference]	1 [Reference]

Abbreviations: NA, not applicable; OR, odds ratio; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

[a]The demographically adjusted OR included age, race, and sex.

[b]The fully adjusted model included demographic variables in addition to smoking status, hypertension, heart failure, myocardial infarction, diabetes, chronic kidney disease, chronic obstructive pulmonary disease, and cancer.

[c]The reference group excluded patients with any history of a schizophrenia spectrum, mood, or anxiety disorder diagnosis or other psychiatric diagnoses listed in eTable 1 of the Supplement.

Select one:

A premorbid diagnosis of an anxiety disorder predicted mortality even after adjusting for other risk factors.

[redacted]

95% of patients with schizophrenia had 1.48-4.80 times higher risk of mortality after COVID-19 infection.

A premorbid diagnosis of a schizophrenia predicted mortality even after adjusting for other risk factors.

Mood disorders had no significant association with mortality after COVID-19 infection.

No one in the reference group died after COVID-19 infection; thus, the Odds Ratio for this group was OR=1

Your answer is correct.

According to the results, when demographic and medical risk factors are adjusted for, only a premorbid diagnosis of a schizophrenia spectrum disorder was significantly associated with mortality. The unadjusted risk seen for mood disorders became insignificant, suggesting that the apparent association with mood disorders and mortality is explained by demographic and other risks in this population. Anxiety disorders had no notable association with mortality before or after adjustment. The reference group has OR=1 because it is the comparator for generating OR; there is a number of deaths in this group as well (n=701).

The correct answer is: A premorbid diagnosis of a schizophrenia predicted mortality even after adjusting for other risk factors.

Question **56**

Correct

Mark 1.00 out of 1.00

Which of the following most increases the risk of reoffending in indecent exposure offenders?

Select one:

- Higher education level
- Older age
- Being married
- Being employed
- 2 previous offences

Your answer is correct.

Risk factors for sexual reoffending (including indecent exposure) include younger age, unemployment, previous offences, and mental disorder. Other risk factors for violent reoffending include single status, lower education level, and substance use disorders.

Ref: Yu R. et al. Prediction of reoffending risk in men convicted of sexual offences: development and validation of novel and scalable risk assessment tools (OxRIS). Journal of criminal justice. 2022 Sep 1;82:101935.

The correct answer is: 2 previous offences

Question **57**

Correct

Mark 1.00 out of 1.00

In the UK, which of the following is characterised by a pattern of acts of assaults, threats, humiliation, degradation, and intimidation that is used to harm, punish, isolate, and/or frighten a victim?

Select one:

- Indecent exposure
- Stalking
- Coercive control
- Insane automatism
- Sane automatism

Your answer is correct.

Coercive control is an act or pattern of acts of assault, threats, humiliation, intimidation, or other abuse that is used to harm, punish, or frighten a victim. This controlling behaviour is intended to make a person dependent by isolating them from support, exploiting them, depriving them of their independence, and regulating their everyday behaviour. Common examples of coercive controlling behaviour are humiliation, degradation, intimidation, threats, and dehumanisation.

Ref: https://www.womensaid.org.uk/information-support/what-is-domestic-abuse/coercive-control/
Lohmann S, Cowlishaw S, Ney L, O'Donnell M, Felmingham K. The trauma and mental health impacts of coercive control: a systematic review and meta-analysis. Trauma, Violence, & Abuse. 2024 Jan;25(1):630-47.

The correct answer is: Coercive control

Question **58**

Correct

Mark 1.00 out of 1.00

Of the following, which is the most likely to be seen in malingering or feigning, as opposed to a case of PTSD?

Select one:

- The patient displays hypervigilance
- The patient is initially reluctant to open up
- The exact same nightmare is described as occurring every night
- There is avoidance of situations reminiscent of the traumatic event
- The patient feels overwhelmed by intense emotions related to the trauma

Your answer is correct.

PTSD is characterised by: 1) re-experiencing in the form of vivid intrusive memories, flashbacks, or nightmares; 2) avoidance (for example, avoidance of situations reminiscent of the event(s)); and 3) persistent perceptions of heightened current threat (for example, as indicated by hypervigilance). Repetitive dreams or nightmares are thematically related to the traumatic event, and re-experiencing is typically accompanied by strong or overwhelming emotions. Re-experiencing in the present can also involve feelings of being overwhelmed or immersed in the same intense emotions that were experienced during the traumatic event.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: The exact same nightmare is described as occurring every night

Question 59

Correct

Mark 1.00 out of 1.00

What neuroimaging finding is most common in individuals with both Down syndrome and Alzheimer dementia?

Select one:

- Cerebellar atrophy
- Enlarged third and fourth ventricles
- Isolated atrophy of the basal ganglia
- Cortical thinning
- Generalised corticocerebellar atrophy

Your answer is correct.

EXPLANATION: The atrophy pattern associated with Alzheimer disease in those with Down syndrome involves posterior dominant cortical thinning with atrophy of the hippocampus, thalamus, and striatum in a similar pattern to what is seen in sporadic Alzheimer disease.

Ref: Fortea J et al. Alzheimer's disease associated with Down syndrome: a genetic form of dementia. The Lancet Neurology. 2021 Nov 1;20(11):930-42.

The correct answer is: Cortical thinning

Question 60

Incorrect

Mark 0.00 out of 1.00

Psychosis is seen in what percentage of people with Fragile X syndrome?

S[redacted]ct one:

>50%

15%

<10%

33%

25%

Your answer is incorrect.

Psychosis is seen in less than 10% of those with Fragile X syndrome (FXS). The syndrome is commonly associated with other psychiatric disorders such as ADHD, learning disorders, OCD, depression, anxiety, and language function disorders.

Ref: Keshtkarjahromi M. et al. Psychosis and catatonia in fragile X syndrome. Cureus. 2021 Jan 21;13(1).

The correct answer is: <10%

Question **61**

Correct

Mark 1.00 out of 1.00

What type of Down syndrome is inherited?

Select one:

- Partial trisomy 21
- Free trisomy 21
- Mosaicism
- Translocation
- Nondisjunction

Your answer is correct.

Translocation of Robertsonian type accounts for trisomy 21 in approximately 3-5% of affected individuals, usually t(14;21) or t(21;21). In translocation, a fusion occurs of two chromosomes, resulting in a total of 46 chromosomes despite the presence of an extra chromosome 21. This is usually inherited (though spontaneous translocations can occur). The translocated chromosome may be found in unaffected parents and siblings. Asymptomatic carriers have only 45 chromosomes but have balanced translocation as they have enough genetic material, albeit in the wrong location. The risk of having another child with Down syndrome if a parent has balanced translocation is estimated to be 15%.

Ref: Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Feb 6;6(1):1-20. Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 101.

The correct answer is: Translocation

Question 62

Incorrect

Mark 0.00 out of 1.00

Which of the following is most likely to be seen in the neonate if paroxetine is taken by the mother late in pregnancy?

Select one:

- Respiratory distress
- Persistent pulmonary hypertension
- Tetralogy of Fallot
- Convulsions
- Irritability

Your answer is incorrect.

The neonate may experience discontinuation symptoms, which are usually mild, such as agitation and irritability, or rarely respiratory distress and convulsions. The risk of discontinuation symptoms is understood to be particularly high with short half-life drugs such as paroxetine and venlafaxine. Paroxetine has been specifically associated with cardiac malformations, particularly after high-dose, first trimester exposure. However, some studies have failed to replicate this finding for paroxetine and have implicated other SSRIs. When taken late in pregnancy, SSRIs may increase the risk of persistent pulmonary hypertension, but the absolute risk is very low.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 689.

The correct answer is: Irritability

Question **63**

Correct

Mark 1.00 out of 1.00

A 62-year-old man has a 1-year history of rigidity and difficulty with his gait, with the recent development of fluctuating cognitive impairment. He also reports seeing intruders in his apartment and is suspicious that one of them has replaced his daughter. Which of the following is the most likely cause of his presentation?

Select one:

[redacted]

Lewy body disease

Alzheimer disease

Vascular disease

Frontotemporal lobar degeneration

Parkinson disease

Your answer is correct.

Dementia with Lewy bodies (DLB) shares many features with Parkinson disease (PD). Patients with DLB exhibit parkinsonism – a masked face, bradykinesia, rigidity, and gait impairment – but little or no tremor. In contrast to PD, where cognitive impairment usually does not arise until 5 years into the illness, cognitive impairment (if not dementia) is characteristically present within a year of onset of DLB. Visual hallucinations plague DLB patients, as well as PD patients, but begin soon after the onset of their illness. In DLB, visions of people or animals and other hallucinations often appear so detailed and vivid that they provoke fear and precipitate confusion. A classic symptom of DLB is Capgras syndrome in which patients believe that an 'imposter' has replaced a close friend or relative.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 126.

The correct answer is: Lewy body disease

Question **64**

Correct

Mark 1.00 out of 1.00

A 68-year-old woman who has been taking flupenthixol and procyclidine for 20 years develops moderate tardive dyskinesia (TD). Which of the following is the only licensed treatment for moderate to severe TD in the UK?

Select one:

- Tetrabenazine
- Amantadine
- Clonazepam
- Reserpine
- Pyridoxine

Your answer is correct.

Tetrabenazine is the only licensed treatment for moderate to severe TD in the UK. Depression, drowsiness, parkinsonism, and akathisia may occur as side effects. Reserpine (similar mode of action) is also effective but is rarely, if ever, used due to its association with depression and suicidality.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 118.

The correct answer is: Tetrabenazine

Question **65**

Correct

Mark 1.00 out of 1.00

Which of the following tests is most likely to be useful in the identification of Korsakoff syndrome?

Select one:

- Grip strength
- Digit span
- Verbal fluency
- Finger tapping
- Delayed recall

Your answer is correct.

Korsakoff syndrome is characterised by the absence of or significant impairment in the ability to lay down new memories (impairment in recent memory is especially prominent), together with a variable length of retrograde amnesia. Working memory (e.g. ability to remember a sequence of numbers) is unimpaired, as is procedural and emotional memory.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 606-607.
Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 277.

The correct answer is: Delayed recall

Question 66

Correct

Mark 1.00 out of 1.00

While taking a dopamine agonist for the treatment of Parkinson disease, a patient develops psychosis. Which of the following should be the first step with respect to medication management?

Select one:

Start olanzapine

Start low-dose quetiapine

Start a cholinesterase inhibitor

Start a trial of clozapine

Decrease the dopamine agonist

Your answer is correct.

With respect to the medication management of psychosis in Parkinson disease, the first step is to consider reducing or stopping anticholinergics and dopamine agonists. Following this, consideration can be given to an atypical antipsychotic (it is probably reasonable to try low-dose quetiapine, which is the best tolerated). Prior to a trial of clozapine, consideration can be given to a cholinesterase inhibitor, particularly if the patient has co-morbid dementia.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 815-816.

The correct answer is: Decrease the dopamine agonist

Question **67**

Correct

Mark 1.00 out of 1.00

How much higher is the risk of dementia at 85 years of age, compared to 65 years of age?

Select one:

- 2 times
- 4 times
- 8 times
- 16 times
- 32 times

Your answer is correct.

Dementia is a progressive syndrome affecting around 5% of those aged >65 years, rising to 20% in the over-80s. The most common cause of dementia is Alzheimer disease, accounting for around 60% of all cases.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 604.

The correct answer is: 4 times

Question 68

Incorrect

Mark 0.00 out of 1.00

Which of the following are included in the pre-treatment stage of dialectical behaviour therapy (DBT)?

Select one:

- Processing of traumatic experiences and establishing future goals
- Reducing life-threatening behaviour and commitment to therapeutic contract
- Processing of traumatic experiences and reducing life-threatening behaviour
- Orientation to therapy and establishing future goals
- Orientation to therapy and commitment to therapeutic contract

Your answer is incorrect.

The pre-treatment phase of DBT includes assessment, orientation to therapy, and commitment to therapeutic contract. Stage 1 focuses on reducing life-threatening behaviour, behaviour which may interfere with the progress of therapy, and behaviour which interferes with the quality of life. Stage 2 focuses on emotional processing of previous traumatic experiences, to target post-traumatic stress-related symptoms such as flashbacks. Stage 3 aims to develop self-esteem and establish future goals.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 917. Zalewski M, et al. Lessons learned conducting dialectical behavior therapy via telehealth in the age of COVID-19. Cognitive and Behavioral Practice. 2021 Nov 1;28(4):573-87.

The correct answer is: Orientation to therapy and commitment to therapeutic contract

Question 69

Correct

Mark 1.00 out of 1.00

Which of the following is considered a primitive defence mechanism?

Select one:

- Compensation
- Reaction Formation
- Sublimation
- Isolation of affect
- Regression

Your answer is correct.

Regression involves responding to stress by reverting to a level of functioning of a previous maturational point (e.g. a teenager sucking their thumb around exam time). Regression is considered a primitive defence mechanism. Reaction formation and isolation of affect are neurotic defence mechanisms. Sublimation and compensation are mature defence mechanisms.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 892-893.

The correct answer is: Regression

Question 70

Correct

Mark 1.00 out of 1.00

Which of the following is considered the first stage of cognitive analytic therapy?

Select one:

Recapitulation

Recognition

Reintegration

Exploration

Reformulation

Your answer is correct.

Cognitive analytic therapy involves three stages. First is reformulation, which involves the identification of key relational patterns and associated ways of acting or coping. Second is recognition, which involves developing the person’s ability to recognise unhelpful patterns. Third is revision, which focuses on developing interventions for or alternatives to the unhelpful patterns that have been identified. During the first, or reformulation, stage of therapy, the emphasis is on collaboratively building a shared narrative concerning the person’s difficulties, which often features several specific reciprocal roles at its core.

Ref: Owen K, Laphan A, Gee B, Lince K. Evaluating cognitive analytic therapy within a primary care psychological therapy service. British Journal of Clinical Psychology. 2023 Jun 14.

The correct answer is: Reformulation

Question **71**

Incorrect

Mark 0.00 out of 1.00

Which of the following types of family therapy introduced the concepts of re-framing and hypothesising?

Select one:

- Strategic
- Narrative
- Psychodynamic
- Solution-focused
- Milan systemic

Your answer is incorrect.

The Milan model introduced the concepts of ‘re-framing’ the problem and hypothesising, as well as ideas about neutrality and curiosity, with the therapist taking more of a ‘not-knowing’, non-expert stance. Importantly, the hypothesis offered is not a reductive speculation, but “suppositions, hunches, maps, explanations or alternative explanations about the family and the 'problem' in its relational context''. Hypothesis offers reasons for the symptom and connective behaviours in a manner that generally includes three or more members of the system.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 719. Barbetta P, Telfener U. The Milan approach, history, and evolution. Family process. 2021 Mar;60(1):4-16.

The correct answer is: Milan systemic

Question 72

Incorrect

Mark 0.00 out of 1.00

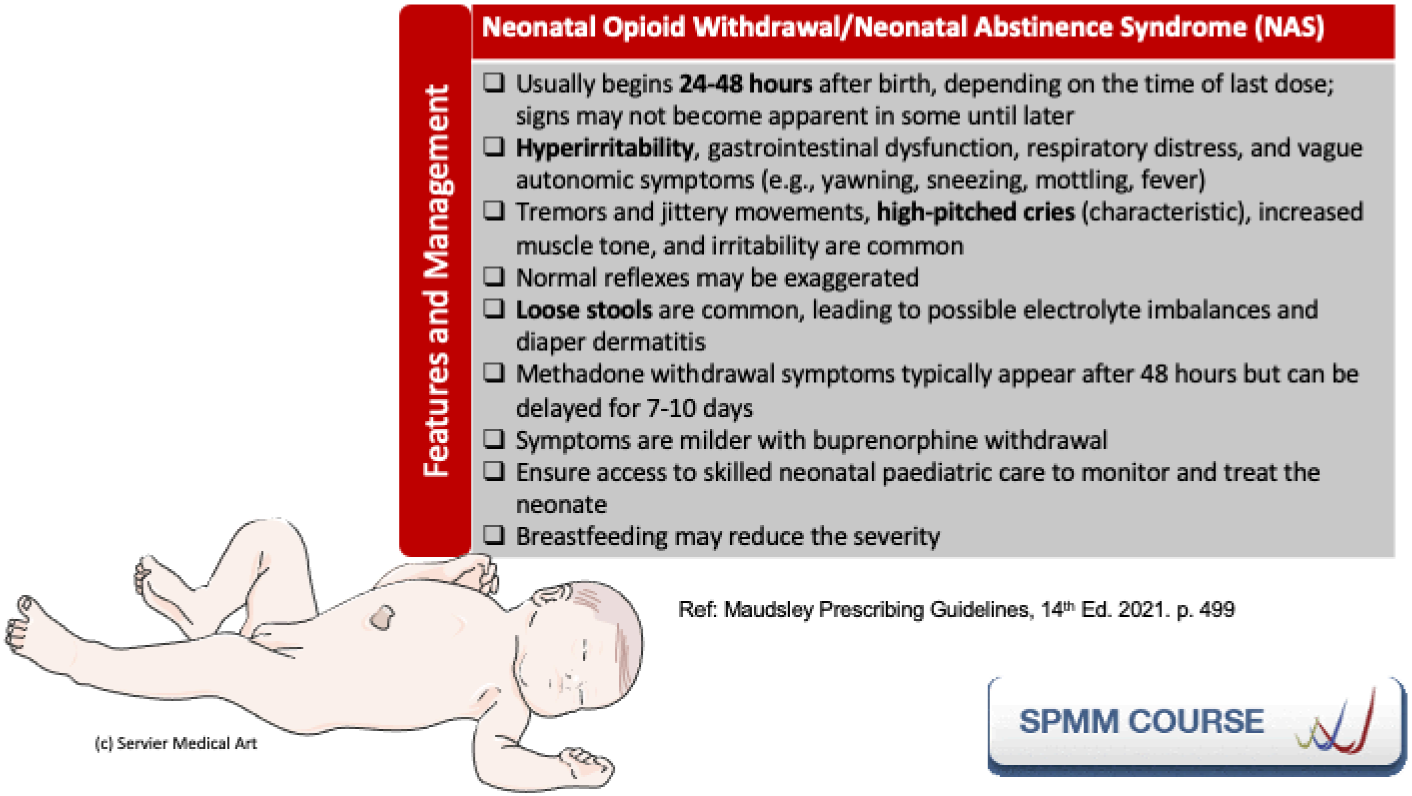
24 hours after birth, a baby is irritable with high-pitched crying. The infant is also noted to be jittery and excessively wakeful. Which of the following substances is the mother most likely to be using?

Select one:

- Opioids
- Cocaine
- Alcohol
- Nicotine
- Cannabis

Your answer is incorrect.

Neonatal abstinence syndrome (NAS) is characterised by a variety of signs and symptoms relating to the autonomic nervous system, gastrointestinal tract, and respiratory system. Infants may have a high-pitched cry, feed hungrily but ineffectively, and be excessively wakeful. Severe NAS is associated with seizures but is uncommon. Breastfeeding may reduce the severity of NAS.



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 499.

The correct answer is: Opioids

Question 73

Incorrect

Mark 0.00 out of 1.00

For what duration are amphetamines usually detectable in the urine?

Select one:

12 hours

48-72 hours

24 hours

7-10 days

21-28 days

Your answer is incorrect.

The duration of detection in the urine depends on half-life, but for most amphetamines, it is typically 48-72 hours.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 531.
Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 631.

The correct answer is: 48-72 hours

Question 74

Correct

Mark 1.00 out of 1.00

A 15-year-old adolescent is seen in an outpatient clinic and diagnosed with mild depression. Which of the following would be the most appropriate first line of management?

Select one:

Watchful waiting

Digital CBT

Group CBT

Interpersonal Therapy

Family Therapy

Your answer is correct.

Based on NICE guidelines, up to 4 weeks of 'watchful waiting' is advised for adolescents with mild depression, during which time contact should be maintained with the family. If symptoms continue, it is advised to offer 2-3 months of individual non-directive supportive therapy, group CBT, or guided self-help. Individual CBT, IPT, or family therapy should be offered for at least 3 months as first-line treatment for moderate to severe depression. This recommendation reflects the evidence that most cases of adolescent depression are time-limited (stably low, or transient) and do not continue to adulthood.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 697.
Weavers, B., et al. The antecedents and outcomes of persistent and remitting adolescent depressive symptom trajectories: a longitudinal, population-based English study. The Lancet Psychiatry, 2021; 8(12), pp.1053-1061.

The correct answer is: Watchful waiting

Question 75

Incorrect

Mark 0.00 out of 1.00

In the management of oppositional defiant disorder, which of the following is a model of parent management training that is effective with adolescents as well as younger children?

Select one:

- Functional family therapy
- Positive parenting programme (Triple P)
- Multisystemic therapy
- Multidimensional therapy in foster care
- Brief strategic family therapy

Your answer is incorrect.

Parent management training (PMT) focuses on parenting skills and includes quality time with the child, as well as differential reinforcement strategies to give proper direction to the child's motivation. Though more effective with smaller children, PMT programmes such as the Triple P model (positive parenting programme) can be effective with adolescents as well. The Triple P program usually involves 10 to 12 sessions lasting 60 to 90 minutes each, where practitioners work with individual families. Originally designed for parents of preschoolers with behavior issues, it has evolved to address various family situations, such as parents of older kids (Teen Triple P) or families dealing with separation (Transition Triple P) or disabilities (Stepping Stones Triple P).

Ref: Booker JA. Oppositional Defiant Disorder. Handbook of Clinical Child Psychology: Integrating Theory and Research into Practice. 2023 Jun 11:857-77.
Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 665-666.

The correct answer is: Positive parenting programme (Triple P)

Question **76**

Correct

Mark 1.00 out of 1.00

On receiver-operating characteristic (ROC) curves, which of the following screening tests has the best cut-off for identifying at-risk drinkers?

Select one:

- CDT
- MCV
- GGT
- AUDIT
- CAGE

Your answer is correct.

EXPLANATION: The Alcohol Use Disorders Identification Test (AUDIT) is the gold standard screening instrument for hazardous drinking in the adult population, for which an abbreviated version has been developed: the AUDIT-Consumption (AUDIT-C). NICE recommends using the AUDIT, or the abbreviated AUDIT-C and the Fast Alcohol Screening Test (FAST).

Ref: Verhoog S et al. The use of the alcohol use disorders identification test–Consumption as an indicator of hazardous alcohol use among university students. European addiction research. 2020;26(1):1-9.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 583.

The correct answer is: AUDIT

Question **77**

Correct

Mark 1.00 out of 1.00

A man in his early thirties takes an overdose of prescribed medication and presents with altered consciousness and seizures. Which of the following is he most likely to have taken?

Select one:

- Methadone
- Buprenorphine
- Tramadol
- Dihydrocodeine
- Morphine

Your answer is correct.

EXPLANATION: Seizures are a common complication of drug intoxication. Antidepressants, diphenhydramine, stimulants (including cocaine and methamphetamine), tramadol, and isoniazid account for the majority of cases. In Australia, tramadol overdose is a particularly common cause of seizures. Unlike cocaine intoxication, heroin and morphine overdoses rarely cause seizures.

Ref: Chen HY et al. Treatment of drug-induced seizures. British journal of clinical pharmacology. 2016 Mar;81(3):412-9.

Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 534-535.

The correct answer is: Tramadol

Question **78**

Correct

Mark 1.00 out of 1.00

According to NICE guidelines, what is the youngest age that a child that may be considered for antidepressant treatment of depression?

Select one:

6 years

4 years

7 years

8 years

5 years

Your answer is correct.

EXPLANATION: NICE guidelines cover the identification and management of depression in children and young people aged 5-18 years. Following multidisciplinary review, fluoxetine may be cautiously considered for the treatment of moderate to severe depression in a child of 5-11 years if they are unresponsive to psychological therapy, although the evidence for fluoxetine's effectiveness in this age group is not established. There has been no research investigating antidepressant medication use in preschool children, and medications are not recommended for this age group.

Ref: National Institute for Health and Clinical Excellence. The NICE guideline on the identification and management of depression in children and young people. 2019.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 545.

The correct answer is: 5 years

Question 79

Correct

Mark 1.00 out of 1.00

According to ICD-11, in order to diagnose a child with autism spectrum disorder, they must have demonstrated developmental issues prior to what age?

Select one:

- 4 years
- ICD-11 does not specify
- 3 years
- 2 years
- 5 years

Your answer is correct.

EXPLANATION: Autism spectrum disorder (ASD) can present clinically at all ages, including during adulthood, and is a lifelong disorder. Characteristic features may emerge during infancy, although they may only be recognised as indicative of ASD in retrospect. It is usually possible to make the diagnosis of ASD during the preschool period (up to 4 years), especially in children exhibiting generalised developmental delay. However, the manifestations and impacts of ASD are likely to vary according to age, intellectual and language abilities, co-occurring conditions, and environmental context.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: ICD-11 does not specify

Question 80

Correct

Mark 1.00 out of 1.00

A 17-year-old girl presents with recurrent episodes of overeating and loss of control over food consumption. She has recently gained weight and her BMI is 22. She is interested in psychological therapies. Which of the following is a first-line treatment for her condition?

Select one:

- Interpersonal therapy
- Family therapy
- Cognitive analytic therapy
- Dialectical behaviour therapy
- Cognitive behavioural therapy

Your answer is correct.

EXPLANATION: First-line treatment of binge eating disorder is guided self-help and up to 20 sessions of adapted CBT.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 418.

The correct answer is: Cognitive behavioural therapy

Question **81**

Correct

Mark 1.00 out of 1.00

Which of the following is a pre-requisite for providing a psychiatric report in criminal proceedings?

Select one:

- Physical examination
- Interviewing the patient
- Getting patient consent
- Receiving instructions
- Mental state examination

Your answer is correct.

EXPLANATION: The request for psychiatric assessment should indicate the legal issues towards which the psychiatrist should direct the assessment. In many cases the instructions are not specific, but the main issues to consider are usually: fitness to plead; criminal responsibility; the presence of a mental disorder and whether assessment and/or treatment under compulsion (or otherwise) is required; and the risk the person poses. With respect to the interview itself, it is good practice and advisable to gain written consent from the person to access their records and to prepare the report. If the person refuses to be interviewed, this should be respected and reported to the person requesting the report.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 129, 764-765.

The correct answer is: Receiving instructions

Question **82**

Correct

Mark 1.00 out of 1.00

The prevalence of depression in older adults living in institutions is

Select one:

[redacted]

Similar to those living in the community

Higher than those living in the community

Higher than those in acute hospitals

Lower than those living in the community

Identical to those with subclinical depressive symptoms

Your answer is correct.

EXPLANATION: The prevalence of depression is 8.7% (95% CI 7.3-10.2) in older adults living in the community in England and Wales, and as high as 27% for those living in institutions. Studies of older adults in acute hospitals show that the prevalence of depression is 29%.

Ref: Lilford P, Hughes JC. Epidemiology and mental illness in old age. BJPsych Advances. 2020 Mar;26(2):92-103.

The correct answer is: Higher than those living in the community

Question 83

Correct

Mark 1.00 out of 1.00

Which of the following is true regarding the progression of mild cognitive impairment (MCI) to dementia?

Select one:

- The progression from MCI to dementia follows a predictable course
- Acetylcholinesterase inhibitors prevent progression of MCI to dementia
- Those with MCI are a heterogenous group with variable prognosis
- The vast majority of those with MCI ultimately progress to dementia
- People with MCI will not return to normal levels of cognition

Your answer is correct.

EXPLANATION: Mild cognitive impairment (MCI) is hypothesised to represent a pre-clinical stage of dementia but forms a heterogenous group with variable prognosis. Although around 10% of elderly individuals with MCI will progress to dementia in one year, others will never develop dementia and some return to normal levels of cognition. A Cochrane review assessing the safety and efficacy of acetylcholinesterase inhibitors in MCI found there was very little evidence that they affect progression to dementia or cognitive test scores.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 625.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 159.

The correct answer is: Those with MCI are a heterogenous group with variable prognosis

Question **84**

Correct

Mark 1.00 out of 1.00

Which of the following is characteristic of late-onset schizophrenia (LOS) and very-late-onset schizophrenia-like psychosis (VLOS), as compared to early-onset schizophrenia (EOS)?

S[redacted]ct one:

- Higher rates of and more severe persecutory delusions
- Male preponderance
- More negative symptoms
- Higher genetic risk
- Prominent formal thought disorder

Your answer is correct.

EXPLANATION: Critical differences exist in the clinical presentation of LOS and VLOS compared with the early-onset type. Researchers have generally found a female preponderance, lower genetic risk (reflected by a lower incidence of family history of psychosis), and higher rates of and more severe paranoid symptoms and persecutory delusions in LOS and VLOS patients. Partition delusions - concerning the unwanted permeability of borders - are highly prominent. On the other hand, formal thought disorder is usually absent. Furthermore, there are fewer negative symptoms such as affective flattening in LOS and VLOS compared to EOS.

Ref: Suen YN et al. Late-onset psychosis and very-late-onset-schizophrenia-like-psychosis: an updated systematic review. International Review of Psychiatry. 2019 Aug 18;31(5-6):523-42.

Van Assche L et al. The neuropsychology and neurobiology of late-onset schizophrenia and very-late-onset schizophrenia-like psychosis: a critical review. Neuroscience & Biobehavioral Reviews. 2017 Dec 1;83:604-21.

The correct answer is: Higher rates of and more severe persecutory delusions

Question **85**

Correct

Mark 1.00 out of 1.00

Which of the following personality disorders is associated with death by suicide in older adults?

[redacted]lect one:

Narcissistic

Avoidant

Antisocial

Dependent

Histrionic

Your answer is correct.

EXPLANATION: Obsessive-compulsive and avoidant personality disorders have been implicated in death by suicide in older adults, although this association may be confounded by depression. An inability to adapt to the changes occurring in late life may help to explain the association between suicide in old age and higher conscientiousness as well as obsessive-compulsive and avoidant personality disorders. Cluster B personality disorders and their defining constructs play a smaller role in late-life suicide compared to suicide in younger adults.

Ref: Szücs A et al. Personality and suicidal behavior in old age: a systematic literature review. Frontiers in psychiatry. 2018 May 7;9:128.

The correct answer is: Avoidant

Question **86**

Correct

Mark 1.00 out of 1.00

Most of the heritability in frontotemporal dementia is accounted for by what type of mutations?

Select one:

- Autosomal dominant
- X-linked recessive
- Autosomal recessive
- X-linked dominant
- Point

Your answer is correct.

EXPLANATION: Frontotemporal dementia (FTD) is a highly heritable group of neurodegenerative disorders, with around 30% of patients having a strong family history. Most of that heritability is accounted for by autosomal dominant mutations in the chromosome 9 open reading frame 72 (C9orf72), progranulin (GRN), and microtubule-associated protein tau (MAPT) genes, with mutations more rarely see in several other genes. Behavioural variant FTD (bvFTD) is the most common diagnosis in each of the genetic groups, although in C9orf72 carriers, amyotrophic lateral sclerosis either alone, or with bvFTD, is also common.

Ref: Greaves CV, Rohrer JD. An update on genetic frontotemporal dementia. Journal of neurology. 2019 Aug;266(8):2075-86.

The correct answer is: Autosomal dominant

Question **87**

Correct

Mark 1.00 out of 1.00

What is the most important indicator of cognitive performance in depression?

Select one:

- Male sex
- Late age of depression onset
- Current depression severity
- Early age of depression onset
- Historic depression severity

Your answer is correct.

EXPLANATION: Cognitive impairment may occur in major depressive disorder at any age, and different patterns of cognitive difficulties in later-onset depression (LOD) compared with early onset depression (EOD) have been reported. However, results have been conflicting, with greater executive dysfunction demonstrated in patients with LOD than EOD in some studies, with others finding the reverse. Similarly, several studies have shown that patients with LOD are more impaired in memory function, specifically in episodic memory, while others have had opposite results, or found similar patterns of cognitive impairment in LOD and EOD patients. The most important indicator of cognitive performance in depression appears to be current, rather than historic, depression severity.

Ref: Eraydin IE et al. Investigating the relationship between age of onset of depressive disorder and cognitive function. International journal of geriatric psychiatry. 2019 Jan;34(1):38-46.

Alexopoulos GS. Mechanisms and treatment of late-life depression. Translational psychiatry. 2019 Aug 5;9(1):1-6.

The correct answer is: Current depression severity

Question 88

Incorrect

Mark 0.00 out of 1.00

Compared to the depression of younger adults, which of the following responds less well to antidepressants?

Select one:

- Depression-executive dysfunction
- Vascular depression and depression-executive dysfunction
- Vascular depression
- Late life depression
- Late life depression, vascular depression, and depression-executive dysfunction

Your answer is incorrect.

EXPLANATION: Late life depression overall, and depression-executive dysfunction (DED) and vascular depression especially, respond less well to antidepressants than depression of younger adults. Late life depression has a less favourable response to antidepressants in part because large subgroups (i.e., DED syndrome, vascular depression) have a poor response to these agents. The depression profile of DED is characterised by anhedonia, psychomotor retardation, pronounced disability, lack of insight, and suspiciousness but less prominent depressive ideation and a mild vegetative syndrome. This presentation is consistent with disruption of frontal-subcortical networks.

Ref: Alexopoulos GS. Mechanisms and treatment of late-life depression. Translational psychiatry. 2019 Aug 5;9(1):1-6.

The correct answer is: Late life depression, vascular depression, and depression-executive dysfunction

Question **89**

Correct

Mark 1.00 out of 1.00

An 80-year-old man with late-onset psychosis has a history of falls due to postural hypotension. Which of the following drugs is most likely to cause orthostatic hypotension?

Select one:

- Lurasidone
- Amisulpride
- Quetiapine
- Olanzapine
- Haloperidol

Your answer is correct.

EXPLANATION: Orthostatic hypotension (postural hypotension) is one of the most common cardiovascular adverse effects of antipsychotics. Antipsychotics with a high affinity for postsynaptic alpha-1-adrengeric receptors are most frequently implicated. Among the SGAs, the reported incidence is highest with clozapine (24%), quetiapine (27%), and iloperidone (19.5%). It is lowest with lurasidone (<2%) and asenapine (<2%). Data is limited for FGAs, but low-potency phenothiazines (e.g. chlorpromazine) are more likely to cause orthostatic hypotension.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 160-161.

The correct answer is: Quetiapine

Question **90**

Correct

Mark 1.00 out of 1.00

A 60-year-old man with Parkinson disease and depression starts spending large amounts of money on gambling. His distressed wife reports that in the past, he only gambled on occasion, and never spent much money at all. Which of the following drugs is most likely to account for his current presentation?

Select one:

- Propranolol
- Ropinirole
- Diazepam
- Dothiepin
- Duloxetine

Your answer is correct.

EXPLANATION: Impulse control disorders (ICD) are common in patients with Parkinson disease (PD). This can include compulsive gambling, shopping, eating, and hypersexual behaviours. These can occur in isolation or together. In Parkinson disease, they are associated with dopamine agonist therapy, such as ropinirole. Left unrecognised and untreated, impulse control disorders can have very significant ramifications for personal relationships and finances. Individuals at higher risk for developing impulse control disorders in Parkinson disease are men, those with certain personality traits or psychiatric disturbance (such as impulsivity, novelty-seeking, anxiety and depression), younger patients, smokers, and those taking high doses of dopamine agonists for a longer duration.

Ref: Sarangi A et al. Treatment and Management of Sexual Disinhibition in Elderly Patients With Neurocognitive Disorders. Cureus. 2021 Oct 3;13(10).

Jones S et al. Neuropsychiatric symptoms in Parkinson's disease: aetiology, diagnosis and treatment. BJPsych Advances. 2020:1-10.

The correct answer is: Ropinirole

Question **91**

Correct

Mark 1.00 out of 1.00

Overall, what percentage of Alzheimer disease patients carry an ApoE4 allele?

Select one:

- >50%
- 25%
- 33%
- 5%
- 10%

Your answer is correct.

EXPLANATION: Overall, more than 50% of Alzheimer disease patients carry one E4 allele. Not only does the E4 allele put the individual at risk for developing the disease, but it also hastens the appearance of the symptoms. For example, those carrying one E4 allele show Alzheimer disease symptoms 5-10 years earlier, and those with two E4 alleles show the symptoms 10-20 years earlier. On the other hand, having one or even two E4 alleles is neither necessary nor sufficient for developing Alzheimer disease.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 123.

The correct answer is: >50%

Question **92**

Correct

Mark 1.00 out of 1.00

For the past 5 months, a 78-year-old man with dementia has displayed agitation and behavioural problems. He has been taking risperidone 1mg twice a day for the past few months. In addition, he was prescribed citalopram 20mg a few days ago. His behavioural problems and agitation are unchanged. What should be the next step in his management?

Select one:

- Slow withdraw of risperidone
- Increase risperidone to 1.5mg bd
- Increase citalopram to 40mg
- Trial of low dose PRN haloperidol
- Trial of regular, low dose haloperidol

Your answer is correct.

EXPLANATION: Risperidone and haloperidol are the only drugs licensed in the UK for the management of non-cognitive symptoms associated with dementia. Due to the serious adverse effects of haloperidol, risperidone is the agent of choice. Risperidone is licensed up to 1mg twice a day, although the optimal dose in dementia has been found to be 500mg twice a day (1mg daily). Risperidone is specifically indicated for short-term treatment (up to 6 weeks) of persistent aggression. Review of the drug needs to be done at 4-6 weeks (or earlier in inpatients), then at 3 months and every 6 months. Consideration should be given to stopping the antipsychotic at each review. In this particular case, given the ineffectiveness of risperidone at maximum dose for a few months, it should be withdrawn.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 647-648.

The correct answer is: Slow withdraw of risperidone

Question 93

Incorrect

Mark 0.00 out of 1.00

In large epidemiological studies, which of the following is the biggest modifiable risk factor in later life contributing to population attributable risk (PAR) for dementia?

S[redacted]ct one:

- Physical inactivity
- Air pollution
- Smoking
- Alcohol misuse
- Social isolation

Your answer is incorrect.

EXPLANATION: A growing body of evidence supports several potentially modifiable risk factors for dementia. Risk factors in early life (less education), midlife (hypertension, obesity, hearing loss, traumatic brain injury, and alcohol misuse), and later life (smoking, depression, physical inactivity, social isolation, diabetes, and air pollution) can contribute to increased dementia risk. Of the later life risk factors, smoking has the highest population attributable fraction. Stopping smoking, even in later life, ameliorates this risk.

Ref: Livingston G et al. Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. The Lancet. 2020 Aug 8;396(10248):413-46.

The correct answer is: Smoking

Question **94**

Correct

Mark 1.00 out of 1.00

Which of the following is an irreversible cause of dementia?

Select one:

- Chronic subdural haematoma
- Wilson disease
- Normal pressure hydrocephalus
- Motor neuron disease
- Pellagra

Your answer is correct.

EXPLANATION: Normal pressure hydrocephalus, chronic subdural haematoma, and pellagra (niacin deficiency) are all reversible causes of cognitive impairment/dementia. Early diagnosis and treatment can also reverse the manifestations of Wilson disease, including cognitive impairment/dementia. No current treatment can cure or even arrest motor neuron disease/amyotrophic lateral sclerosis. About 10-20% of patients with motor neuron disease meet the criteria for dementia. This group generally displays some clinical and pathological features of frontotemporal dementia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 129, 154-155.

Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 69,433.

The correct answer is: Motor neuron disease

Question **95**

Correct

Mark 5.00 out of 5.00

Findings on neuroimaging studies

For each of the following diseases, choose the characteristic neuroimaging finding.

1. Normal pressure hydrocephalus [redacted]
2. Alzheimer disease [redacted]
3. Dementia with Lewy bodies [redacted]
4. Huntington disease [redacted]
5. Parkinson disease [redacted]

CORRECT ANSWERS:

1. Normal pressure hydrocephalus – CT/MRI show ventricular dilation, particularly of the temporal horns, and thinning of the cortex
2. Alzheimer disease – SPECT shows temporal and posterior parietal hypoperfusion
3. Dementia with Lewy bodies – CT/MRI show relative sparing of medial temporal lobes; frequent periventricular lucencies on MRI
4. Huntington disease – CT/MRI show atrophy of the basal ganglia, with ‘boxing’ of the caudate and dilatation of the ventricles
5. Parkinson disease – PET shows asymmetrically decreased dopamine activity in the basal ganglia

EXPLANATION:

- In normal pressure hydrocephalus, CT/MRI show ventricular dilation, particularly of the temporal horns, and thinning of the cortex.
- In Alzheimer disease, CT shows cortical atrophy, especially over parietal and temporal lobes, and ventricular enlargement. MRI shows atrophy of the grey matter. SPECT shows temporal and posterior parietal hypoperfusion.
- In dementia with Lewy bodies, CT/MRI show relative sparing of medial temporal lobes in most cases. Moderate increases in deep white matter lesions and frequent periventricular lucencies are seen on MRI.
- In Huntington disease, CT/MRI show atrophy of the basal ganglia, with 'boxing' of the caudate and dilatation of the ventricles. PET shows decreased metabolism in the basal ganglia.
- In Parkinson disease (PD), PET using radiolabeled fluorodeoxyglucose or fluorodopa usually shows asymmetrically decreased dopamine activity in the basal ganglia in pre-symptomatic individuals and in those with overt PD.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 154, 157, 162.
Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 130, 421.

Question 96

Partially correct

Mark 5.00 out of 6.00

Prevention in mental health

Choose the most appropriate term(s) from the options for each prevention scenario given below. Each option can be used more than once or not at all.

1. For individuals identified as having an "at-risk mental state" for psychosis, a prevention strategy is implemented offering Cognitive Behavioural Therapy (CBT) (choose TWO) Indicated prevention ✔ Secondary prevention ✘

2. A school program aimed at providing educational and support services to low-income families to reduce the risk factors associated with mental health problems in children (choose TWO) Selective prevention ✔ Primary prevention ✔

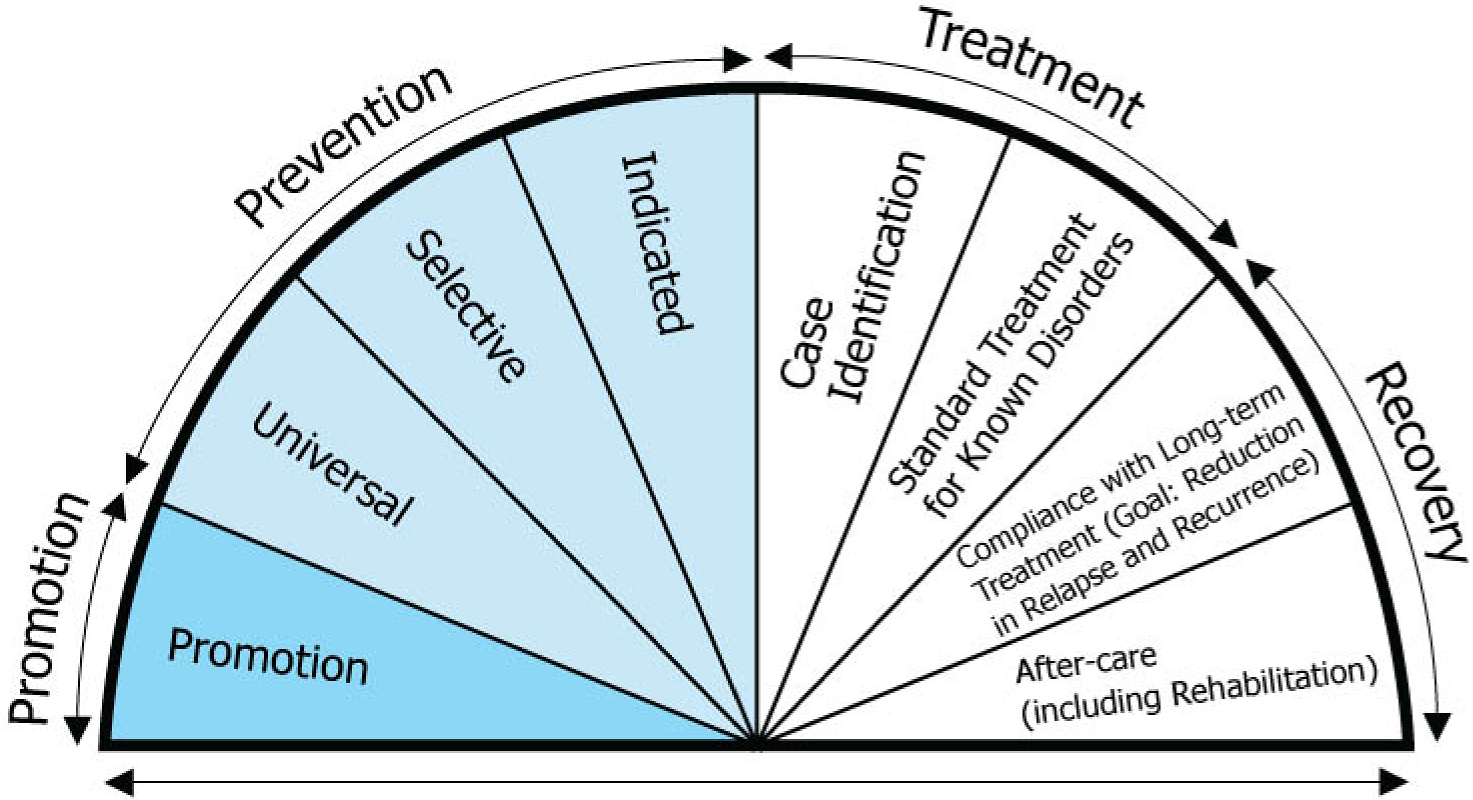
3. A one-day school-wide retreat program is designed to prevent bullying among students (choose TWO) Universal prevention ✔ Primary prevention ✔

CORRECT ANSWERS:

1. For individuals identified as having an "at-risk mental state" for psychosis prevention strategy is being implemented via offering Cognitive Behavioural Therapy (CBT) (choose TWO) - Primary prevention; Indicated prevention
2. A school program aimed at providing educational and support services to low-income families to reduce the risk factors associated with mental health problems in children. (choose TWO) - Primary prevention; Selective prevention
3. A one-day schoolwide retreat program is designed to prevent bullying among students. (choose TWO) - Primary prevention; Universal prevention

EXPLANATION:

Prevention strategies in mental health are categorized into various levels, including primary (BEFORE the onset of an illness), secondary (as soon as the illness begins), and tertiary (targeting those with established conditions).



Primary prevention strategies target a population or community to prevent the initial occurrence of a health condition. These can be selective, as in the school program for low-income families that aims to reduce risk factors before any specific mental health problems develop, or universal as in the program aimed at reducing bullying behaviour among all students irrespective of age, gender or prior reports of bullying behaviour. Indicated primary prevention strategies target individuals who are at heightened risk of developing a specific condition. In this case, individuals with an "at-risk mental state" for psychosis fall into this category, and providing CBT at this stage can help prevent the onset of full-blown psychosis. Promotion refers to strategies that create supportive environment and conditions to improve overall health and not focussed on any specific problem group or behaviour. As we move from promotion to tertiary preventions (e.g., rehabilitation), the number of direct beneficiaries will reduce but the specificity of the interventions will increase.

Ref: Figure from Dr. Audrey Begun https://ohiostate.pressbooks.pub/substancemisusepart1/chapter/ch-2-name-5/

Question 97

Correct

Mark 3.00 out of 3.00

Management of medication side effects

Choose the best medication to be added in the management of each of the following scenarios.

1. A patient taking clozapine gains substantial weight. Metformin ✔

2. A patient with schizophrenia and diabetes develops subjective restlessness in his legs. Propranolol ✔

3. A patient receiving treatment for schizophrenia experiences drooling and tightening of the facial muscles. Propantheline ✔

CORRECT ANSWERS:

1. A patient taking clozapine gains substantial weight - Metformin
2. A patient with schizophrenia and diabetes develops subjective restlessness in his legs - Propranolol
3. A patient receiving treatment for schizophrenia experiences drooling and tightening of the facial muscles - Propantheline

EXPLANATION:

Metformin is now considered to be the drug of choice for the prevention and treatment of antipsychotic-induced weight gain, although GLP-1 agonists may ultimately prove more effective and better tolerated.

There is limited evidence on the benefit-risk balance for any pharmacological treatment for akathisia, even those most commonly used, such as switching to an antipsychotic medication with a lower liability for the condition, or adding a beta-adrenergic blocker, a 5-HT2A antagonist or anticholinergic agent. The first-choice medication to add is usually low-dose propranolol, provided there are not contraindications (e.g. asthma, bradycardia, hypotension, etc.)

There are no medications licenced for the treatment of antipsychotic (especially clozapine)-induced hypersalivation, and many of the relevant published studies have limitations that preclude any confident treatment recommendations. The evidence, such as it is, tends to favour anti-muscarinic agents such as propantheline and diphenhydramine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 114-115, 126, 227-229.

Question 98

Correct

Mark 4.00 out of 4.00

Management of medication side effects

Choose the best medication to be added in the management of each of the following scenarios.

1. A patient taking clozapine gains substantial weight. Metformin

2. A patient with schizophrenia and diabetes develops subjective restlessness in his legs. Propranolol

3. A patient receiving treatment for schizophrenia experiences drooling and tightening of the facial muscles. Propantheline

4. A patient with longstanding schizophrenia develops involuntary, repetitive movements of her tongue and lips. Tetrabenazine

EXPLANATION: Metformin is now considered to be the drug of choice for the prevention and treatment of antipsychotic-induced weight gain, although GLP-1 agonists may ultimately prove more effective and better tolerated.

There is limited evidence on the benefit-risk balance for any pharmacological treatment for akathisia, even those most commonly used, such as switching to an antipsychotic medication with a lower liability for the condition, or adding a beta-adrenergic blocker, a 5-HT2A antagonist, or anticholinergic agent. The first-choice medication to add is usually low-dose propranolol, provided there are not contraindications (e.g. asthma, bradycardia, hypotension, etc.)

There are no medications licenced for the treatment of antipsychotic (especially clozapine)-induced hypersalivation, and many of the relevant published studies have limitations that preclude any confident treatment recommendations. The evidence, such as it is, tends to favour anti-muscarinic agents such as propantheline and diphenhydramine.

Tetrabenazine (reversible VMAT2 inhibitor) is the only licensed treatment for moderate to severe TD in the UK. Depression, drowsiness, parkinsonism, and akathisia may occur. CYP2D6 genotyping may be required when higher doses are used. Reserpine is also effective but is rarely, if ever, used. Reserpine produces irreversible and nonselective, inhibition of both VMAT1 and VMAT2, resulting in blood pressure reduction, notable issues with gastric motility, impotence, and risk of prolonged depression, though some early evidence indicated a positive effect on psychotic symptoms per se.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 114-115, 118, 126, 227-229. Caroff SN. Recent advances in the pharmacology of tardive dyskinesia. Clinical Psychopharmacology and Neuroscience. 2020 Nov 11;18(4):493.

CORRECT ANSWERS:
1. – Metformin
2. – Propranolol
3. – Propantheline
4. – Tetrabenazine

Question 99

Partially correct

Mark 5.00 out of 8.00

Prevention in mental health

Choose the most appropriate term(s) for each prevention scenario. Each option can be used more than once or not at all.

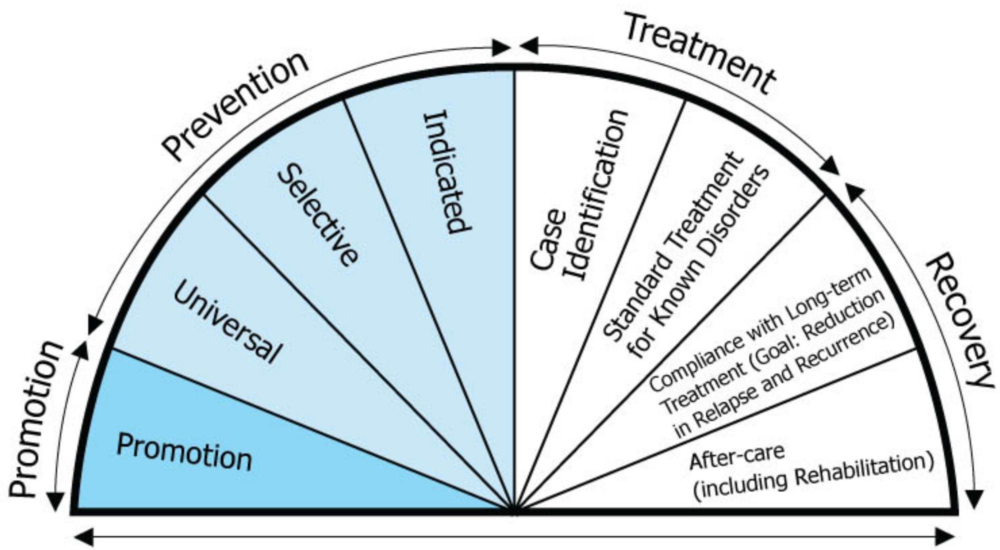
1. A boy identified as having an "at-risk mental state" for psychosis is offered Cognitive Behavioural Therapy (CBT). (choose TWO) Indicated prevention ✔ Secondary prevention ✘

2. Patients with bipolar I disorder who have attempted suicide are offered lithium with the aim of preventing further suicide attempts. (choose TWO) Secondary prevention ✘ Indicated prevention ✘

3. A one-day schoolwide retreat program is designed to prevent bullying among students. (choose TWO) Primary prevention ✔ Universal prevention ✔

4. A suicide helpline is established and advertised to the general public. (choose TWO) Secondary prevention ✔ Universal prevention ✔

EXPLANATION:



Prevention strategies in mental health are categorized into various levels, including primary (before the onset of an illness), secondary (as soon as the illness begins), and tertiary (targeting those with established conditions). Primary prevention strategies target a population or community to prevent the initial occurrence of a health condition. These can be selective, as in a school program for low-income families that aims to reduce risk factors before any specific mental health problems develop, or universal as in a program aimed at reducing bullying behaviour among all students irrespective of age, gender, or prior reports of bullying behaviour. Indicated primary prevention strategies target individuals who are at heightened risk of developing a specific condition. Individuals with an "at-risk mental state" for psychosis fall into this category, and providing CBT at this stage can help prevent the onset of full-blown psychosis. Promotion refers to strategies that create supportive environments and conditions to improve overall health and are not focussed on any specific problem group or behaviour. As we move from promotion to tertiary preventions (e.g., rehabilitation), the number of direct beneficiaries will reduce but the specificity of the interventions will increase.

Ref: Figure from Dr. Audrey Begun https://ohiostate.pressbooks.pub/substancemisusepart1/chapter/ch-2-name-5/ , Singh V, Kumar A, Gupta S. Mental health prevention and promotion—A narrative review. Frontiers in Psychiatry. 2022 Jul 26;13:898009.

CORRECT ANSWERS:
1. – Indicated, Primary.
2. – Selective, Tertiary.
3. – Primary, Universal.
4. – Secondary, Universal.

Question **100**

Partially correct

Mark 2.00 out of 3.00

Effects of drugs in pregnancy

Which drug, when taken by a mother while pregnant, is most likely to be implicated in each case scenario?

1. 1-year-old child with cleft palate and delayed milestones Sodium valproate ✔

2. 10-year-old child with hyperactivity and distinct facial features, including small eye fissures, an underdeveloped philtrum, and thin upper lip Sodium valproate ✘

3. Jittery newborn with overarousal and tachypnoea 2 hours after delivery Cocaine ✔

EXPLANATION: There is strong evidence that valproate is associated with increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language, lower intellectual ability), and autism.

Clinical features of fetal alcohol spectrum disorder include microcephaly, abnormal facial features (small eye fissures, epicanthic folds, short palpebral fissure, small maxillae and mandibles, underdeveloped philtrum, cleft palate, thin upper lip), growth deficits, mild intellectual disability, and associated behavioural problems (such as hyperactivity and sleep problems).

Stimulant use (such as cocaine) prior to delivery is associated with a neonatal withdrawal syndrome including symptoms of agitation, vomiting, and tachypnoea.

Ref: McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.
National Institute for Health and Clinical Excellence. Antenatal and postnatal mental health: clinical management and service guide. Updated

edition 2020.
Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 690-691.
Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019. p. 818.

CORRECT ANSWERS: (1) – Sodium valproate; (2) – Alcohol; (3) – Cocaine

Question **101**

Correct

Mark 3.00 out of 3.00

ICD-11 personality disorders qualifiers (I)

Identify the most appropriate ICD-11 personality disorder qualifier applicable to each case scenario.

1. A woman comes to clinic at the insistence of her new partner who is concerned that the woman has been rapidly fired from several jobs. During the appointment, the woman is easily distracted, constantly on her phone, says she doesn't care about those 'boring' jobs, and impulsively books an expensive flight for an international holiday. Disinhibition ✓

2. A reserved 45-year-old man lives with his mother and spends most of his time on the computer. He works from home and seldom leaves the house. He has never had an intimate relationship and has no friends. Detachment ✓

3. A 48-year-old man enjoys making denigrating comments about those around him. He believes that he is very smart and savvy, and 'collects' young women on social media who seem to admire him. Though he portrays himself as caring, he takes pleasure in inflicting harm and can quickly become violent. Dissociality ✓

EXPLANATION: Common manifestations of disinhibition include impulsivity, irresponsibility, recklessness, and lack of planning. Individuals high on disinhibition also have difficulty staying focused on tasks that require sustained effort. They quickly become bored or frustrated and are easily distracted.
Common manifestations of detachment include social detachment (avoidance of social interactions, lack of friendships, and avoidance of intimacy) and emotional detachment (reserve, aloofness, and limited emotional expression and experience). These individuals seek out employment that does not involve interactions with others and they have few to no friends.
Common manifestations of dissociality include self-centredness (e.g., sense of entitlement, expectation of others' admiration, positive or negative attention-seeking behaviours, concern with own's own needs, desires, and comfort and not those of others) and lack of empathy (i.e., being deceptive, manipulative, and exploitative of others, being mean and physically aggressive, callousness in response to others' suffering, and ruthlessness in obtaining one's goals).

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)
Bach B. et al. Entering the Brave New World of ICD-11 Personality Disorder Diagnosis. Frontiers in Psychiatry. 2021 Nov 17;12:793133.

CORRECT ANSWERS:
1. – Disinhibition
2. – Detachment
3. – Dissociality

Question 102

Partially correct

Mark 2.00 out of 3.00

ICD-11 personality disorders qualifiers (II)

Identify the most appropriate ICD-11 personality disorder qualifier applicable to each case scenario.

1. A 46-year-old woman works long hours at the office every day, redoing the tasks of others because they do not meet her high standards. She stubbornly insists on following set schedules, even when unforeseen circumstances arise. Anankastia ✔

2. A 38-year-old man is almost always anxious and pessimistic. He has a strong tendency to ruminate about his past mistakes, as well as potential future problems. He frequently gets upset in meetings and usually needs to leave the room to calm down. Disinhibition ✘

3. A 28-year-old woman engages in repetitive self-harm. She has had many boyfriends, but the relationships have been brief, and have usually ended in screaming fights. In times of stress, she dissociates and experiences brief hallucinations. Borderline pattern ✔

EXPLANATION: Common manifestations of anankastia include perfectionism, as well as emotional and behavioural constraint. These individuals may redo the work of others because it does not meet their perfectionistic standards. They are inflexible and lack spontaneity, stubbornly following set schedules and adhering to plans. They often perseverate, having difficulty disengaging from tasks because they are perceived as not yet perfect down to the last detail.

Common manifestations of negative affectivity include experiencing a broad range of negative emotions with a frequency and intensity out of proportion to the situation; emotional lability and poor emotion regulation; negativistic attitudes; low self-esteem and self-confidence; and mistrustfulness. These individuals can ruminate over their shortcomings or past mistakes, over real or perceived threats/slights/insults, or over future problems. Once upset, they have difficulty regaining their composure.

The borderline pattern specifier is characterised by a pervasive pattern of instability of interpersonal relationships, self-image, and affects, and marked impulsivity. There may be recurrent episodes of self-harm. Displays of temper may include yelling or screaming, throwing or breaking things, and/or getting into physical fights. These individuals may experience transient dissociative symptoms or psychotic-like features in situations of high affective arousal.

Of note, both ICD-11 and DSM-5 have five broad trait domains, but share only four (negative affectivity, detachment, disinhibition, and a fourth called antagonism in DMS-5 or dissociality in ICD-11). So, each model poses a fifth trait domain that the other does not. The ICD-11 trait model does not include the psychoticism domain (concerning proneness to eccentric or delusional behaviours, cognitions, and thought processes). In turn, the DSM-5 trait model does not include the domain of anankastia (concerning perfectionism, rigidly sticking to the norm and obligations, or emotional and behavioural constraint).

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)
Bach B. et al. Entering the Brave New World of ICD-11 Personality Disorder Diagnosis. Frontiers in Psychiatry. 2021 Nov 17;12:793133.
Strus W. et al. Anankastia or Psychoticism? Which one is better suited for the fifth trait in the pathological Big Five: Insight from the Circumplex of Personality Metatraits perspective. Frontiers in Psychiatry. 2021 Oct 14;12:1764.

CORRECT ANSWERS:
1. – Anankastia
2. – Negative affectivity
3. – Borderline pattern

Question 103

Partially correct

Mark 4.00 out of 6.00

Definitions in Forensic Psychiatry

Match each forensic term with its definition.

1. Actus reus The (voluntary) act of crime
2. Mens rea The (evil) intent of crime
3. Culpable homicide Crime committed without intent
4. Involuntary Manslaughter
5. Voluntary Crime committed with intent
6. Indictable offence A serious offence tried in Crown court

CORRECT ANSWERS:

1. Actus reus – The (voluntary) act of crime
2. Mens rea – The (evil) intent of crime
3. Culpable homicide – Manslaughter
4. Involuntary – Crime committed without intent
5. Voluntary – Crime committed with intent

6. Indictable offence – A serious offence tried in Crown court

EXPLANATION:

According to criminal law, committing a socially harmful act is not the sole criterion of whether a crime has been committed. Instead, the objectionable act must have two components: actus reus and mens rea. Actus reus refers to the (voluntary) act of crime. Mens rea refers to the (evil) intent of crime. An evil intent cannot exist when an offender's mental status is so deficient, so abnormal, or so diseased to have deprived the offender of the capacity for rational intent. Neither behaviour, however harmful, nor the intent to harm, alone, is grounds for criminal action.

Culpable homicide (manslaughter) may be voluntary or involuntary. When voluntary, killing is done with the intent to murder, but partial defence applies (e.g. suicide pact, severe provocation). When involuntary, the conduct was either grossly negligent (given the risk of death), or the conduct was an unlawful act involving a danger or harm that resulted in death. An indictable offence is a serious offence tried in Crown court.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 843.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 732.

Question 104

Correct

Mark 5.00 out of 5.00

Statistical Concepts and Definitions

You are reviewing research papers in the field of psychiatry. Each paper mentions a statistical concept or term. Match the statistical concepts or terms below with correct descriptions. Each term can be used more than once or not at all.

1. The probability of making an incorrect rejection of a null hypothesis when it is true. [redacted]
2. The probability of correctly detecting a true effect or difference in a study. [redacted]
3. The level of statistical significance used to determine whether to reject the null hypothesis. [redacted]
4. The probability of failing to detect a true effect or difference when it exists. [redacted]
5. A measure of the likelihood that the results of a study occurred by chance. [redacted]

CORRECT ANSWERS:

1. The probability of making an incorrect rejection of a null hypothesis when it is true. - Type 1 Error
2. The probability of correctly detecting a true effect or difference in a study. - Power
3. The level of statistical significance used to determine whether to reject the null hypothesis. - Significance Level (α)
4. The probability of failing to detect a true effect or difference when it exists. - Type 2 Error
5. A measure of the likelihood that the results of a study occurred by chance. - p-value

EXPLANATION:

- The probability of making an incorrect rejection of a null hypothesis when it is actually true is referred to as "Type 1 Error". This error occurs when a statistical test incorrectly indicates a significant result when there is no real effect or difference.
- The probability of correctly detecting a true effect or difference in a study is known as "Power". Power represents the ability of a statistical test to correctly identify a significant result when a true effect exists. Higher power indicates a better chance of detecting real effects.
- The level of statistical significance used to determine whether to reject the null hypothesis is referred to as the "Significance Level (a)". Commonly used significance levels include 0.05 (5%) and 0.01 (1%).
- The probability of failing to detect a true effect or difference when it exists is known as "Type 2 Error". This error occurs when a statistical test fails to identify a significant result when there is a real effect. Type 2 Error is the complement of Power (i.e., 1- Power = type 2 error rate).
- A measure of the likelihood that the results of a study occurred by chance is represented by the "p-value". The p-value quantifies the probability of obtaining observed results (or more extreme results) if the null hypothesis is true. Smaller p-values suggest stronger evidence against the null hypothesis.

Question **105**

Correct

Mark 4.00 out of 4.00

Theme: Costs and economic studies
Lead in: You are conducting an economic study comparing the cost-effectiveness of medication-based treatment to non-medication-based treatment (residential care in a therapeutic open community) for chronic PTSD. For each of the question below, choose the appropriate description from the provided list.

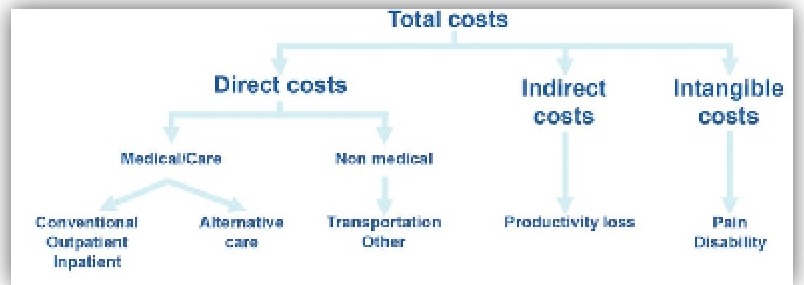
1. You want to describe the additional cost incurred for the medication-based treatment over the non-medication-based treatment. [redacted]

2. A junior doctor is temporarily reassigned to the therapeutic open community unit that is providing the non-medication-based treatment. As this unit is at a geographically distant location, the hospital is paying for the commuting expenses of the doctor. Which cost concept best represents the financial impact of this process? [redacted]

3. Patients receiving the therapeutic open community care are required to be absent from work due the need for residential stay, resulting in a loss of productivity and wages. What type of cost is associated with the days lost to sickness? [redacted]

4. 4.During the trial, patients who receive medications that result in excessive sleep report being stigmatised by their family members for being inactive. Some of them find this distressing and believe that this contributes to difficulties in making living arrangements. What type of cost is associated with this phenomenon? [redacted]

Explanation:



- Incremental cost refers to the additional cost incurred when comparing two alternative treatments or interventions. In this case, it quantifies the extra cost associated with using medication-based treatment compared to non-medication-based treatment.

  Direct cost represents the specific, measurable expenses incurred because of a particular activity in the course of provision of care. In this scenario, the additional commuting expenses directly relate to the junior doctor's temporary reassignment, making it a direct cost for the health system. Indirect cost refers to the costs that are not directly tied to a specific activity but arise because of that activity. In this case, the days lost to sickness result in a loss of productivity and wages for the patient, which are indirect costs of the treatment, borne by the patients. Intangible cost is a 'conceptual' cost, not something that is readily quantifiable in euros or pounds but are still significant and impactful. It represents costs that are challenging to quantify in monetary terms because they involve non-material aspects, such as emotional distress or stigma. The economic impact of stigma associated with mental health conditions often falls into the category of intangible costs.

  The attached figure is from Fautrel B, Boonen A, de Wit M, Grimm S, Joore M, Guillemin F. Cost assessment of health interventions and diseases. RMD open. 2020 Nov 1;6(3):e001287.

Question 106

Correct

Mark 6.00 out of 6.00

Evidence-based psychiatry concepts

In a research study comparing rTMS and ECT for the treatment of major depressive disorder (MDD), the response rates in each treatment arm are provided to calculate the treatment effect. Response Rate in rTMS Group: 60%; Response Rate in ECT Group: 75%. For each question below, choose the correct option(s) from the list provided.

1. The difference in risk of achieving treatment response between two treatment groups, expressed as an absolute percentage (Choose TWO) [redacted]

2. The number of patients needed to be treated with ECT over rTMS to achieve one additional treatment response (Choose TWO)

[redacted]

3. The measure of the likelihood of achieving treatment response with ECT compared to rTMS, based on response rates (Choose TWO)

[redacted]

CORRECT ANSWERS:

1. The difference in risk of achieving treatment response between two treatment groups, expressed as an absolute percentage (Choose TWO) - Absolute risk reduction; 15%
2. The number of patients needed to be treated with ECT over rTMS to achieve one additional treatment response (Choose TWO) - Number Needed to Treat; 7
3. The measure of the likelihood of achieving treatment response with ECT compared to rTMS, based on response rates (Choose TWO) - Relative Risk; 1.25

EXPLANATION:

- Absolute Risk Reduction (ARR) is the difference in risk of achieving treatment response between the two treatment groups, which can be calculated by subtracting the response rate in the rTMS group from the response rate in the ECT group (75% - 60% = 15%).
- Number Needed to Treat (NNT) represents the number of patients needed to be treated with one therapy (ECT) over the other (rTMS) to achieve one additional treatment response, which can be calculated as the reciprocal of the ARR (1/ARR = 1/0.15 = 6.67, rounded to 7 patients).
- Relative Risk (RR) is a measure of the likelihood of achieving treatment response in one therapy group (ECT) compared to the other (rTMS), calculated as the ratio of response rates (RR = Response Rate in ECT / Response Rate in rTMS = 75% / 60% = 1.25).